Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 6 – Software User Guides

Automatic Attendance System Based on Real-Time Location and Face Recognition

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Table of Contents

Definitions & Acronyms **5**
I. Record of Changes **6**
II. Release Package & User Guides **8**
1. Deliverable Package 8
2. Installation Guides 8
2.1 System Requirements 8
2.2 Installation Instructions 9
2.2.1 Install Android 9
2.2.2 Install Front-end Admin Web 14
2.2.3 Back-end Installation 14
3. User Manual 15
3.1 Website 15
3.1.1 Overview 15
3.1.2 Authentication 15
3.1.2.1 Sign in 16
3.1.2.2 Sign out 17
3.1.3 Dashboard 17
3.1.3.1 View Dashboard 17
3.1.4 Room Management 20
3.1.4.1 Manage Rooms 20
3.1.5 Course Management 24
3.1.5.1. Manage Course 24
3.1.6 Class Management 29
3.1.6.1 Manage Classes 29
3.1.6.2 Add New Classes 33
3.1.6.3 Manage Class 35
3.1.6.4 Manage Enrollments 37
3.1.6.5 Manage Schedules 44
3.1.6.6 Manage Sessions 50
3.1.6.7 Manage Session 54
3.1.6.8 Edit Attendance 55
3.1.7 User Management 56
3.1.7.1 Manage Users 56
3.1.7.2 View User Details 63
3.1.8 Active Device Management 64
3.1.8.1 Manage Active Devices 64
3.1.9 Device Request Management 66
3.1.9.1 Manage Device Requests 66
3.1.10 FaceID Management 68
3.1.10.1 View Active FaceIDs 68
3.1.11 Configuration Management 69

3.1.11.1 Manage Attribute Definitions....69
3.1.11.2 Manage Settings....75
3.2 Application - Lecture....79
3.2.1 Overview....79
3.2.1 Authentication....80
3.2.1.1 Sign in....80
3.2.1.2 Log out....81
3.2.2 HomePage....82
3.2.2.1 View HomePage....82
3.2.3 Schedule Management....83
3.2.3.1 View teaching schedule....83
3.2.3.2 View Calendar....85
3.2.3.3 View Class Session....86
3.2.3.4 FaceId request....89
3.2.3.5 End the session....90
3.2.3.6 History....92
3.2.4 Report Management....94
3.2.4.1 View My report....94
3.2.4.2 View report detail....96
3.2.4.3 Update attendance....98
3.2.5 Profile Management....99
3.2.5.1 View lecturer profile....99
3.2.5.2 Notification setting....101
3.2.5.3 Device Management....103
3.3 Application - Student....107
3.3.1 Overview....107
3.3.1 Authentication....107
3.3.1.1 Sign in....107
3.2.1.2 Log out....108
3.2.2 HomePage....109
3.2.2.1 View HomePage....109
3.2.3 Schedule Management....110
3.2.3.1 View schedule....110
3.2.3.2 View Final attendance....112
3.2.3.3 View Calender....113
3.2.3.4 FaceId request....114
3.2.4 Report Management....115
3.2.4.1 View report....115
3.2.4.2 View report detail....116
3.2.5 Profile Management....117
3.2.5.1 View student profile....117
3.2.5.2 Notification setting....118

3.2.5.3 Device Management..... 121
3.2.5.4 Face Id..... 124

List of Figures

Figure No.	Title	Page
Figure 1		
Figure 2		
Figure 3		
Figure 4		

Table 1 - List of Figures

List of Tables

Table No.	Title	Page
Table 1	List of Figures	5
Table 2	List of Tables	5
Table 3	Definitions & Acronyms	6
Table 4	Record of Changes	7
Table 5		
Table 6		
Table 7		
Table 8		
Table 9		
Table 10		
Table 11		

Table 2 - List of Tables

Definitions & Acronyms

No	Acronyms & Abbreviations	Definition
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Table 3 - Definitions & Acronyms

I. Record of Changes

Date	A* M, D	In charge	Change Description
7/8/2025	A	DungND	
14/8/2025	A	DungND	
15/8/2025	A	DungND	
17/8/2025	M	DungND	
18/8/2025	M	DungND	
19/8/2025	M	DungND	

*A - Added M - Modified D - Deleted

II. Release Package & User Guides

1. Deliverable Package

No	Deliverable Item	URL(s)	File Type
1	**Documents**		
1.1	Project Tracking	SEP490_G16_Project Tracking.xlsx	.xlsx
1.2	Project Weekly Report	SEP490_G16_Project_Weekly_Re…	.xlsx
1.3	Project Introduction	SEP490_G16_Report1_Project Int…	.pdf
1.4	Project Management Plan	SEP490_G16_Report2_Project M…	.pdf
1.5	Software Requirements Specification	SEP490_G16_Report3_Software_…	.pdf
1.6	Software Design Description	SEP490_G16_Report4_Software_…	.pdf
1.7	Software Testing Document	SEP490_G16_Report5_Test Docu…	.pdf
1.8	Software User Guides	SEP490_G16_Report6_Software …	.pdf
1.9	Unit Test	Report5.1_Unit Test	.xlsx
1.10	Integration Test	Report5.2_Integration Test	.xlsx
1.11	System Test	Report5.3_System Test	.xlsx
1.12	Acceptance Test	Acceptance Testing (Responses)	.xlsx
1.13	Final Project Report	SEP490_G16_Report7_Final Proje…	.pdf
1.14	Presentation		.pdf
2	**Source Code**		
2.1	API	Github	.zip
2.2	Website	Github	.zip
2.3	Mobile	Github	.zip
3	**Database**		
3.1	Database Script(s)	Database Scripts	.sql

Table 3 - Definitions & Acronyms

2. Installation Guides

2.1 System Requirements

For Develop Android Application:

- **RAM:** At least 16.00GB
- **Operating System:** Windows 10/11 or Ubuntu 22.04 LTS 64-bit
- **Software:** Android Studio
- **Technologies:** Java Android
- **Database Management Tool:** Datagrip/Dbeaver

For Develop Front-end Admin Web:

- **RAM:** At least 8.00GB
- **Operating System:** Windows 10/11 or Ubuntu 22.04 LTS 64-bit
- **Software:** Visual Studio Code, Webstorm
- **Technologies:** ReactJS, Next.js, Git
- **Database Management Tool:** Datagrip/Dbeaver

For Develop Back-end:

- **RAM:** At least 16.00GB
- **Operating System:** Windows 10/11 or Ubuntu 22.04 LTS 64-bit
- **Technologies:** ASP.NET (dotnet core 8.x)
- **Software:** Visual Studio Code, Jetbrain Rider

- **Database Management Tool:** Datagrip/Dbeaver

2.2 Installation Instructions

2.2.1 Install Android

APK Link: https://www.mediafire.com/file/gjoxiyuvjnkb2xu/zentry.apk/file

Step 1: Download the APK
Download the APK file from the link above. Once completed, the file will appear in the Downloads folder with the name **zentry.apk** (150 MB).



Figure 1: Downloaded file zentry.apk displayed in the Downloads list.

Step 2: Start the Installation
Tap on the downloaded file. A system prompt will appear asking:
"Do you want to install this application?"
Select **Install** to continue.

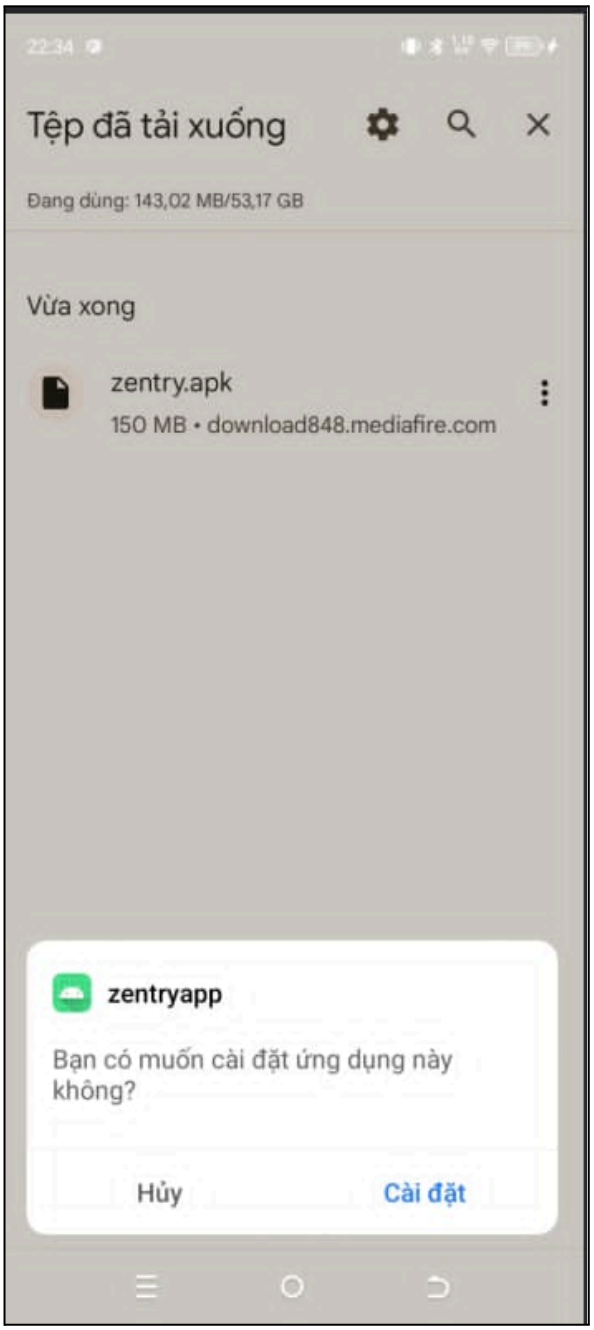


Figure 2: Installation confirmation dialog for zentry.apk.

Step 3: Installing the Application
The installation process will begin. Wait until the progress bar is completed.

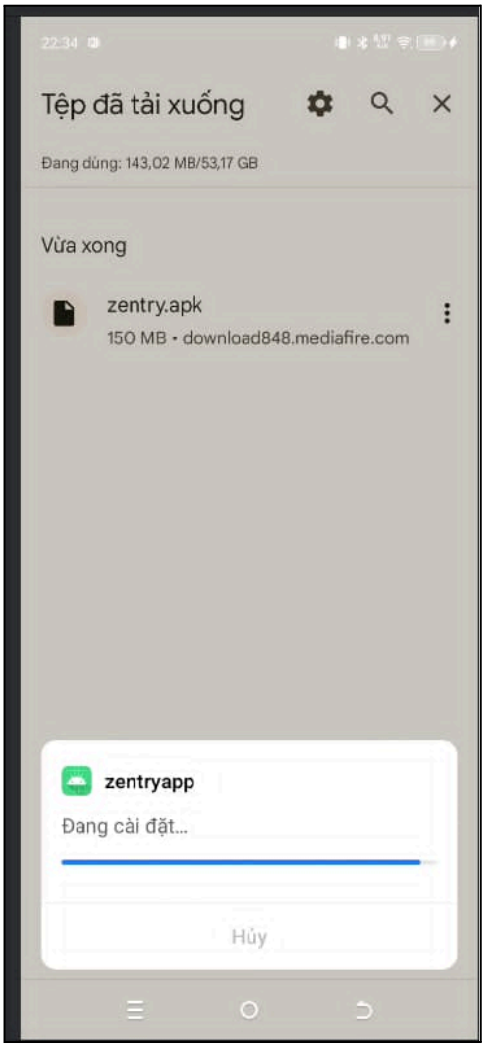


Figure 3: Installation in progress for the Zentry app.

Step 4: Installation Completed
When the installation is finished, the system will display the message:
"App installed."
You may now select:

- **Open** → to launch the app immediately.
- **Done** → to finish and return to the Downloads screen.

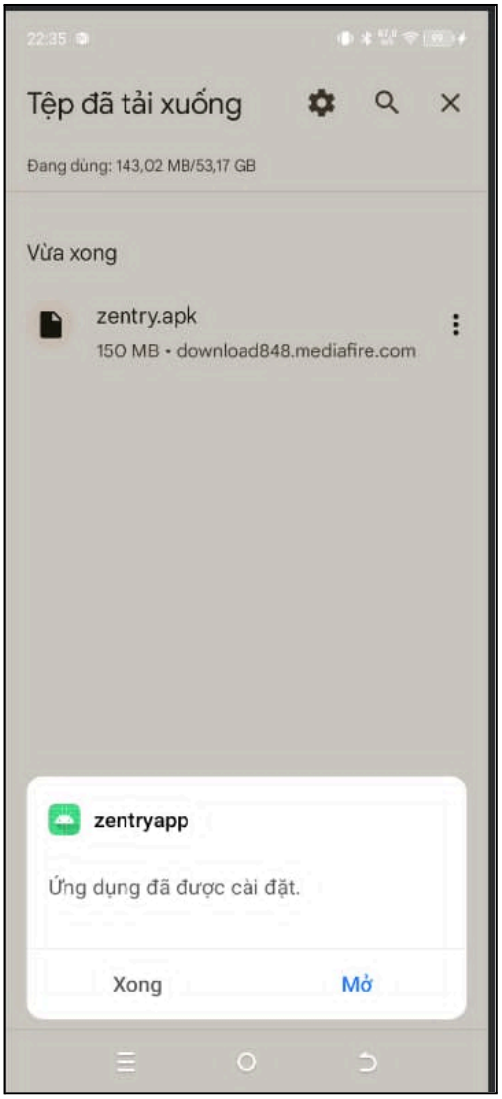


Figure 4: Installation completed notification.

Step 5: Grant App Permissions
Upon first launch, the Zentry app may request access permissions, such as location. You can select from the following options:

- **Allow only while using the app** (recommended)
- **Allow only this time**
- **Deny**

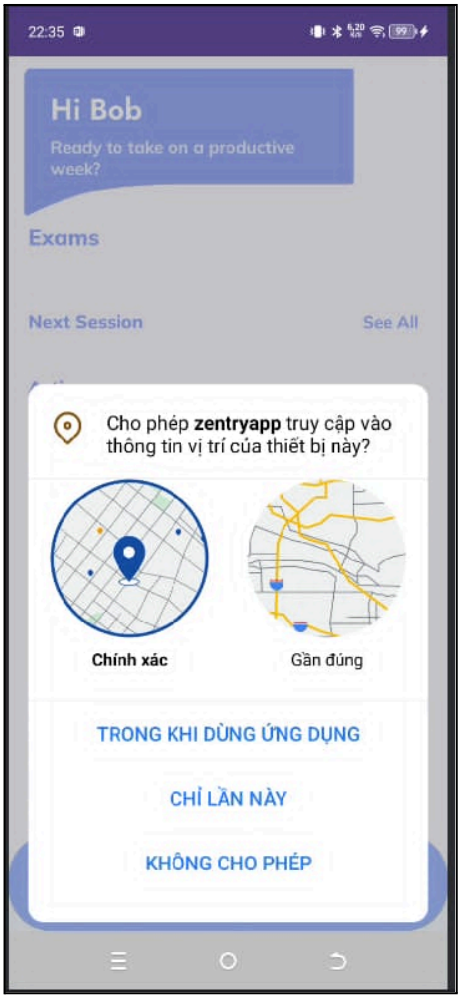


Figure 5: *Location permission request for the Zentry app.*

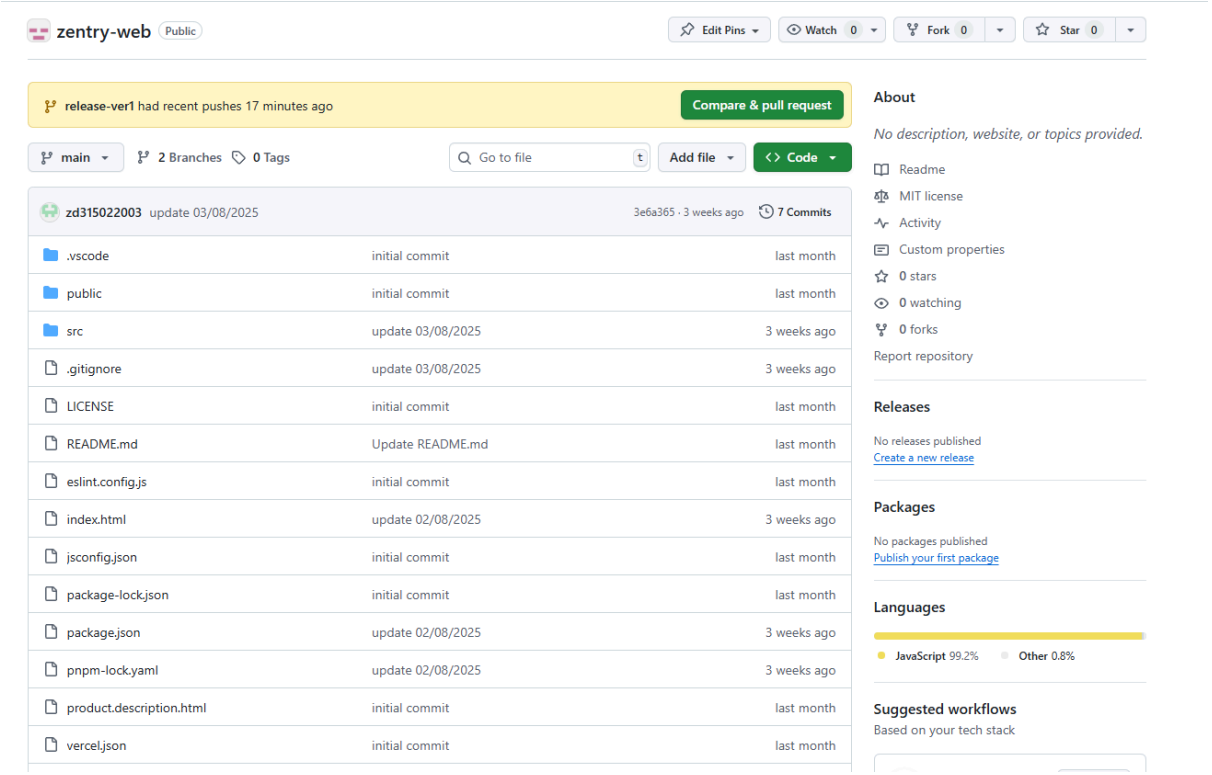
After completing these steps, the **Zentry app** will be installed and ready to use on your device.

2.2.2 Install Front-end Admin Web

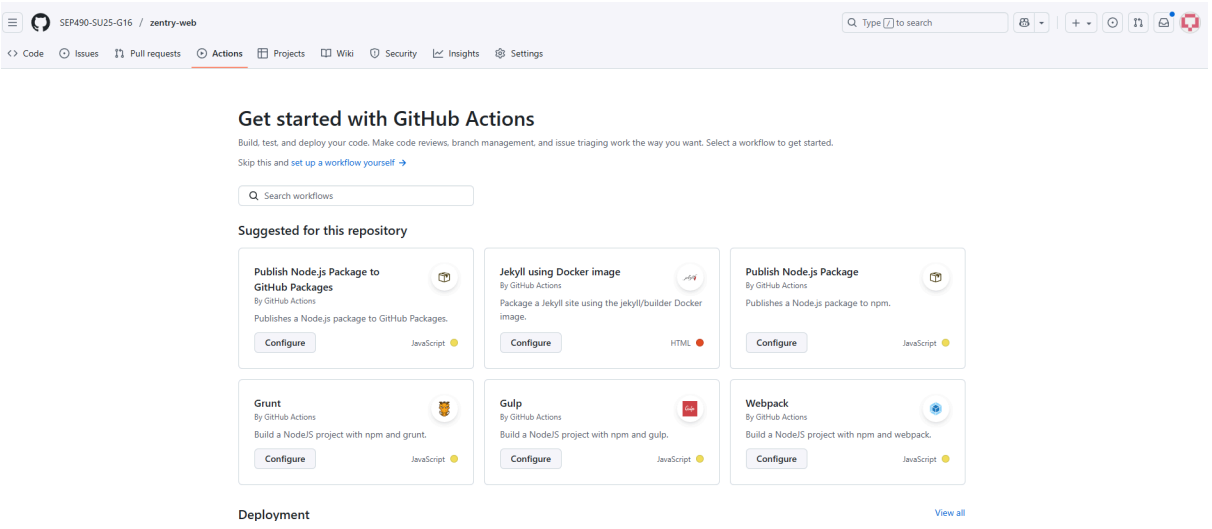
Source code: https://github.com/SEP490-SU25-G16/zentry-web

Steps to install:

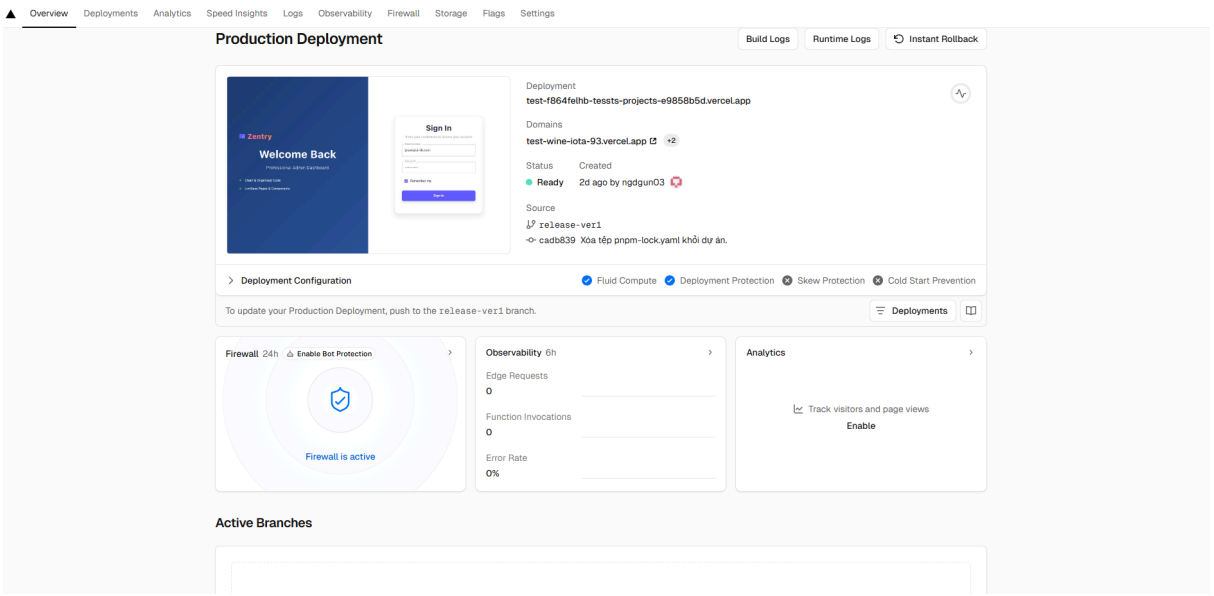
- Step 1: Prepare source code of website on GitHub

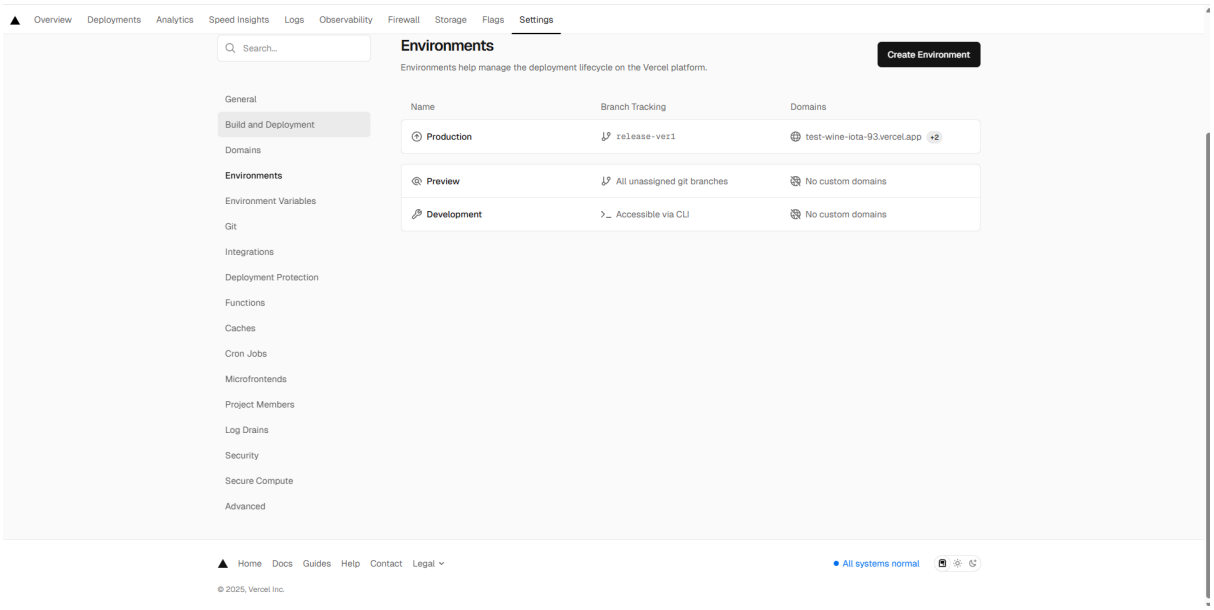


- Step 2: Create pipeline for GitHub action to automatically deploy to Vercel hosting



- Step 3: Deploy on Vercel hosting and check status. To deploy a website on Vercel: prepare your project (Node.js, dependencies), push the code to a Git repo, connect Vercel to your Git provider, configure build settings (framework, build command, output directory), deploy the site, test using the provided URLs, and optionally set up a custom domain or environment variables.





2.2.3 Back-end Installation

1. Install Docker
2. Install docker-compose. Refer to Install | Docker Docs for information on how to install.
3. Clone github repository: https://github.com/SEP490-SU25-G16/zentry-server.git
4. cd to the root directory of the project.
5. Create .env file environment variables if needed or update directly in docker-compose.yml.

```
# ASP.NET Core
ASPNETCORE_ENVIRONMENT=Development

# PostgreSQL
POSTGRES_DB=zentry
POSTGRES_USER=admin
POSTGRES_PASSWORD=pass
```

```
# ConnectionStrings
FACEID_CONNECTION=Host=postgres;Port=5432;Database=zentry;Username=admin;Password=pass;
DEFAULT_CONNECTION=Host=postgres;Port=5432;Database=zentry;Username=admin;Password=pass;

# Redis
REDIS_CONNECTION=redis:6379

# MongoDB
MONGO_CONNECTION=

# RabbitMQ
RABBITMQ_CONNECTION=amqp://admin:pass@rabbitmq:5672/

# JWT
JWT_SECRET=a-super-secret-key-that-is-long-enough-for-hs256-algorithm
JWT_ISSUER=https://zentry.local
JWT_AUDIENCE=https://zentry.local
JWT_EXPIRATION_MINUTES=60
```

6. Update file docker-compose.yml
7. Start the services

```
docker-compose up -d

# view services
docker ps
```

3. User Manual

3.1 Website

3.1.1 Overview

The **Zentry Web-based Administration Portal** is designed for administrators to manage and monitor the overall attendance system.
It provides centralized tools for:

- **Dashboard**: View overall statistics of rooms, courses, classes, devices, and face IDs.
- **Schedule Management**: Create and manage rooms, courses, classes, sessions, and schedules.
- **User Management**: Add, update, and manage lecturers, students, and administrators.
- **Device & FaceID Management**: Approve, reject, or monitor devices and face ID requests.
- **Configuration Management**: Manage system-wide settings and attribute definitions.

This portal is optimized for desktop browsers and ensures that administrators can configure the system efficiently, validate attendance data, and maintain security across all modules.

3.1.2 Authentication

3.1.2.1 Sign in

Description: The **Sign In** screen allows users to enter their login credentials (Email and Password) to access the Zentry Professional Admin Dashboard. There is also a “Remember me” option to save login information for future visits.

Instruction:

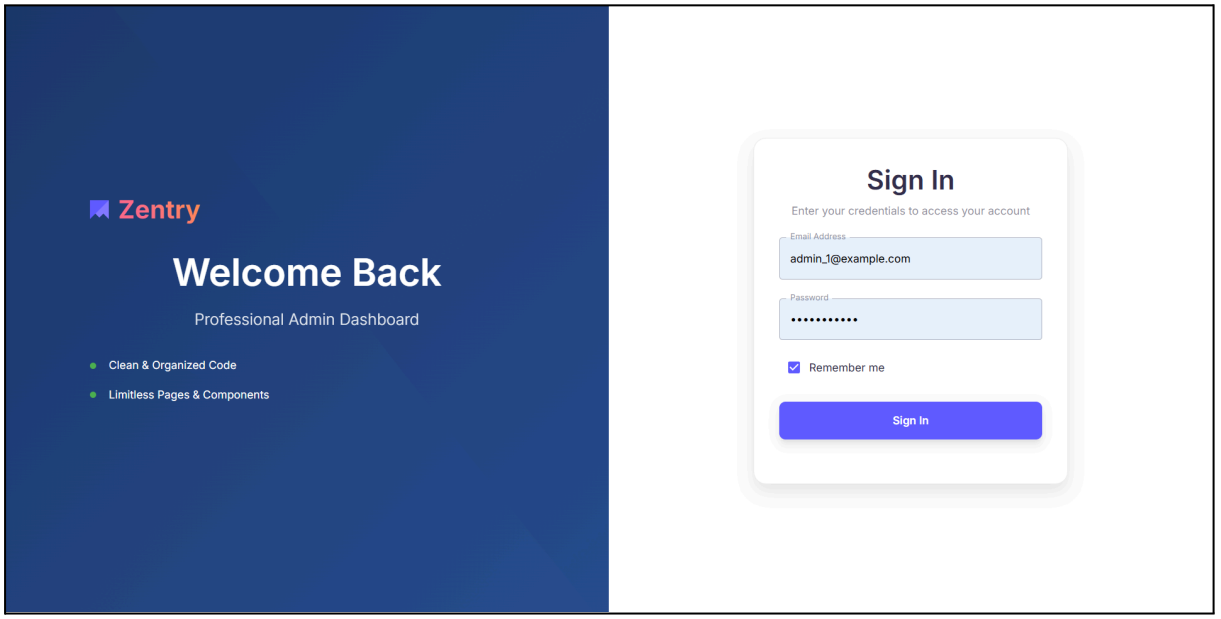


Figure xx - Sign in screen

- **Step 1**: In the **Email Address** field, enter the email address registered with the system.
- **Step 2**: In the **Password** field, enter the corresponding account password.
- **Step 3**: (Optional) Check the **Remember me** box if you want the system to automatically save your login information for the next session.
- **Step 4**: Click the **Sign In** button to submit your credentials.
    - If the information is valid: the system will navigate to the main dashboard page.

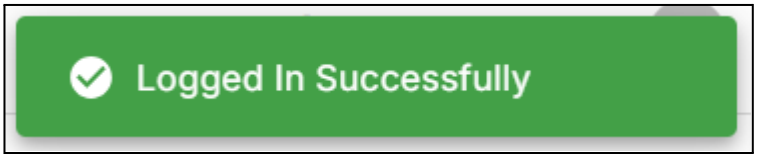


Figure xx - Sign in success message

- If the information is invalid: the system will display an error message and prompt you to re-enter.



Figure xx - Sign in failure message

3.1.2.2 Sign out

Description: The **Sign Out** function allows the user to securely log out of the Zentry Professional Admin Dashboard. This ensures that no unauthorized person can access the account after the session ends.

Instruction:

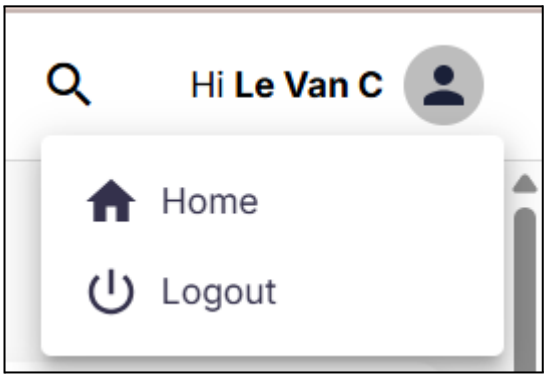


Figure xx - Logout dropdown

- **Step 1**: Click on the **user profile icon** located at the top-right corner of the screen.
- **Step 2**: In the dropdown menu, select **Logout**.
- **Step 3**: The system will log you out and redirect you to the **Sign In** screen. Ensure you have saved all work before logging out to avoid data loss.

3.1.3 Dashboard

Description: The **Dashboard** screen provides an overview of key statistics, upcoming schedules, and quick navigation to important modules within the Zentry Professional Admin Dashboard. It displays summary cards, a calendar, performance charts, registered statistics, and the top 5 most active courses.

Instruction:

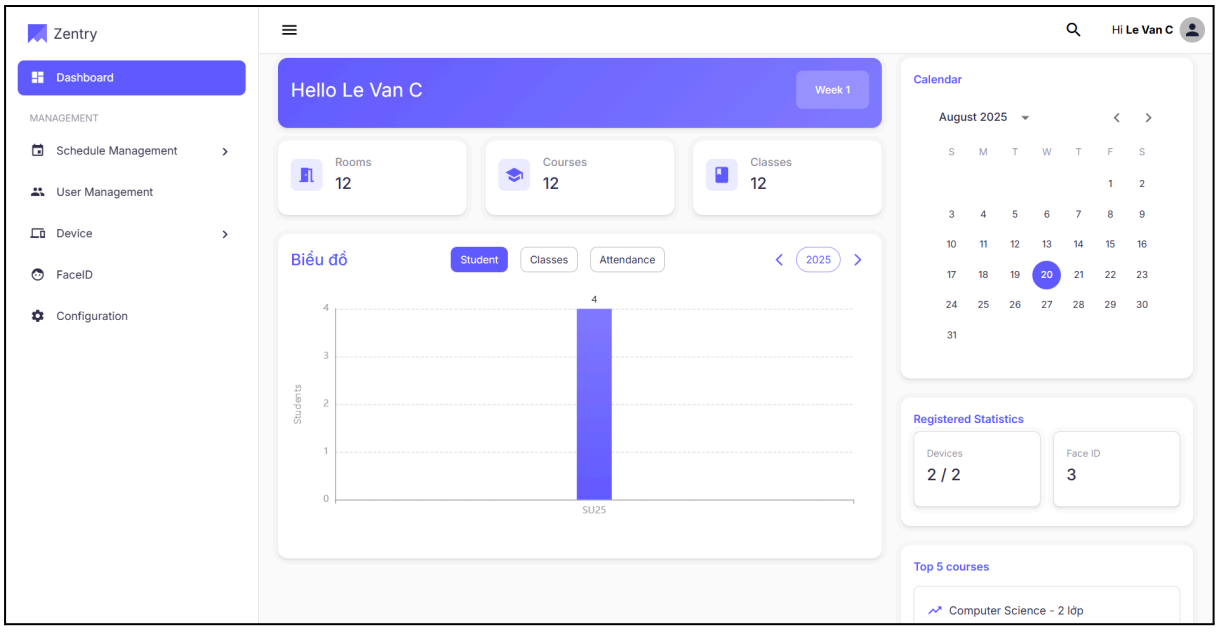


Figure xx - Dashboard screen

- **Step 1**: **View Summary Cards**

- **Rooms**: Displays the total number of rooms in the system.
- **Courses**: Displays the total number of courses available.
- **Classes**: Displays the total number of classes scheduled.



Figure xx - Summary cards

- **Step 2**: **View Chart**

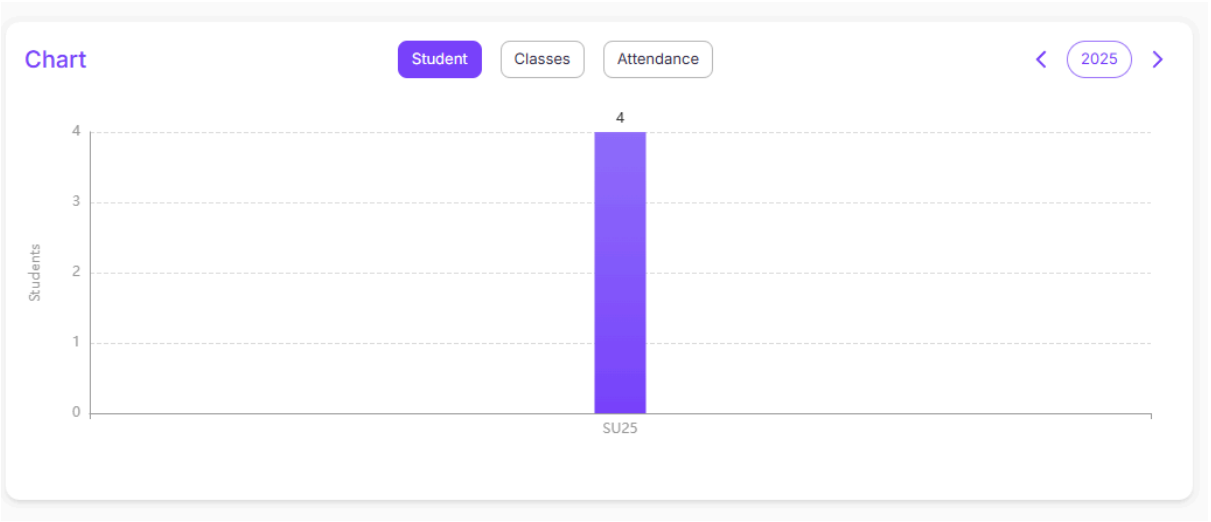


Figure xx - Dashboard chart

- Use the tabs (**Student**, **Classes**, **Attendance**) to switch between chart types.
- The chart shows data trends for the selected category and year.
- Navigate between years using the left and right arrow icons.

- **Step 3**: **Check Calendar**

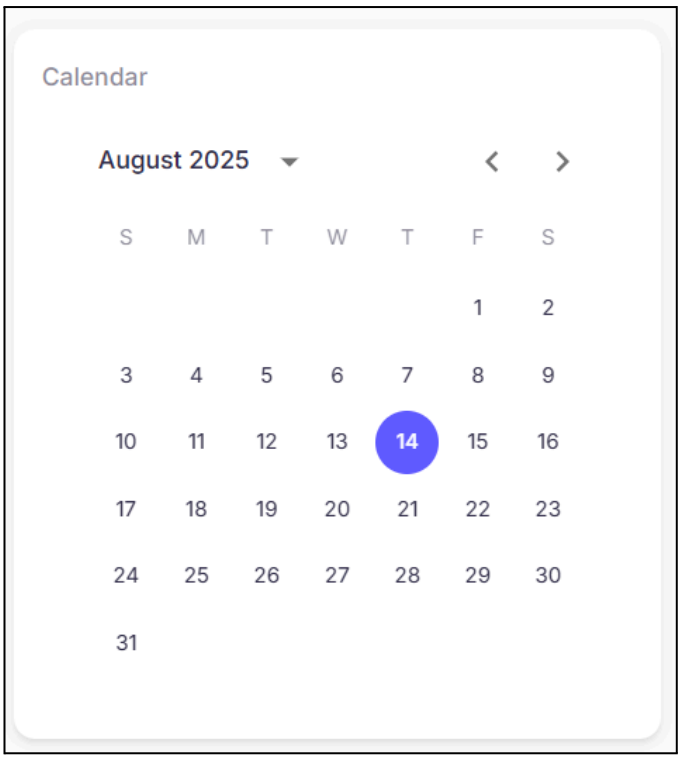


Figure xx - Calendar

- The calendar shows the current month with the current date highlighted.
- Navigate between months using the arrow buttons.

- **Step 4**: **Review Registered Statistics**

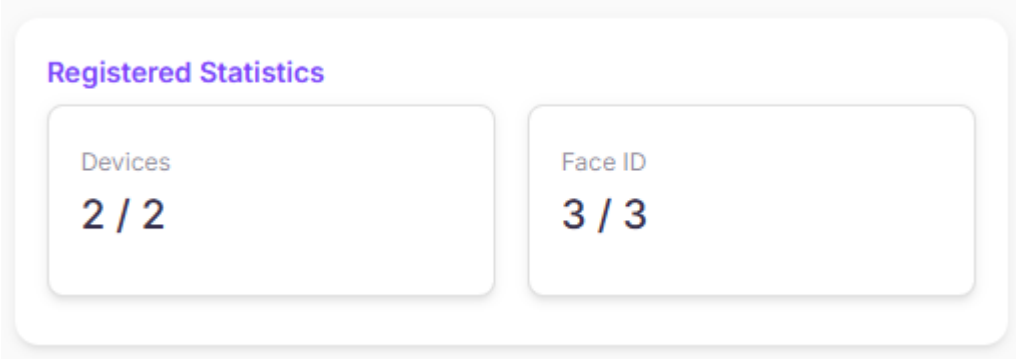


Figure xx - Registered statistics scorecards

- **Devices**: Displays the total number of devices registered in the system.
- **Face ID**: Displays the total number of registered Face IDs.

- **Step 5**: **View Top 5 Courses**

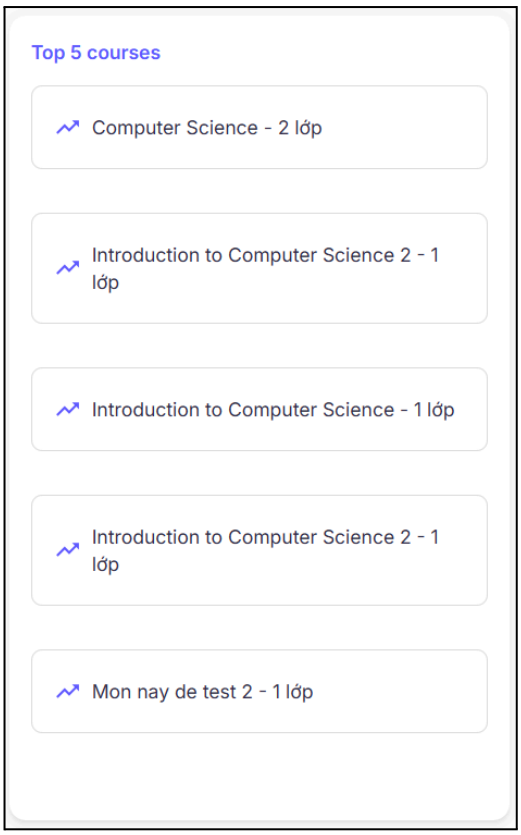


Figure xx - Top 5 courses with the most number of classes of current semester

- Shows the most active or popular courses along with the number of classes each course has.
- Courses are listed in order of activity, from most to least.

3.1.4 Room Management

View Rooms

Description: The **Rooms Management** screen allows users to view all rooms in the facility. Each row displays the room name, building, capacity, creation date, and available actions (edit or delete). Pagination is available for navigation between multiple pages of rooms.

Instruction:

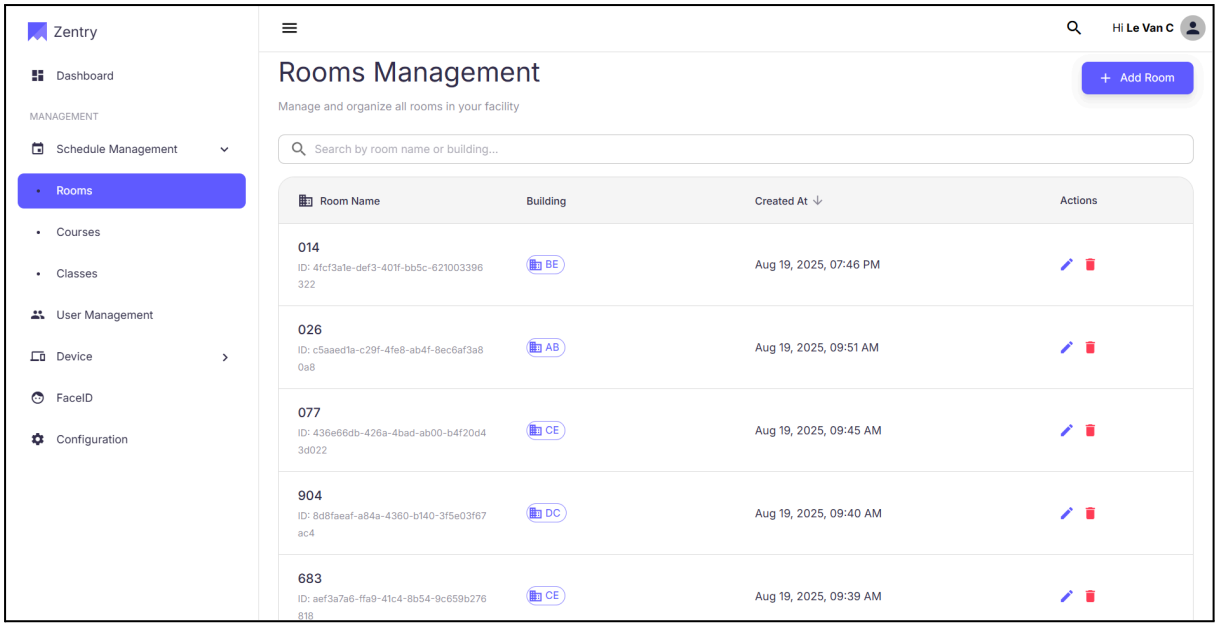


Figure xx - Room management screen

- **Step 1**: In the sidebar menu, click **Schedule Management** to expand the submenu.
- **Step 2**: Click **Rooms** to open the Rooms Management page.
- **Step 3**: Review the table displaying all existing rooms (click up/down arrow to sort), including:
    - **Room Name** and unique ID.
    - **Building** where the room is located.
    - **Created At** date and time.
    - **Actions** (Edit, Delete).
- **Step 4**: To search, type a room name or building code in the **search bar** above the table. The list updates automatically.
- **Step 5**: If the number of rooms exceeds the page size, use the **pagination controls** at the bottom to navigate:

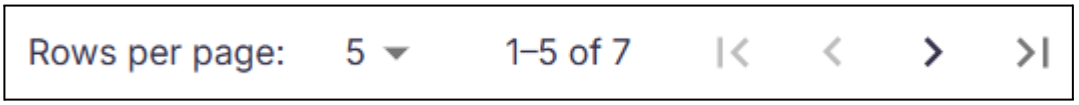


Figure xx - Pagination controls

- Change **Rows per page** to show more or fewer items.
- Use **Next** (>) or **Previous** (<) arrows to move between pages.

Add Room

Description: This function allows users to create a new room record by entering its details into a form.

Instruction:

- **Step 1**: From the Rooms Management page, click the **Add Room** button in the top-right corner.

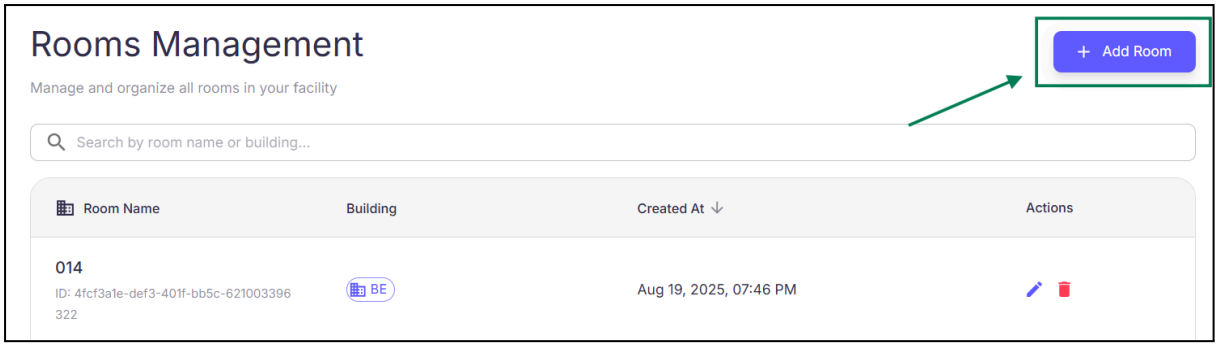


Figure xx - Add room button

- **Step 2**: In the **Add New Room** dialog:
  - Enter the **Room Name**.
  - Enter the **Building** name.
  - Enter the **Capacity** (number of people).

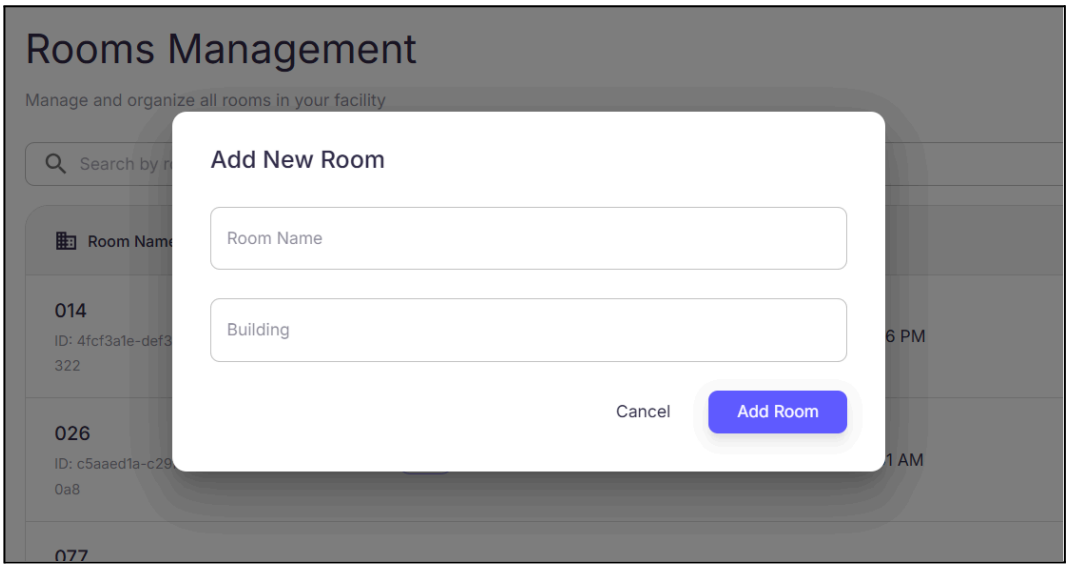


Figure xx - Add new room dialog

- **Step 3**: Click **Add Room** to save the new room, or **Cancel** to stop the action.
- **Step 4**: System response:
  - **If successful**: A green message appears — *"Room created successfully"*. The new room is added to the list.

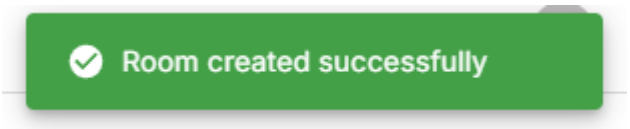


Figure xx - Add room success message

  - **If unsuccessful**: The system shows an error message (e.g., missing information).

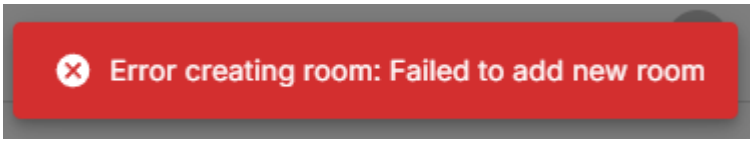


Figure xx - Add room failure message

Edit Room

Description: This function allows users to update the details of an existing room.

Instruction:

- **Step 1**: Locate the room you want to edit in the table.
- **Step 2**: Click the **Edit icon** (pencil) in the **Actions** column.

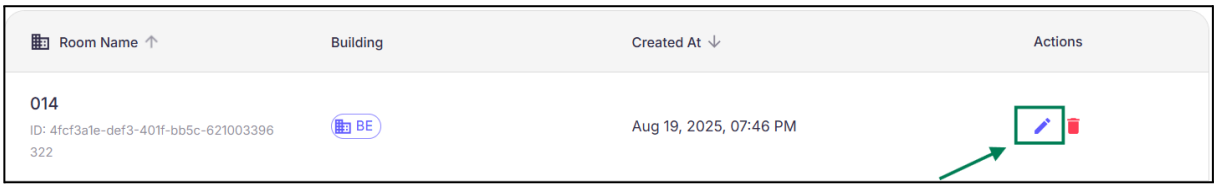


Figure xx - Edit room icon

- **Step 3**: In the **Edit Room** dialog, modify one or more fields:
  - **Room Name**
  - **Building**

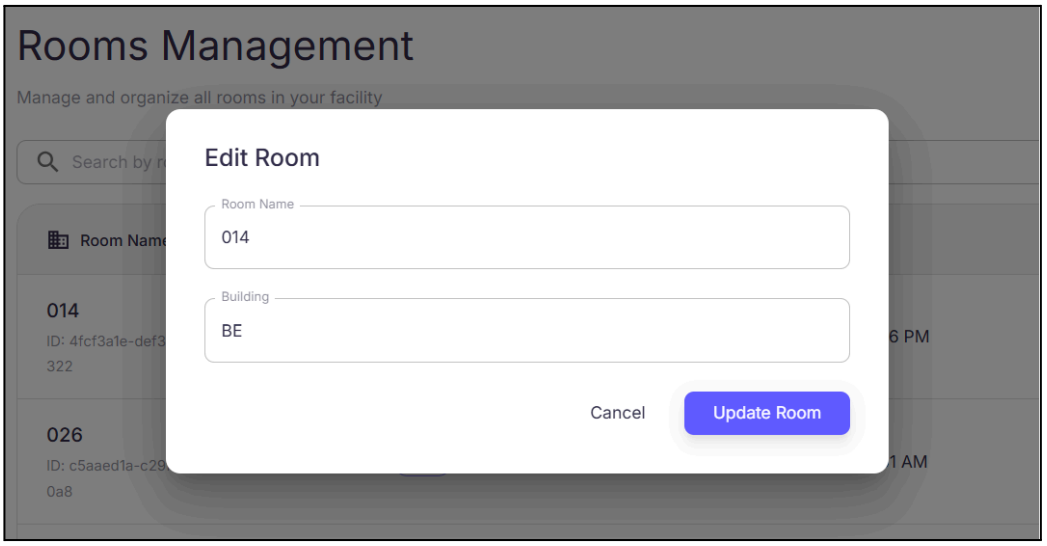


Figure xx - Edit room dialog

- **Step 4**: Click the **Update Room** to save changes, or **Cancel** to stop the action.
- **Step 5**: System response:
  - **If successful**: A green message appears — *"Room edited successfully"*.



Figure xx - Edit room success message

- **If unsuccessful**: An error message will appear.

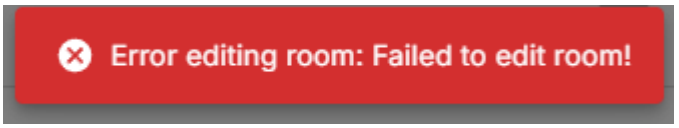


Figure xx - Edit room failure message

Delete Room

Description: This function removes a room from the system. Deletion is not allowed if the room is assigned to a class in the current semester.

Instruction:

- **Step 1**: Locate the room you want to delete in the table.
- **Step 2**: Click the **Delete icon** (trash bin) in the **Actions** column.



Figure xx - Delete icon

- **Step 3**: In the confirmation dialog, review the warning message.

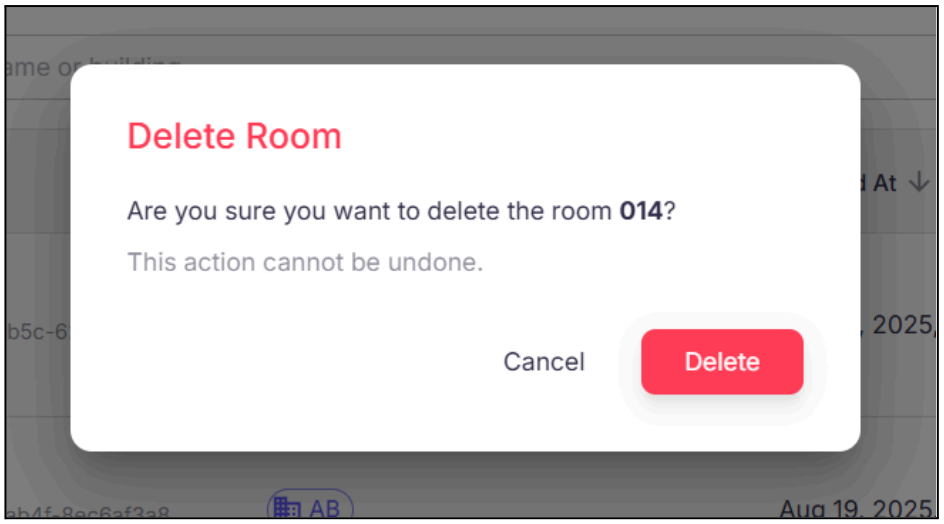


Figure xx - Delete room dialog

- **Step 4**: Click **Delete** to confirm, or **Cancel** to stop the action.
- **Step 5**: System response:
  - **If successful**: The room is removed and a green message appears — *"Room deleted successfully"*.



Figure xx - Delete room success message

- **If unsuccessful**: The system shows an error message.



Figure xx - Delete room failure message

3.1.5 Course Management

View Courses

Description: The **Courses Management** screen allows users to view all courses in the institution. Each course entry displays its code, name, description, and provides action buttons for editing or deleting the course. Pagination and search features are available to make navigation easier.

Instruction:

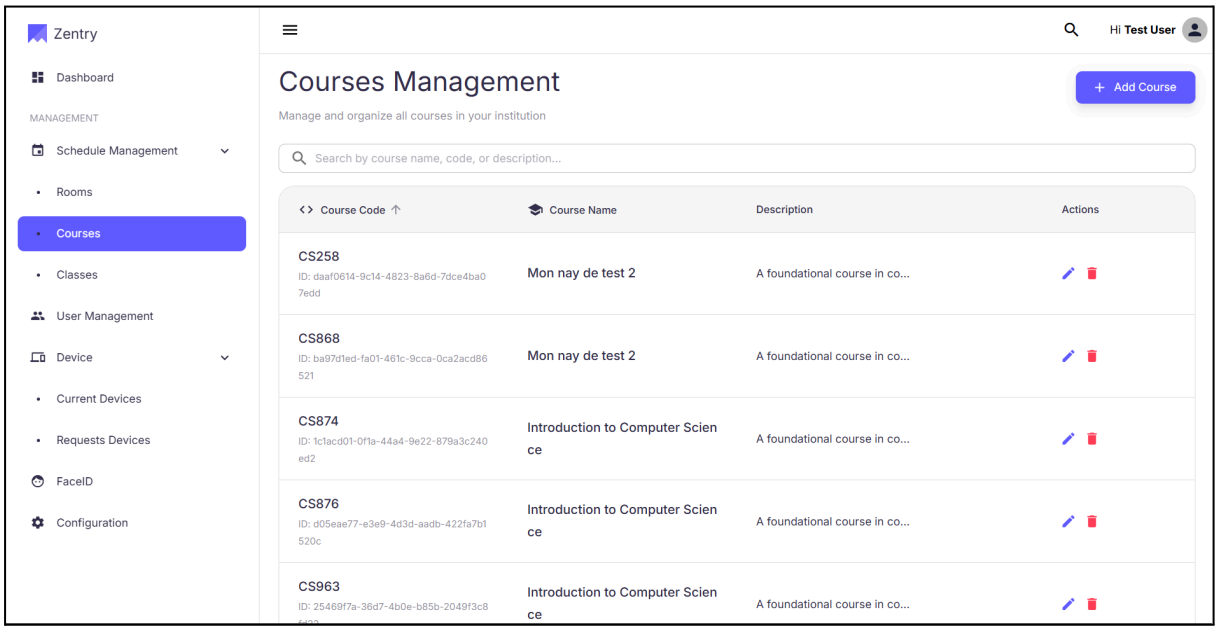


Figure xx - Course management screen

- **Step 1**: In the sidebar menu, click **Schedule Management** to expand the submenu.
- **Step 2**: Click **Courses** to open the Courses Management page.
- **Step 3**: Review the table displaying all existing courses, including:
  - **Course Code** and unique ID.
  - **Course Name**.
  - **Description** (brief overview of the course).
  - **Actions** (Edit, Delete).
- **Step 4**: To search, type a **course code**, **name**, or **description** keyword in the **search bar** above the table. The list will update automatically.

- **Step 5**: If the number of courses exceeds the page size, use the **pagination controls** at the bottom:
    - Adjust **Rows per page** to change how many records are shown.
    - Use **Next** (>) or **Previous** (<) arrows to navigate between pages.

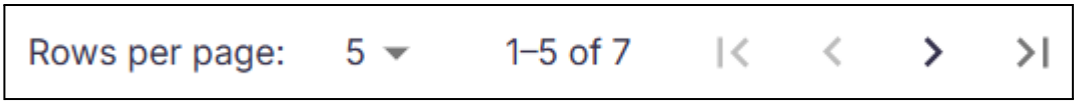


Figure xx - Pagination controls

Add Course

Description: This function allows users to create a new course by providing its code, name, and description.

Instruction:

- **Step 1**: On the Courses Management page, click the **Add Course** button at the top-right corner.

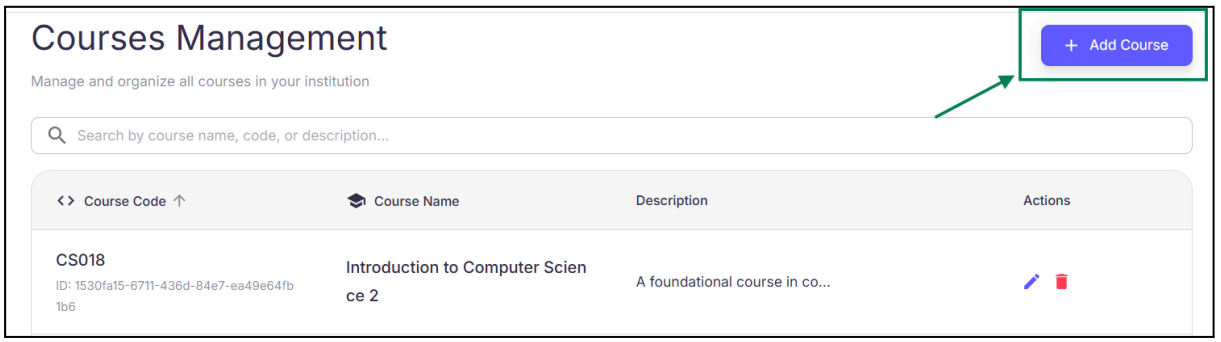


Figure xx - Add course button

- **Step 2**: In the **Add New Course** dialog:
    - Enter the **Course Code** (e.g., CS101).
    - Enter the **Course Name**.
    - Enter a brief **Description** of the course.

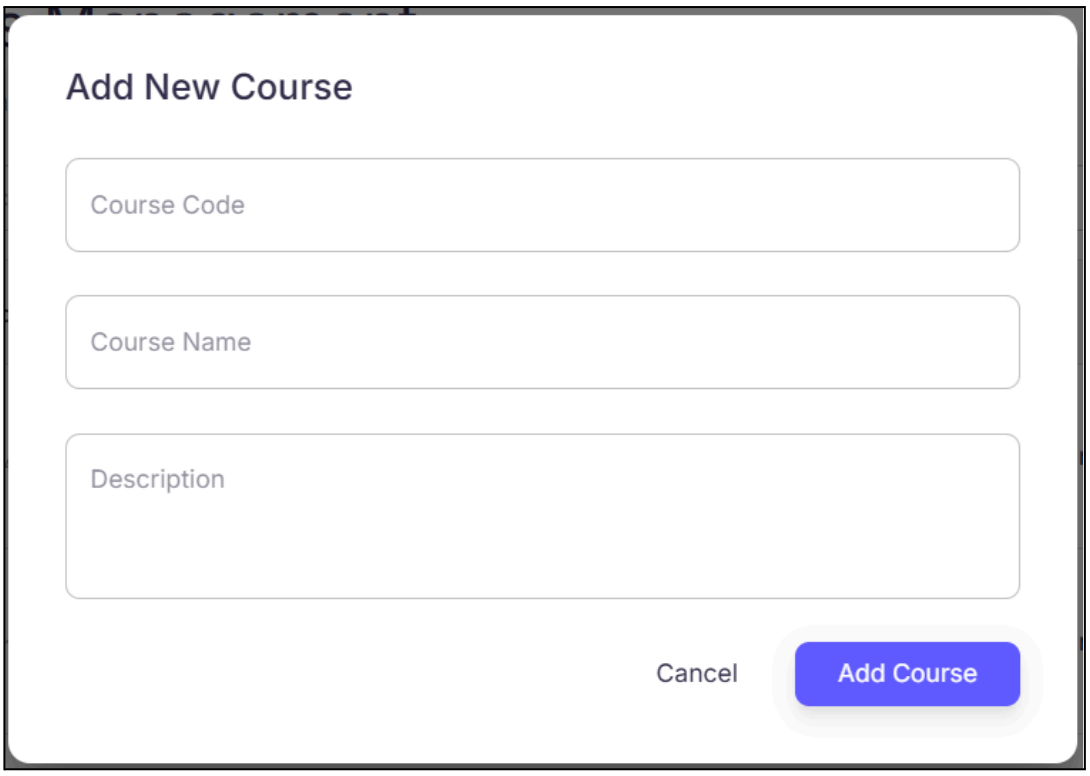


Figure xx - Add new course dialog

- **Step 3**: Click **Add Course** to save the new course, or **Cancel** to stop the action.
- **Step 4**: System response:

  If successful: A green notification will appear — *"Course created successfully"*, and the course will be added to the table.

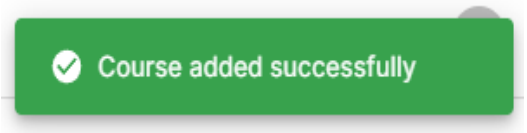


Figure xx - Add new course success message

If unsuccessful: An error message will appear (e.g., missing or invalid information).



Figure xx - Add new course failure message

Edit Course

Description: This function allows users to modify an existing course's details.

Instruction:

- **Step 1**: Locate the course you wish to edit in the table.
- **Step 2**: Click the **Edit icon** (pencil) in the **Actions** column.

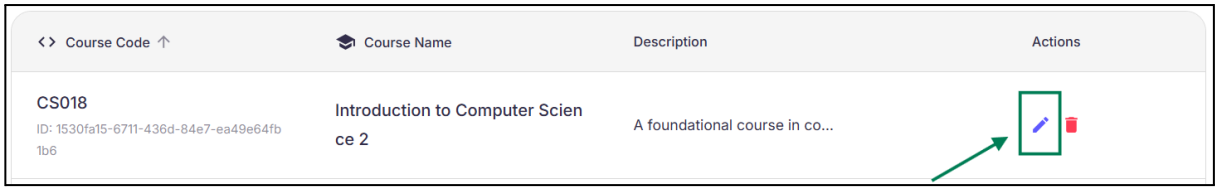


Figure xx - Edit course icon

- **Step 3**: In the **Edit Course** dialog, update one or more fields:
  - **Course Code**
  - **Course Name**
  - **Description**

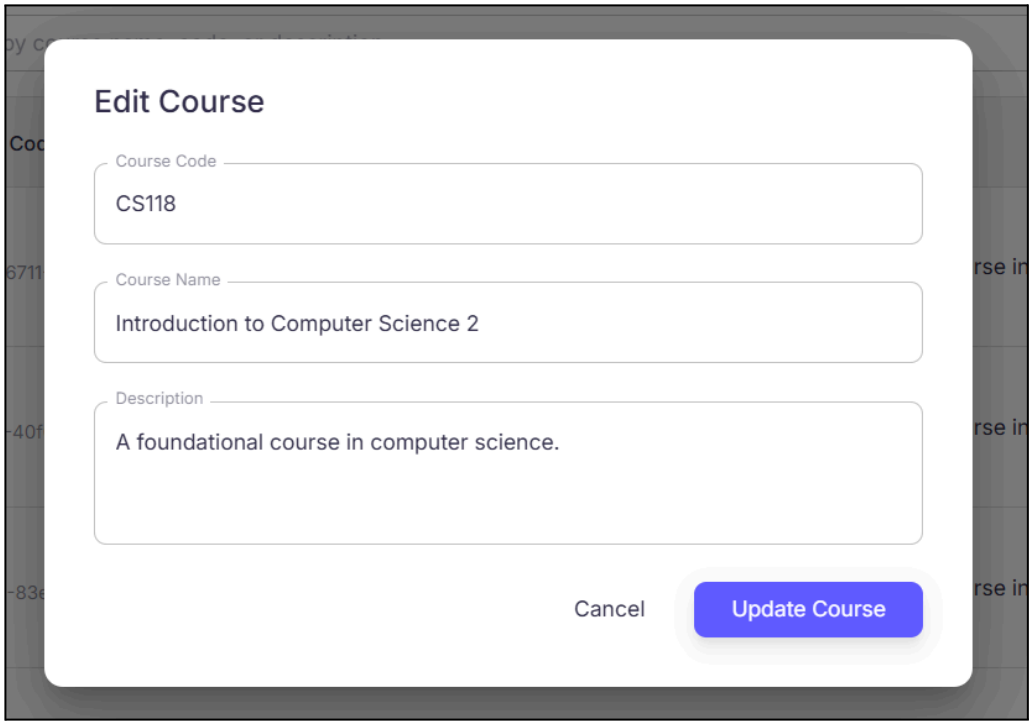


Figure xx - Edit course dialog

- **Step 4**: Click **Update Course** to save changes, or **Cancel** to stop the action.
- **Step 5**: System response:
  - **If successful**: A green notification will appear — *"Course edited successfully"*.

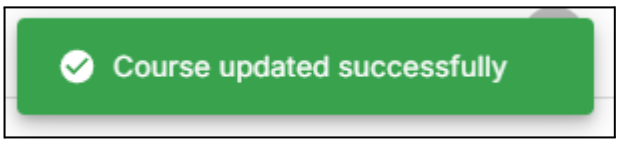


Figure xx - Edit course success message

  - **If unsuccessful**: An error message will appear.

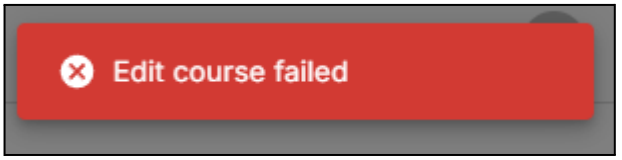


Figure xx - Edit course failure message

Delete Course

Description: This function allows users to remove a course from the system.

Instruction:

- **Step 1**: Locate the course you wish to delete in the table.
- **Step 2**: Click the **Delete icon** (trash bin) in the **Actions** column.



Figure xx - Delete course icon

- **Step 3**: In the confirmation dialog, review the warning message.

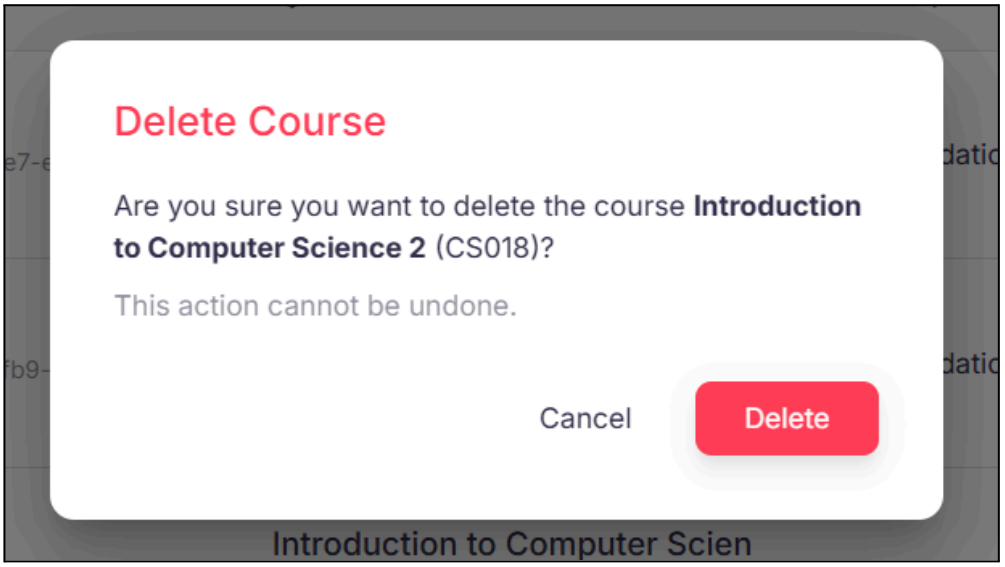


Figure xx - Delete course dialog

- **Step 4**: Click **Delete** to confirm, or **Cancel** to stop the action.
- **Step 5**: System response:
  - **If successful**: The course will be removed and a green notification will appear — *"Course deleted successfully"*.

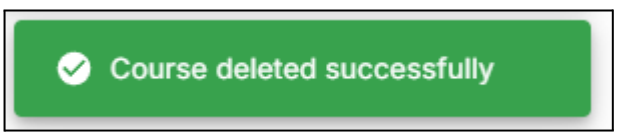


Figure xx - Delete course success message

  - **If unsuccessful**: An error message will appear (e.g., the course is linked to active classes or data constraints).

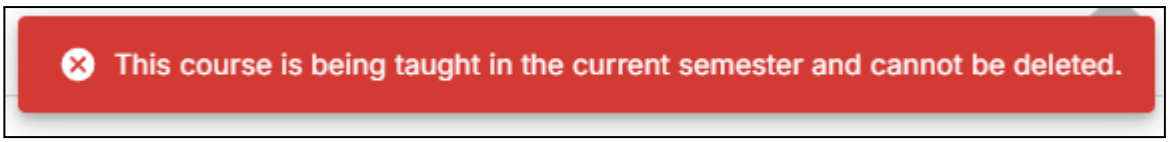


Figure xx - Delete course failure message

3.1.6 Class Management

3.1.6.1 Manage Classes

View Classes

Description: The **Classes Management** screen allows users to view all classes in the institution, organized by semester. Each entry displays the course, assigned lecturer, section code, semester, and actions. The left panel lists available semesters, enabling filtering by a specific term.

Instruction:

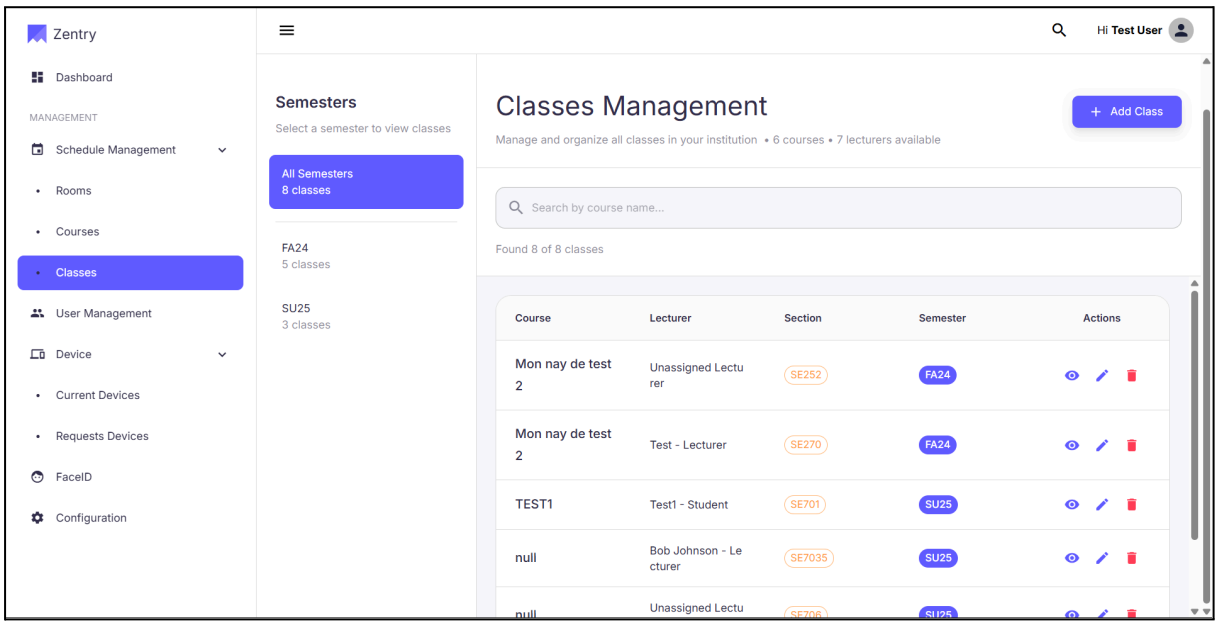


Figure xx - Class management screen

- **Step 1**: In the sidebar menu, click **Schedule Management** to expand the submenu.
- **Step 2**: Click **Classes** to open the Classes Management page.
- **Step 3**: Review the **Semesters** panel on the left:
    - **All Semesters**: Displays all classes in the system.
    - Specific terms (e.g., **FA24**, **SU25**): Shows only classes for that semester. Click a semester name to filter the class list accordingly.
- **Step 4**: Review the class table, which includes:
    - **Course**: The course associated with the class.
    - **Lecturer**: Assigned lecturer's name, or "Unassigned Lecturer" if not yet assigned.
    - **Section**: The section code of the class.
    - **Semester**: The term during which the class is held.
    - **Actions**:
        - **View icon** (eye): Opens the class details page (to be implemented in a future update).

    - **Edit icon** (pencil): Edits the class information.
    - **Delete icon** (trash bin): Deletes the class.
- **Step 5**: Use the **search bar** to find classes by course name.
- **Step 6**: Navigate between pages using the **pagination controls** at the bottom:
    - Change **Rows per page** to adjust the list size.
    - Use **Next** (>) or **Previous** (<) arrows to move between pages.

Edit Class

Description: This function allows users to modify existing class details such as course, section code, and semester.

Instruction:

- **Step 1**: Locate the class you want to edit in the list.
- **Step 2**: Click the **Edit icon** (pencil) in the **Actions** column.

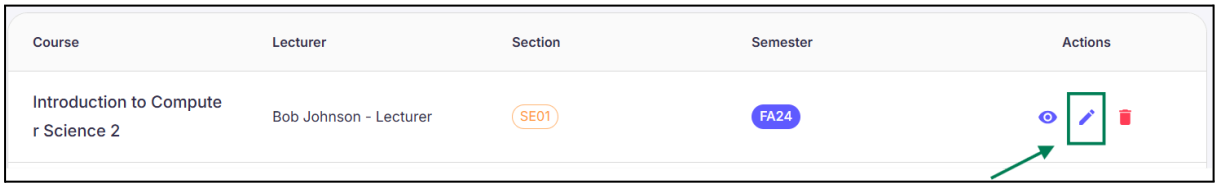


Figure xx - Edit class icon

- **Step 3**: In the **Edit Class** dialog, update one or more of the following:
    - **Course** (select from dropdown).
    - **Section Code**.
    - **Semester**.

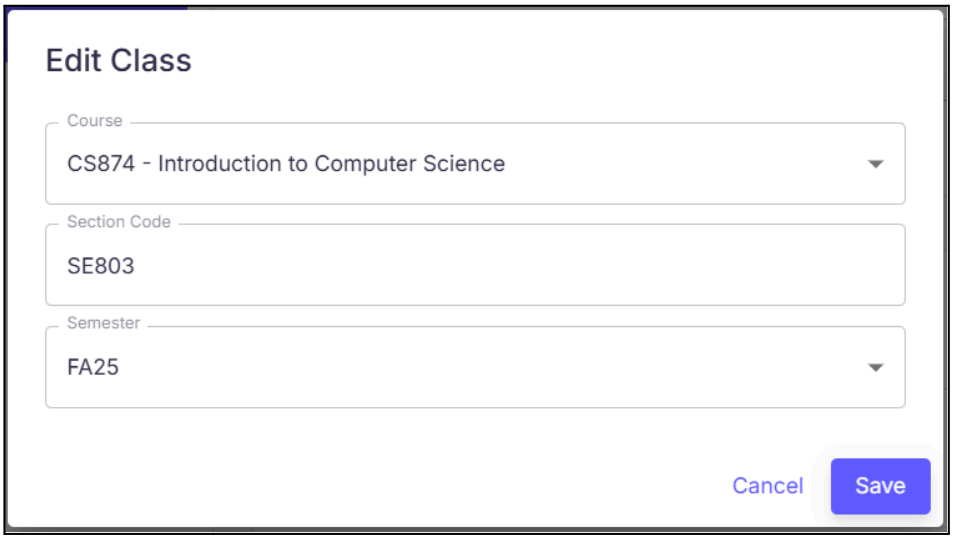


Figure xx - Edit class dialog

- **Step 4**: Click **Save** to apply changes, or **Cancel** to stop the action.
- **Step 5**: System response:
    - **If successful**: A green notification appears — *"Class edited successfully"*.

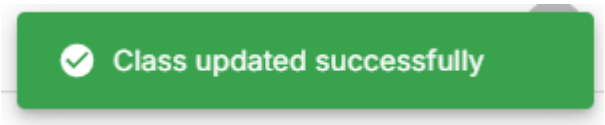


Figure xx - Edit class success message

- **If unsuccessful**: An error message appears.



Figure xx - Edit class failure message

Delete Class

Description: This function removes a class from the system.

Instruction:

- **Step 1**: Locate the class you want to delete in the list.
- **Step 2**: Click the **Delete icon** (trash bin) in the **Actions** column.

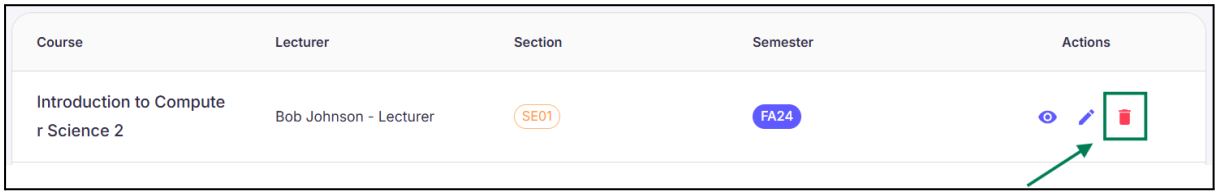


Figure xx - Delete class icon

- **Step 3**: In the confirmation dialog, review the message showing the section code and semester of the class to be deleted.

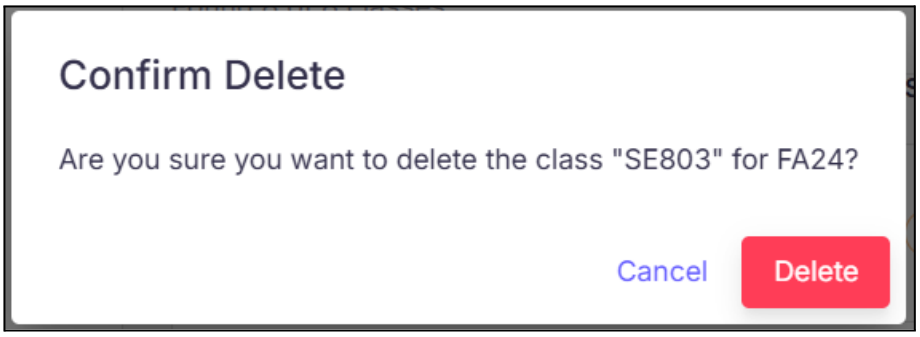


Figure xx - Delete class dialog

- **Step 4**: Click **Delete** to confirm, or **Cancel** to abort.
- **Step 5**: System response:
  - **If successful**: A green notification appears — *"Class deleted successfully"*.

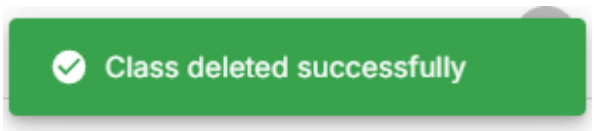


Figure xx - Delete class success message

- **If unsuccessful**: An error message appears if the class cannot be deleted due to related constraints.

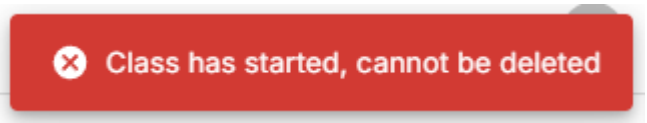


Figure xx - Delete class failure message

3.1.6.2 Add New Classes

Add Class

Description: This function allows users to create a new class by selecting a course, assigning a section code, and choosing a semester.

Instruction:

- **Step 1**: On the Classes Management page, click the **Add Class** button in the top-right corner.

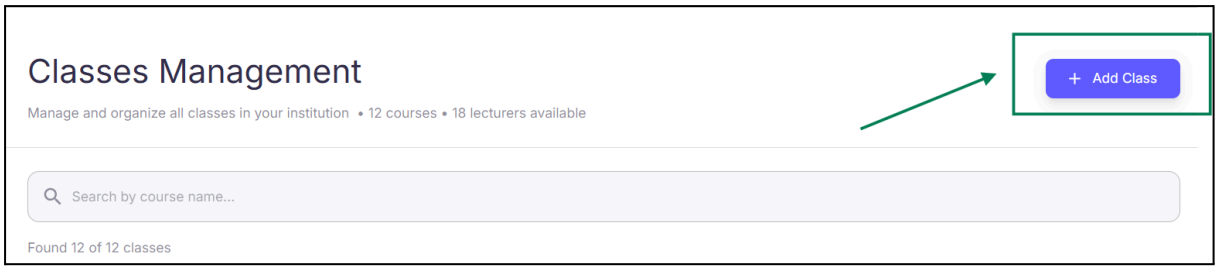


Figure xx - Add class button

- **Step 2**: In the **Add Class** dialog:
  - **Course**: Select from the dropdown list of available courses.
  - **Section Code**: Enter a section code for the class (e.g., SE101).
  - **Semester**: Select the semester from the dropdown list (e.g., FA25, SU25).

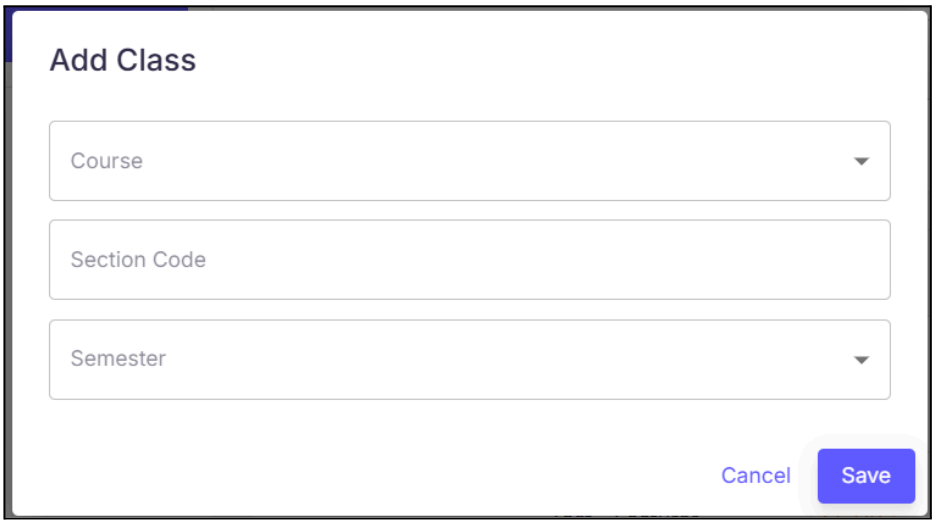


Figure xx - Add class dialog

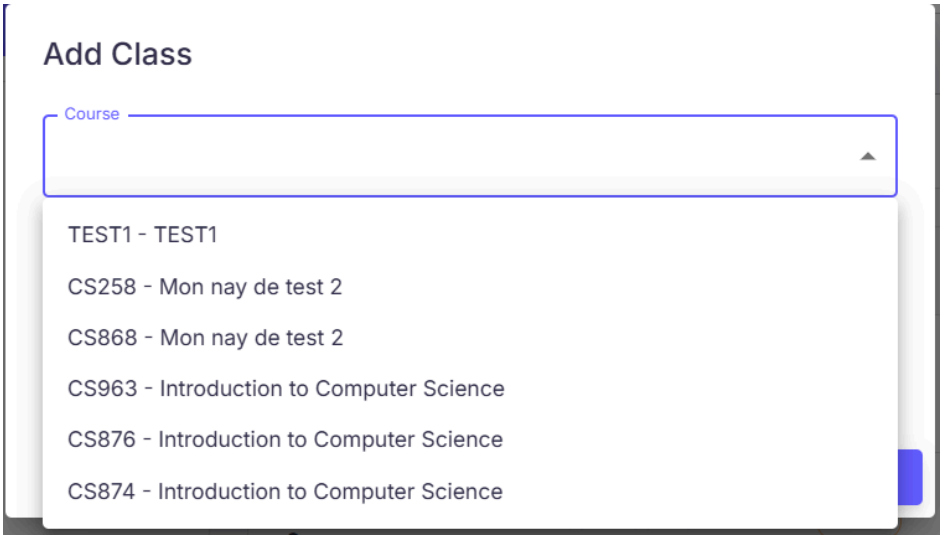


Figure xx - Add class - course selection



Figure xx - Add class - semester selection

- **Step 3**: Click **Save** to add the class, or **Cancel** to stop the action.
- **Step 4**: System response:
    - **If successful**: A green notification appears — *"Class created successfully"*, and the new class appears in the table.



Figure xx - Add class success message

- **If unsuccessful**: An error message appears (e.g., missing or invalid information).



Figure xx - Add class failure message

3.1.6.3 Manage Class

View Class Details

Description: This function allows users to view detailed information about a specific class, including section code, course details, lecturer, semester, enrolled students, class ID, and timestamps.

Instruction:

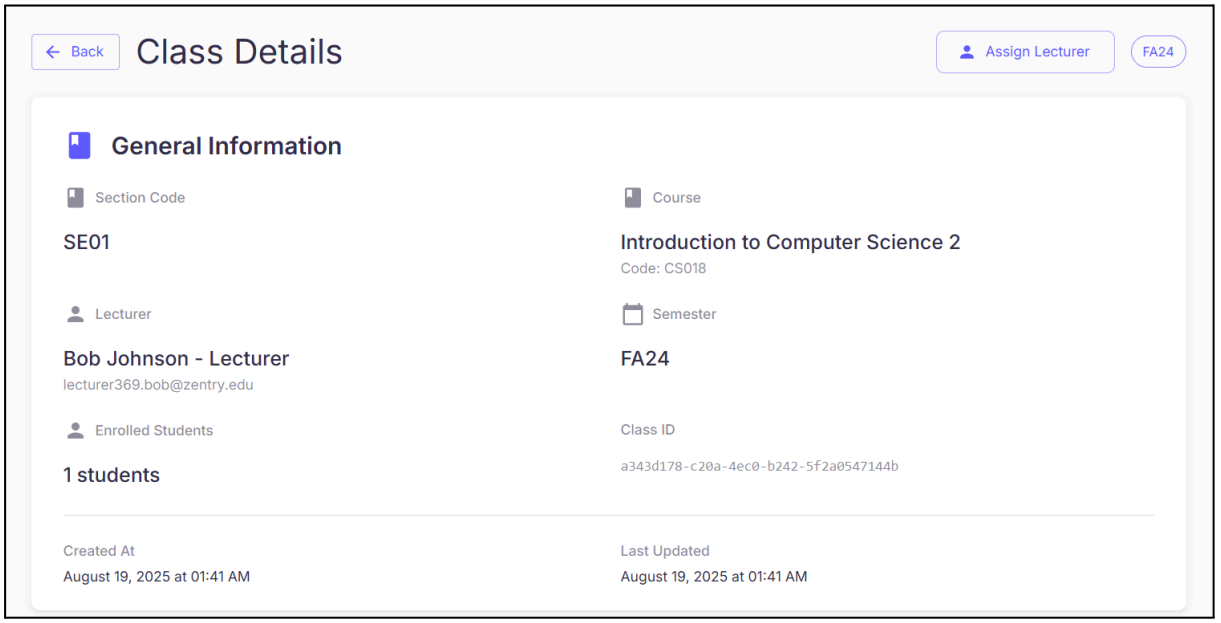


Figure xx - Class detail screen

- **Step 1**: On the Classes Management page, click the **View** icon in the row of the class you want to see.
- **Step 2**: The system navigates to the **Class Details** page. The page contains the following information:
    - **Section Code:** Displays the section code of the class (e.g., SE01).
    - **Course:** Shows the course name and code (e.g., Introduction to Computer Science - CS018).
    - **Lecturer:** Displays the lecturer's name and email address.
    - **Semester:** Shows the semester for the class (e.g., FA24).
    - **Enrolled Students:** Shows the total number of students enrolled.
    - **Class ID:** Unique identifier for the class.
    - **Created At** and **Last Updated:** Timestamps for when the class was created and last modified.
- **Step 3**: Optional actions available on the Class Details page:
    - Click **Assign Lecturer** to change or assign a lecturer to the class.
- **Step 4**: Click the **Back** button to return to the Classes Management page.

Assign Lecturer

Description: This function allows users to assign a lecturer to a specific class by selecting from a list of available lecturers.

Instruction:

- **Step 1**: On the **Class Details** page, click the **Assign Lecturer** button in the top-right corner.



Figure xx - Assign lecturer button

- **Step 2**: In the **Assign Lecturer** dialog, click the dropdown list to view all available lecturers.

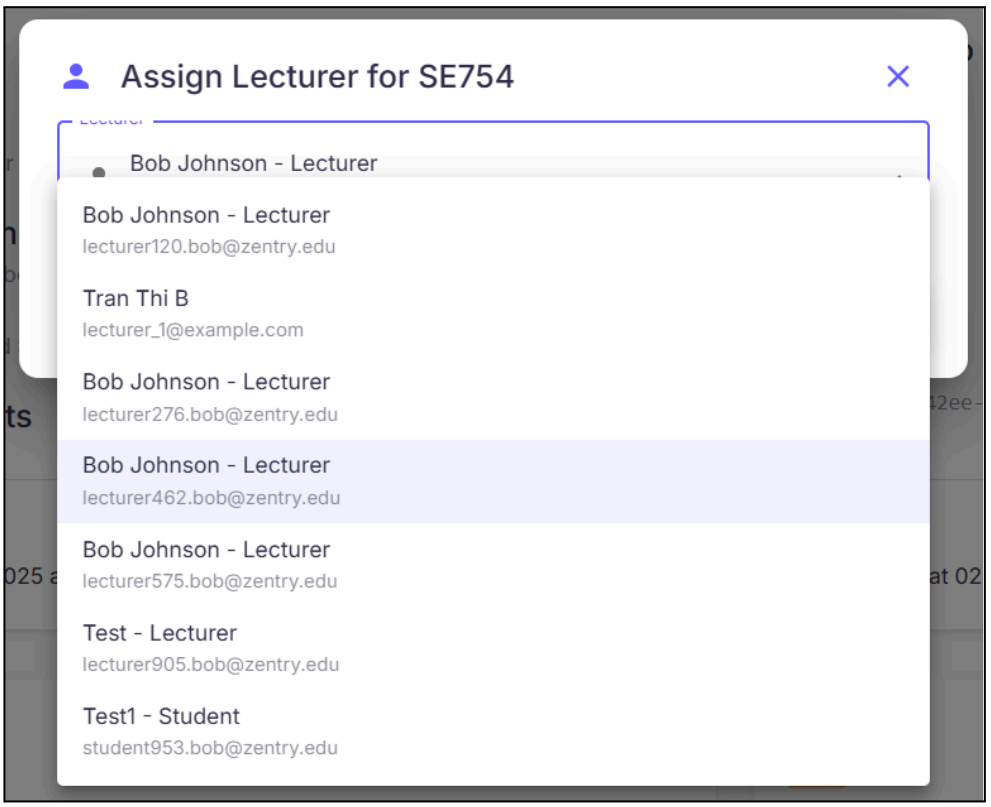


Figure xx - Assign lecturer - lecturer selection

- **Step 3**: Scroll or type to search, then select the desired lecturer from the list.
- **Step 4**: Click the **Assign Lecturer** button to confirm.
- **Step 5**: **System response:**
    - **If successful:** A green notification appears — *"Lecturer assigned successfully"*, and the lecturer's details are updated on the Class Details page.

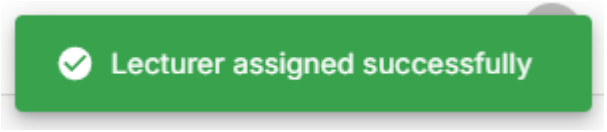


Figure xx - Assign lecturer success message

- **If unsuccessful:** An error message appears (e.g., lecturer not found or system error).

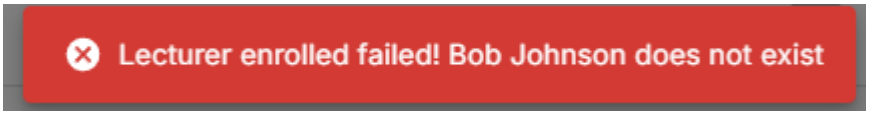


Figure xx - Assign lecturer failure message

3.1.6.4 Manage Enrollments

View Enrolled Student

Description: This function allows users to view the list of students currently enrolled in a specific class. The table displays each student's details, including Student ID, Name, Enrollment Date, and Status.

Instruction:

- **Step 1**: On the **Class Details** page, scroll to the **Enrolled Students** section.
- **Step 2**: Review the table columns:
  - **Student ID:** Unique identifier for each student.
  - **Student Name:** Full name and role of the student.
  - **Enrollment Date:** The date the student was added to the class.
  - **Status:** Current enrollment status (e.g., Active).
  - **Action:** Remove student out of a class

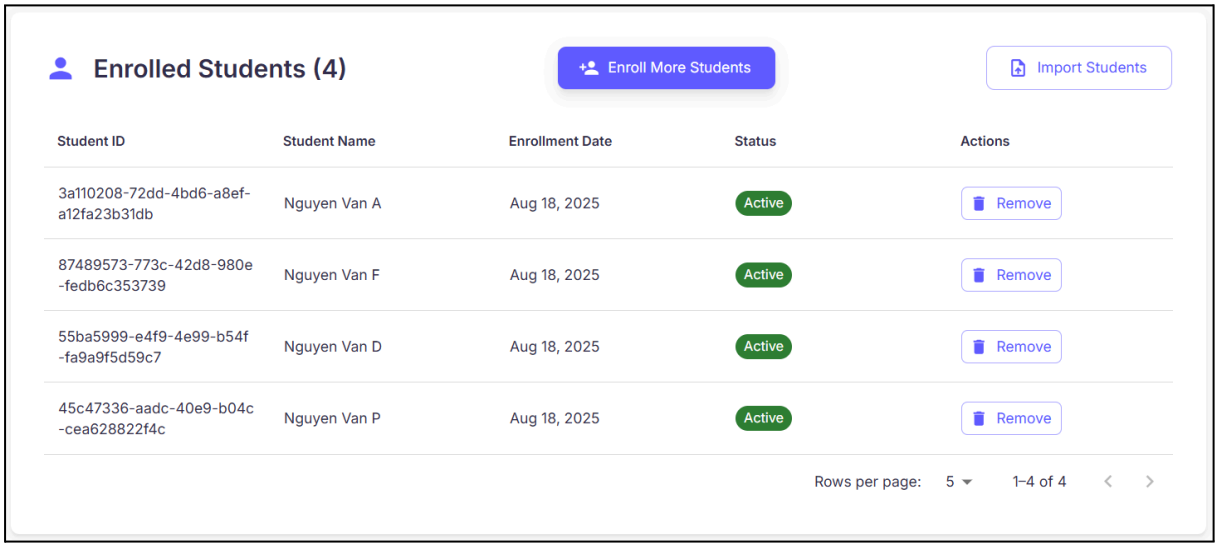


Figure xx - Enrolled students screen

- **Step 3**: Use the **Enroll More Students** button to manually add individual students.
- **Step 4**: Use the **Import Students** button to bulk add students from a CSV file.
- **Step 5**: Use the **Remove** button to remove a specific student from a list.
- **Step 6**: If there are many records, use the pagination controls at the bottom of the table to navigate between pages.
- **Step 7**: System response:
  - Data updates automatically after students are added or removed.

- A green notification appears after a successful action (e.g., “Student added successfully”).

Import Student

Description: This function allows users to import multiple students into a class by uploading a CSV file. The system will parse the file, validate the data, and enroll the students into the selected class.

Instruction:

- **Step 1**: On the **Enrolled Student** list, click the **Import Students** button located in the top-right corner.

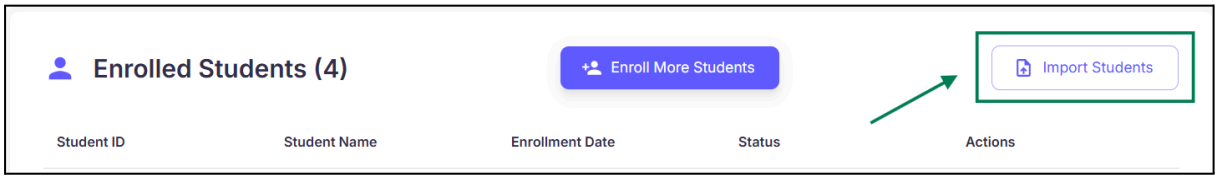


Figure xx - Import students button

- **Step 2**: In the **Import Students** dialog, click **Choose File** and select the CSV file containing the student data. The file must follow the required format:
  - Student ID
  - Full Name
  - Email (optional)
  - Enrollment Date (optional)



Figure xx - Import students dialogs

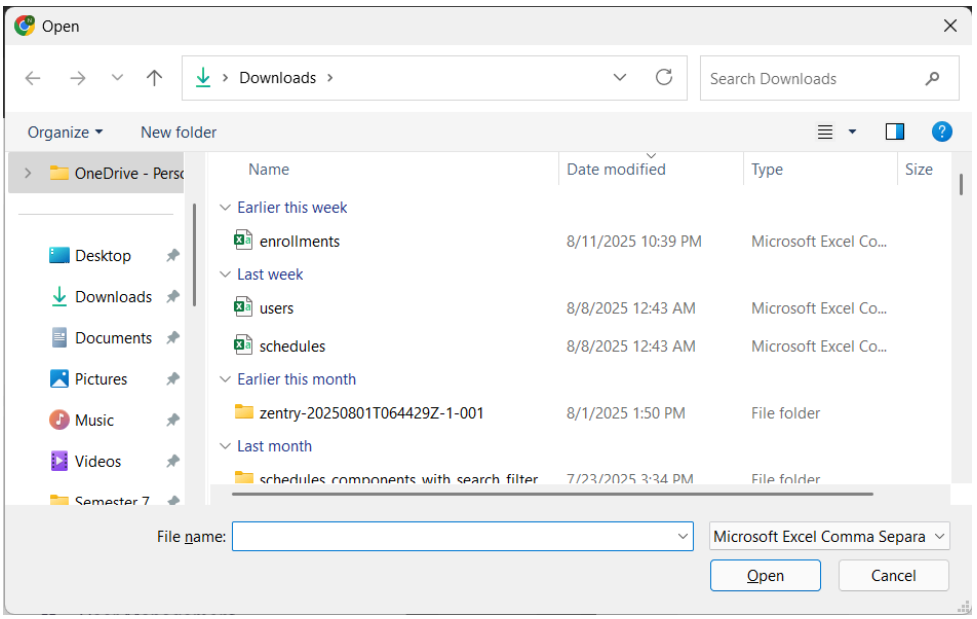


Figure xx - Import students - file selection

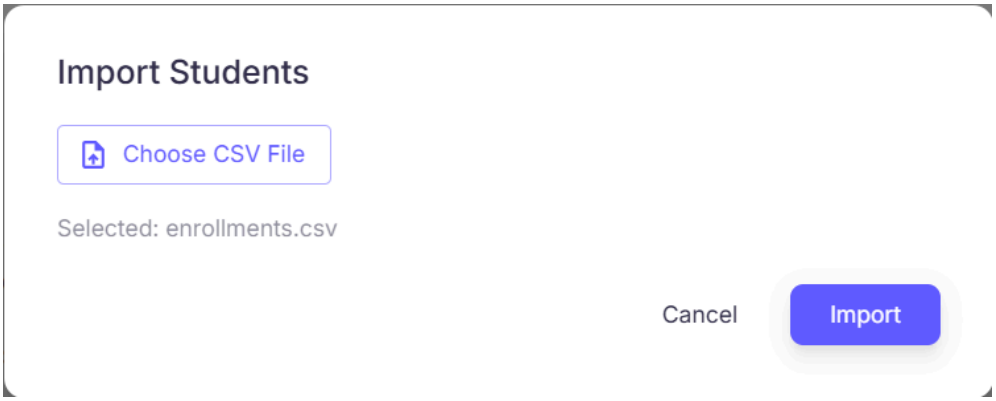


Figure xx - Import students - file selected display

- **Step 3**: Click **Import Students** to upload and process the file.
- **Step 4**: System response:
    - **If successful:** A green notification appears — “Students imported successfully”, and the enrolled students list is updated automatically.

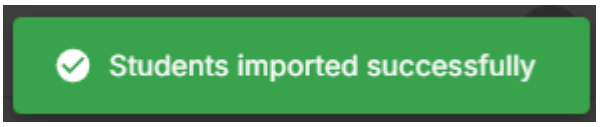


Figure xx - Import students success message

- **If unsuccessful:** An error message appears describing the issue (e.g., missing columns, invalid format).

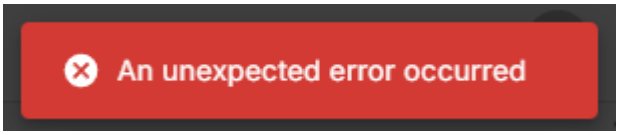


Figure xx - Import students failure message

Enroll Student

Description: This function allows users to add new students to a specific class. Users can manually select students from the available list or bulk import them using a CSV file. The interface supports search, selection, and status display to streamline the enrollment process.

Instruction:

- **Step 1**: On the Class Details page, click the **Enroll More Students** button in the Enrolled Students section.

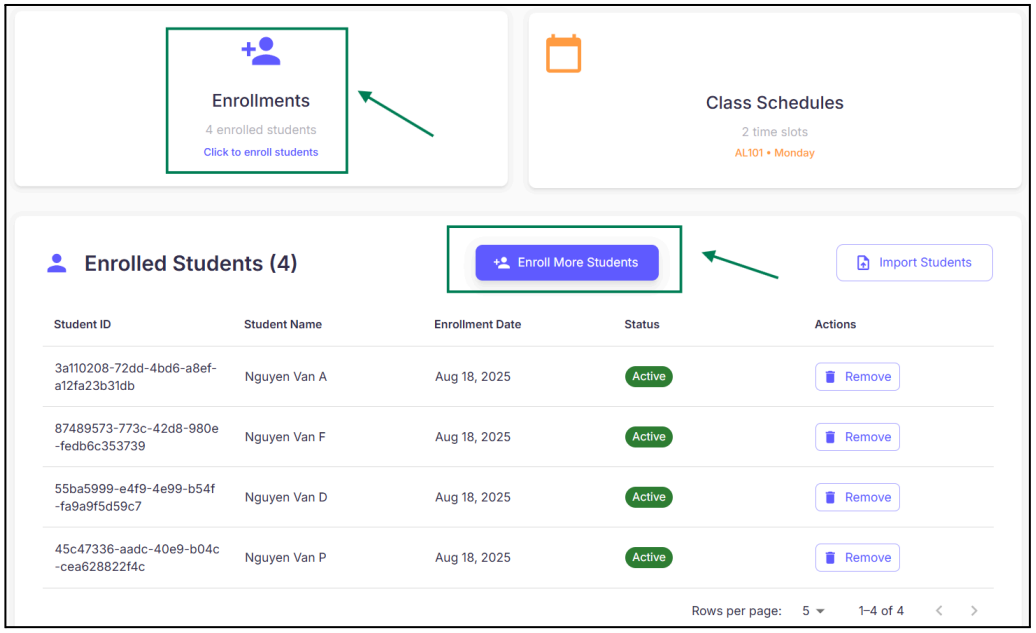


Figure xx - Eroll Student Button

- **Step 2**: In the *Enroll Students in [Class Code]* dialog:
  - **Search Bar:** Enter a student name or ID to filter the list of available students.
  - **Select All Checkbox:** Select or deselect all students in the current page.
  - **Student Table Columns:**
    - **Student ID:** Unique identifier for each student.
    - **Student Name:** Full name and role of the student.
    - **Status:** Current enrollment status (e.g., Active).

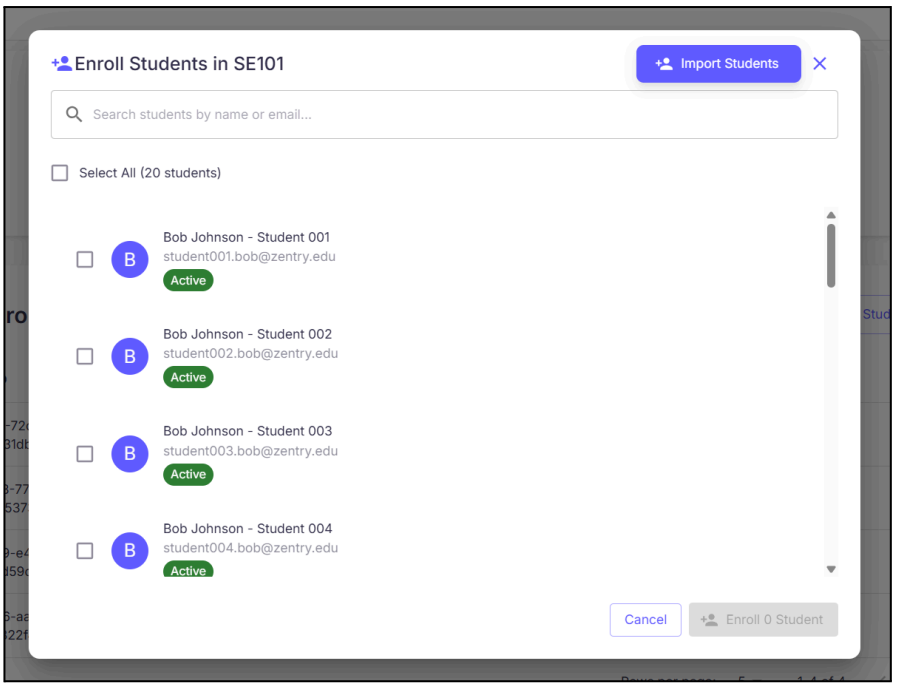


Figure xx - Enroll students dialog

- **Step 3**: Select one or more students to enroll:
    - When no students are selected, the enroll button is disabled.
    - When students are selected, the button label updates to show the number of students to be enrolled (e.g., “Enroll 3 Students”).

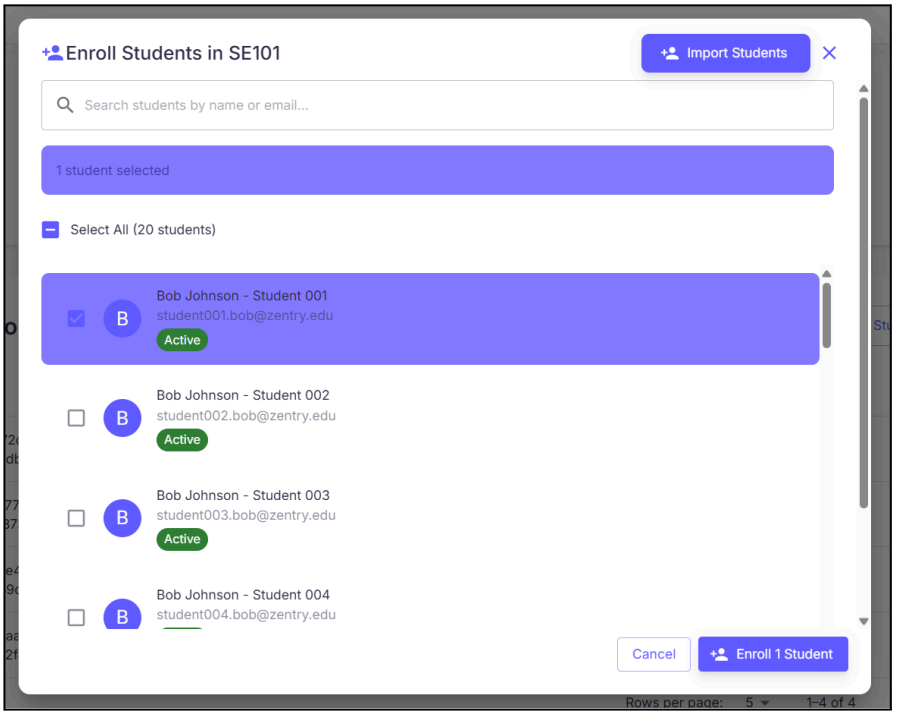


Figure xx - Enroll students - student selection

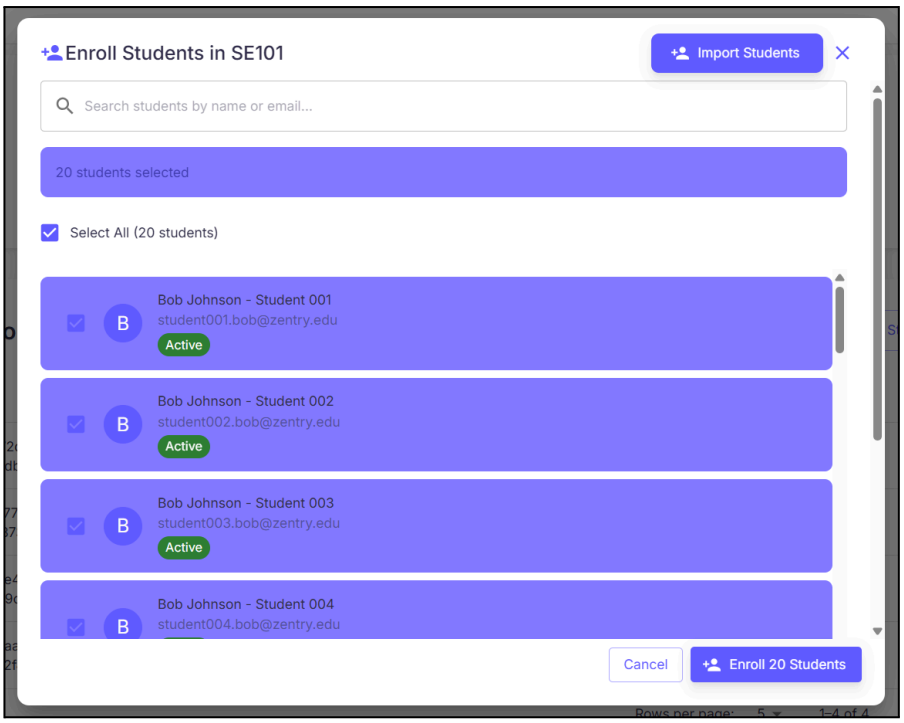


Figure xx - Enroll students - multi selection

- **Step 4**: (Optional) Use the **Import Students** button to bulk upload students from a CSV file.
- **Step 5**: Click the **Enroll X Students** button to confirm.
- **Step 6**: System response:
    - **If successful:** A green notification appears — “Student(s) enrolled successfully”, and the newly added students appear in the Enrolled Students table.

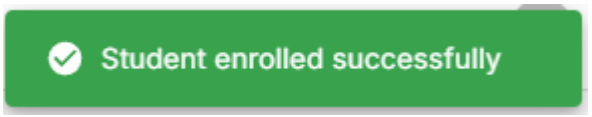


Figure xx - Enroll students success message

- **If unsuccessful:** An error message appears (e.g., “Invalid file format” or “Student already enrolled”).

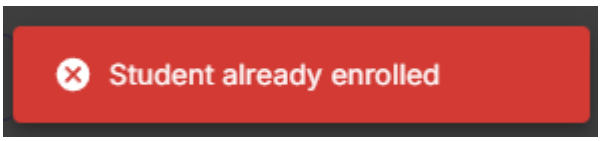


Figure xx - Enroll students failure message

Remove Student

Description: This function allows users to remove one or more students from a specific class. The action is available directly in the Enrolled Students table, with confirmation required to prevent accidental removals. Once removed, students will no longer appear in the enrolled list for that class.

Instruction:

- **Step 1**: On the Class Details page, scroll to the **Enrolled Students** section.
- **Step 2**: Locate the student(s) you want to remove.
    - **Single Remove:** Click the **Remove** icon/button in the *Actions* column for that student.

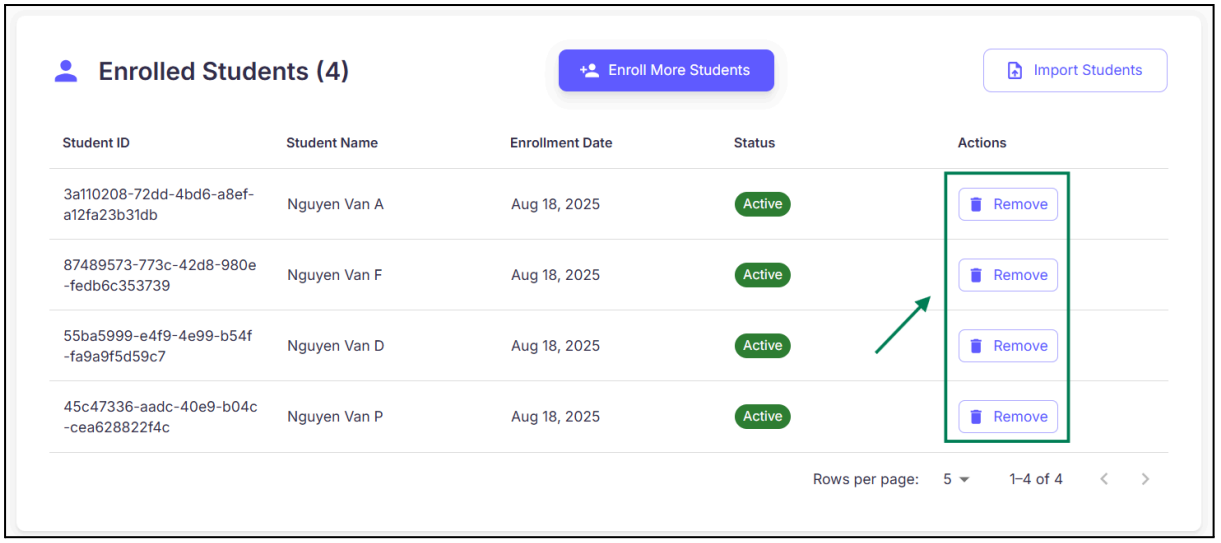


Figure xx - Remove students button

- **Step 3**: The **Confirm Removal** dialog appears with the message:

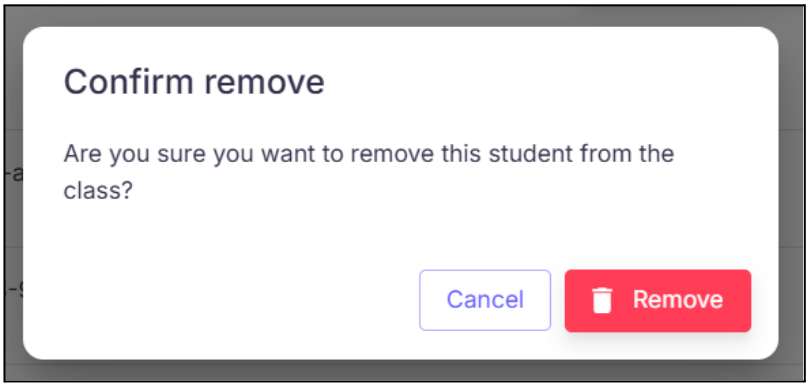


Figure xx - Remove students pop up

- **Step 4**: Click **Remove** to proceed.
- **Step 5**: System response:
    - **If successful:**
        - A green notification appears — “Student(s) removed successfully”.

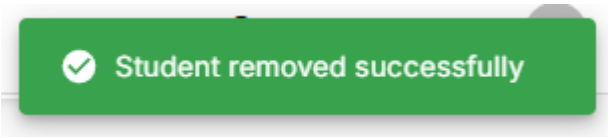


Figure xx - Remove students success message

    - **If unsuccessful:**
        - An error message appears (e.g., “Unable to remove student. Please try again.”).

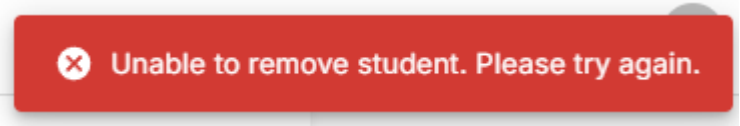


Figure xx - Remove students success message

- **Step 6**: If the list becomes empty after removal, the table displays a placeholder message:

  "No students are currently enrolled in this class."

3.1.6.5 Manage Schedules

View Schedules

Description: This function allows users to view the list of schedules for a specific class. The table displays key details for each schedule, including Week Day, Time, Room, and Date Range.

Instruction:

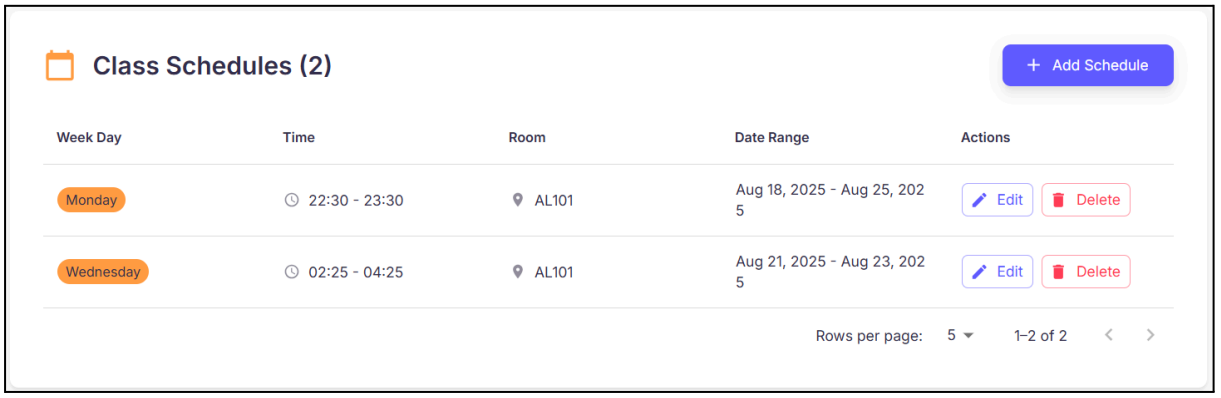


Figure xx - Manage schedules screen

- **Step 1**: On the **Class Details** page, scroll to the **Class Schedules** section.
- **Step 2**: Review the table columns:
  - **Week Day:** The day of the week when the class occurs.
  - **Time:** Start and end time of the class session.
  - **Room:** The room or location where the class will be held.
  - **Date Range:** The start and end date for the scheduled session.
  - **Action:** Edit or Delete Schedule.
- **Step 3**: Use the **Add Schedule** button to create a new schedule for the class.
- **Step 4**: If there are many records, use the pagination controls at the bottom of the table to navigate between pages.
- **Step 5**: System response:
  - The table updates automatically after a schedule is added, edited, or deleted.
  - A green notification appears after a successful action (e.g., "Schedule added successfully").

Add Schedule

Description: This function allows users to add a new schedule to a specific class by specifying the room, weekday, date range, and lesson times.

Instruction:

- **Step 1**: On the **Class Schedules** section of the **Class Details** page, click the **Add Schedule** button in the top-right corner.
- **Step 2**: In the **Add Schedule** dialog, fill in the following fields:
    - **Room:** Select from the dropdown list of available rooms. Each room displays its name, building, and capacity.
    - **Week Day:** Choose the day of the week for the class (e.g., Monday, Thursday).
    - **Start Date** / **End Date:** Select the date range for the schedule within the semester.
    - **Start Time** / **End Time:** Set the start and end time for the lesson.

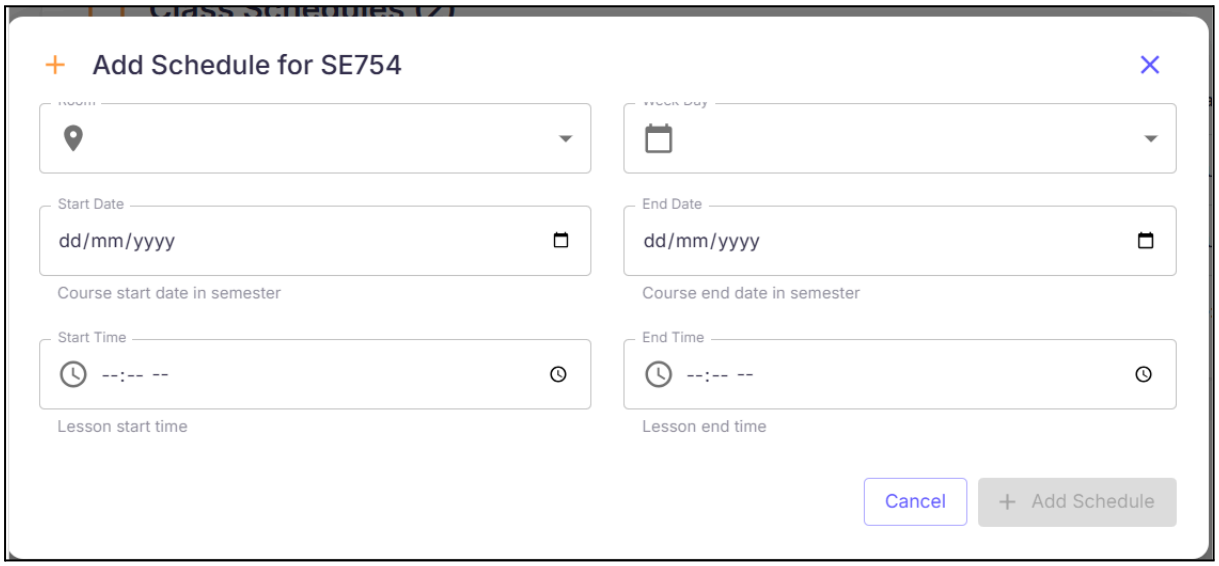


Figure xx - Add schedule dialog

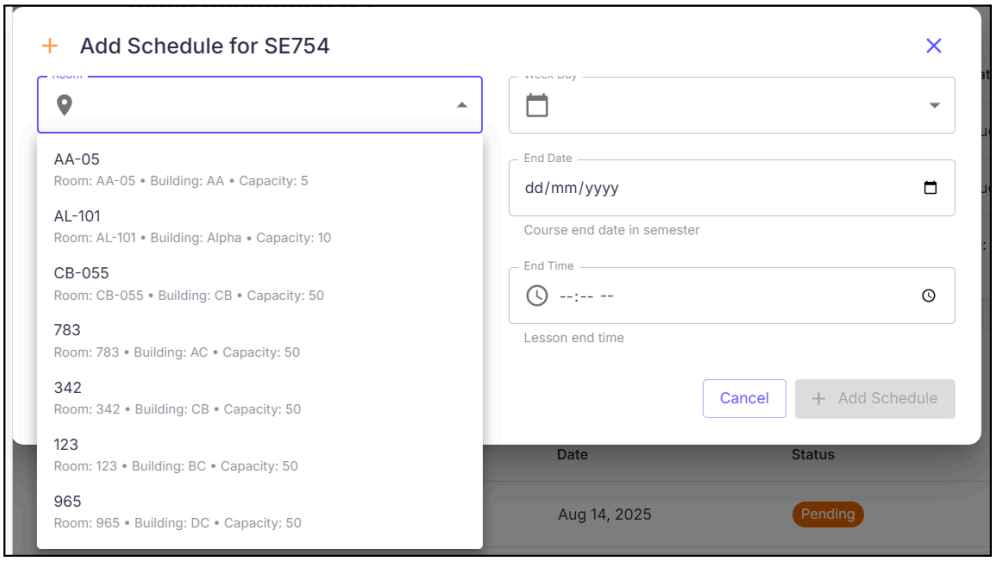


Figure xx - Add schedule - room selection

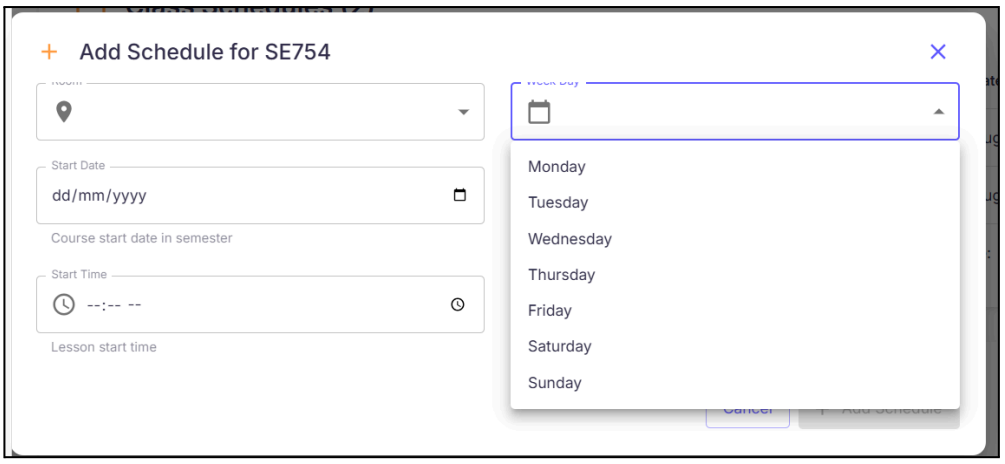


Figure xx - Add schedule - week day selection

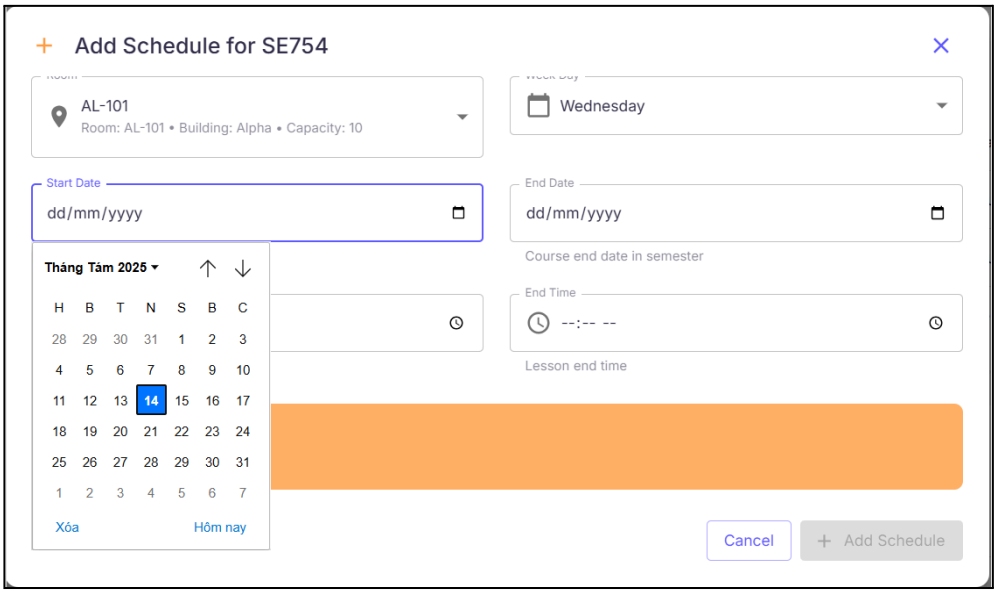


Figure xx - Add schedule - start date selection

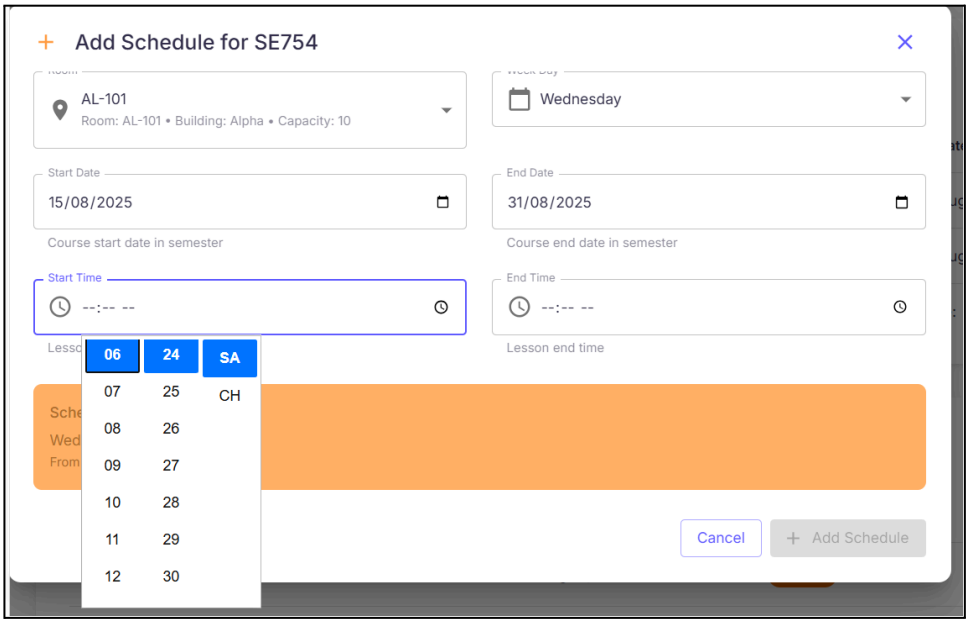


Figure xx - Add schedule - start time selection

- **Step 3**: Review the **Schedule Summary** section for a quick overview of your selections.

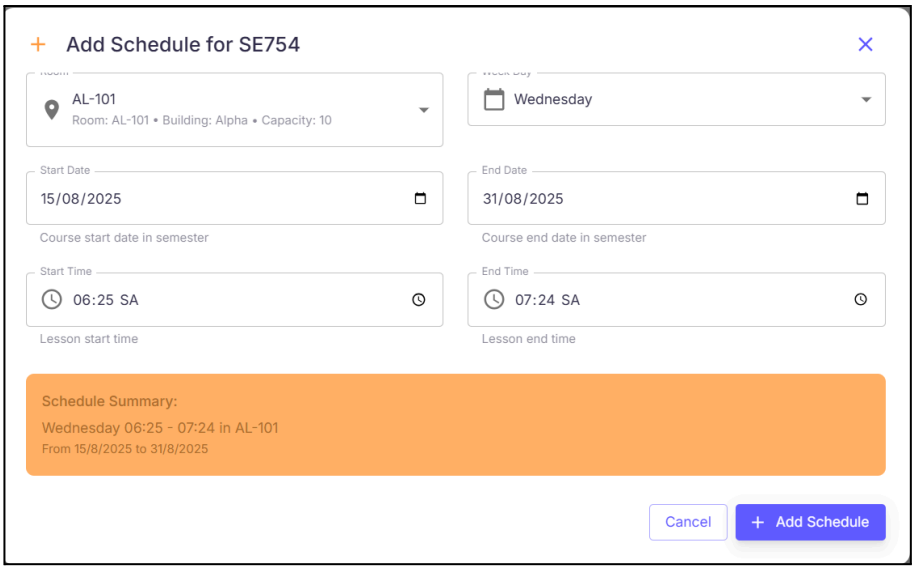


Figure xx - Add schedule - review schedule

- **Step 4**: Click **Add Schedule** to save the schedule.
- **Step 5**: **System response:**
  - **If successful:** The new schedule appears in the Class Schedules table, and a green notification confirms the addition (e.g., “Schedule added successfully”).

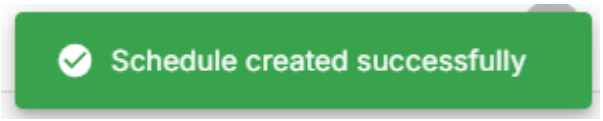


Figure xx - Add schedule success message

  - **If unsuccessful:** An error message appears (e.g., missing or invalid information).



Figure xx - Add schedule failure message

Edit Schedule

Description: This function allows users to modify the details of an existing schedule for a specific class. Users can update information such as room, weekday, date range, and lesson times.

Instruction:

- **Step 1**: On the **Class Schedules** section of the **Class Details** page, locate the schedule you want to edit. Click the **Edit** icon (pencil) in the **Actions** column of that schedule.

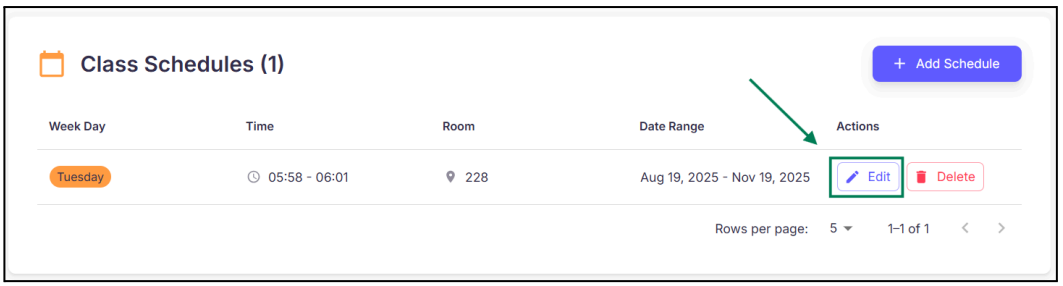


Figure xx -Edit schedule icon

- **Step 2**: In the **Edit Schedule** dialog, update the required fields:
  - **Room:** Select a new room from the dropdown list (if applicable).
  - **Week Day:** Change the weekday for the class (e.g., Monday → Thursday).
  - **Start Date** / **End Date:** Adjust the date range of the schedule.
  - **Start Time** / **End Time:** Modify the lesson start and end times.

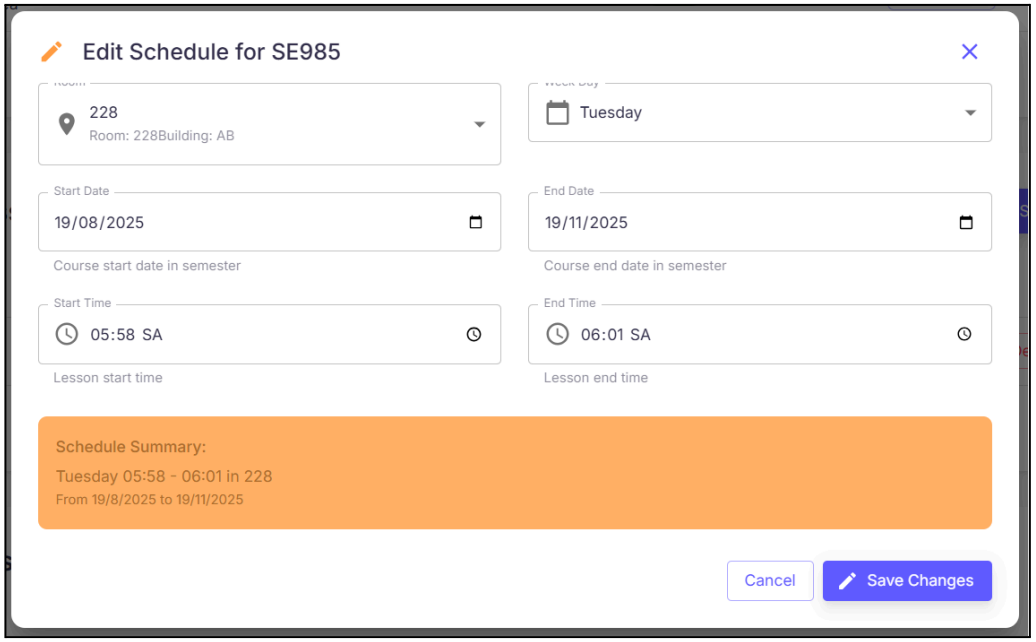


Figure xx -Edit schedule dialog

- **Step 3**: Review the updated schedule information in the **Schedule Summary** section before saving.
- **Step 4**: Click **Save Changes** to apply the updates.
- **Step 5**: **System response:**
  - **If successful:** The Class Schedules table refreshes with the updated information, and a green notification appears (e.g., “Schedule updated successfully”).

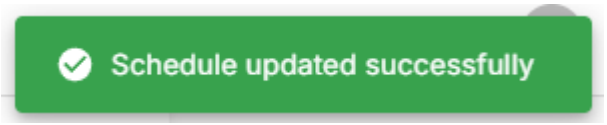


Figure xx -Edit schedule success message

- **If unsuccessful:** An error message appears (e.g., invalid time range, missing information).

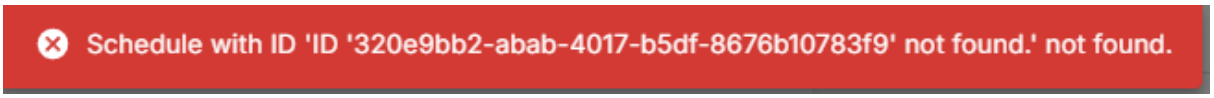


Figure xx -Edit schedule failure message

Delete Schedule

Description: This function allows users to permanently remove an existing schedule from a specific class. It is used when the schedule is no longer applicable or needs to be replaced entirely.

Instruction:

- **Step 1**: On the **Class Schedules** section of the **Class Details** page, locate the schedule you want to delete. Click the **Delete** icon (trash bin) in the **Actions** column for that schedule.

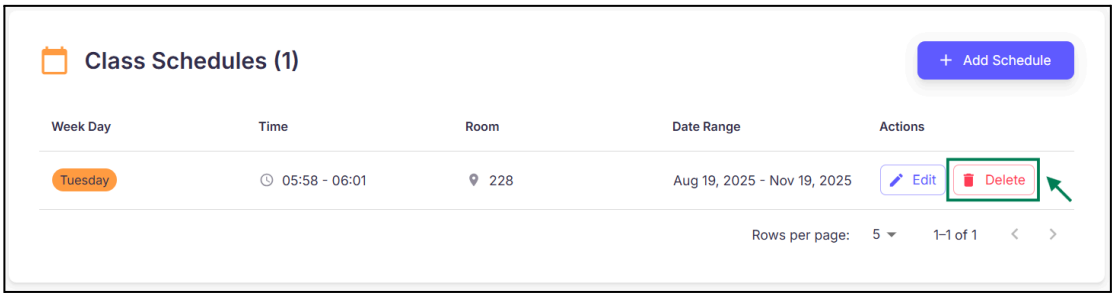


Figure xx - Delete icon

- **Step 2**: A **Delete Confirmation** dialog appears, asking the user to confirm the deletion. The dialog displays the key details of the schedule (Week Day, Time, Room, Date Range) for verification.

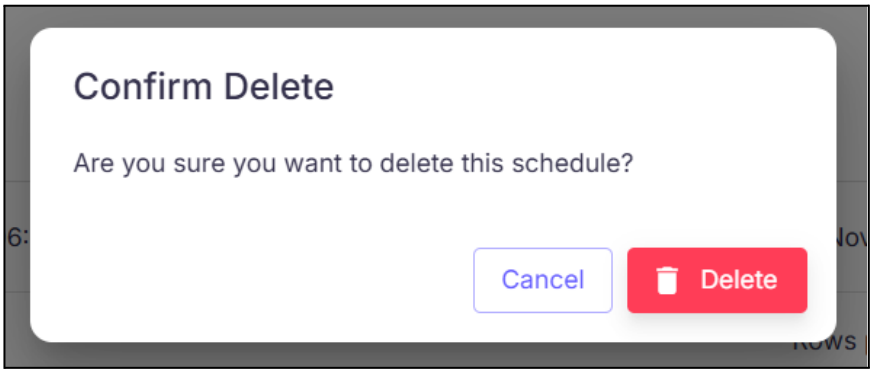


Figure xx - Delete confirm pop up

- **Step 3**: Click **Confirm** to proceed with the deletion. Click **Cancel** to close the dialog without making any changes.
- **Step 4**: **System response:**
  - **If successful:**
    - The schedule is removed from the Class Schedules table.

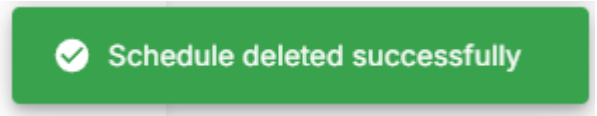


Figure xx - Delete schedule success message

- **If unsuccessful:**



Figure xx - Delete schedule failure message

- **Step 5**: If multiple schedules need to be deleted, repeat Steps 1–4 for each schedule.

3.1.6.6 Manage Sessions

View Sessions

Description: This function allows users to view the list of all sessions associated with a specific class. The table displays essential information for each session, including Week Day, Time, Date, and Status.

Instruction:

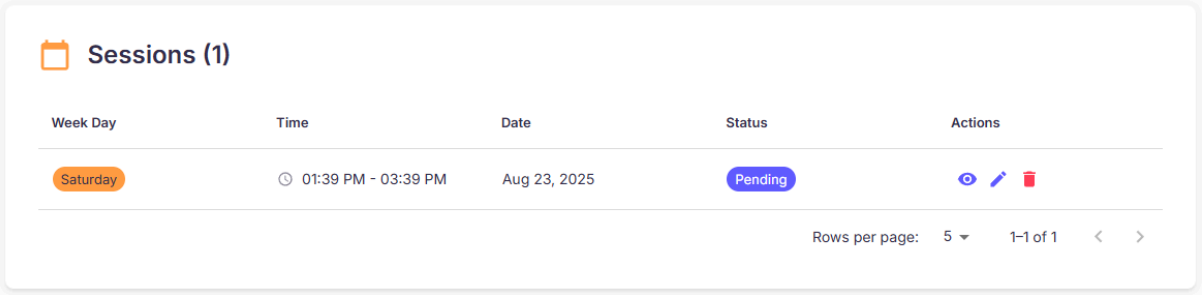


Figure xx - View session screen

- **Step 1**: On the **Class Details** page, scroll to the **Sessions** section.
- **Step 2**: Review the table columns:
  - **Week Day:** Day of the week the session is scheduled.
  - **Time:** Start and end time of the session.
  - **Date:** The specific date of the session.
  - **Status:** Current status of the session (e.g., Pending, Completed, Cancelled).
  - **Actions:** Allows the user to view more details about the session. Edit and even delete a specific session.
- **Step 3**: To view more details for a specific session, click the **View Details** button in the **Actions** column.
- **Step 4**: Use the pagination controls at the bottom of the table to navigate between pages if there are many records.
- **Step 5**: **System response:**
  - Data is automatically updated when new sessions are added or existing sessions are updated.

- Clicking **View Details** opens the **Session Details** page, showing full session information and related actions.

Edit Session

Description: This function allows users to modify the details of an existing session. Updates may include changing the Week Day, Time, Date, Status, or associated Room. This is useful for correcting scheduling errors or accommodating timetable changes.

Instruction:

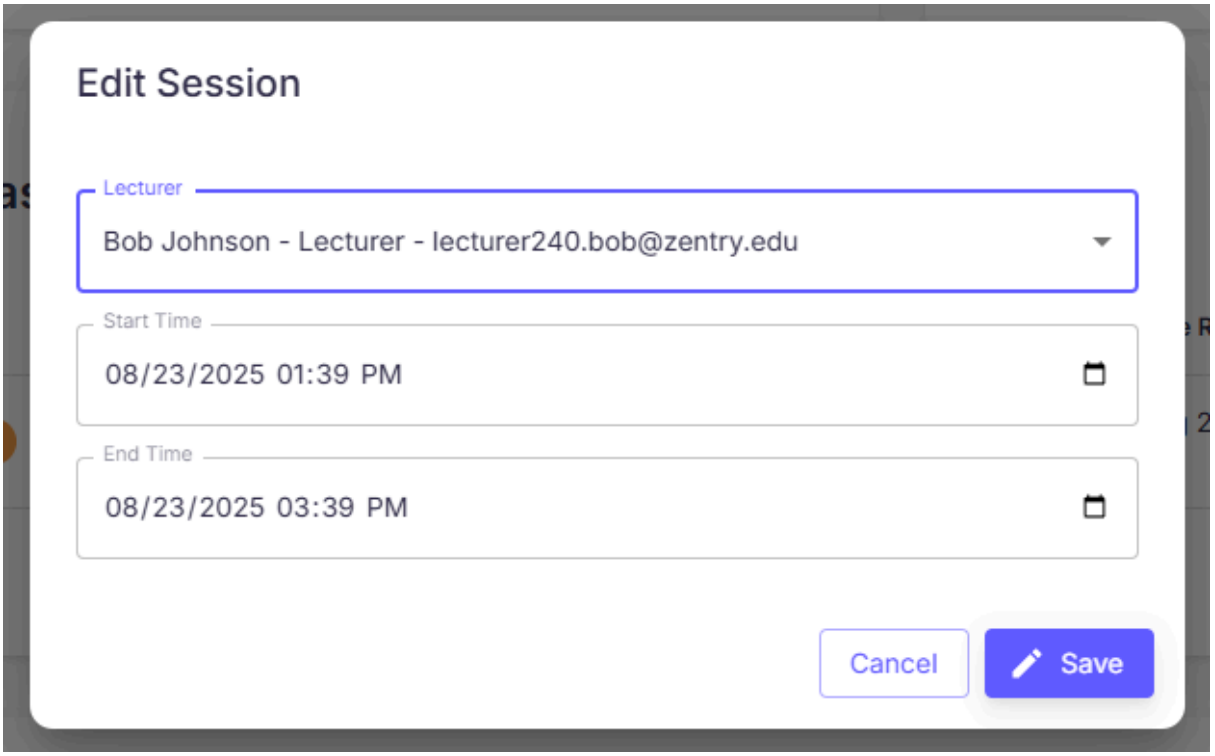
- **Step 1**: On the **Sessions** table in the **Class Details** page, locate the session you want to edit and click the **Edit** button in the **Actions** column.
- 

Figure xx - Edit session icon

- **Step 2**: The system opens the **Edit Session** form in a modal dialog or a dedicated page.
- **Step 3**: Update the required fields:
  - **Week Day:** Select a new day from the dropdown list (e.g., Monday, Tuesday, etc.).
  - **Time:** Adjust the start and end times using the time picker.
  - **Date:** Select a new date from the date picker.
  - **Room:** Choose the room from the list of available rooms.
  - **Status:** Select the current session status (e.g., Pending, Completed, Cancelled).

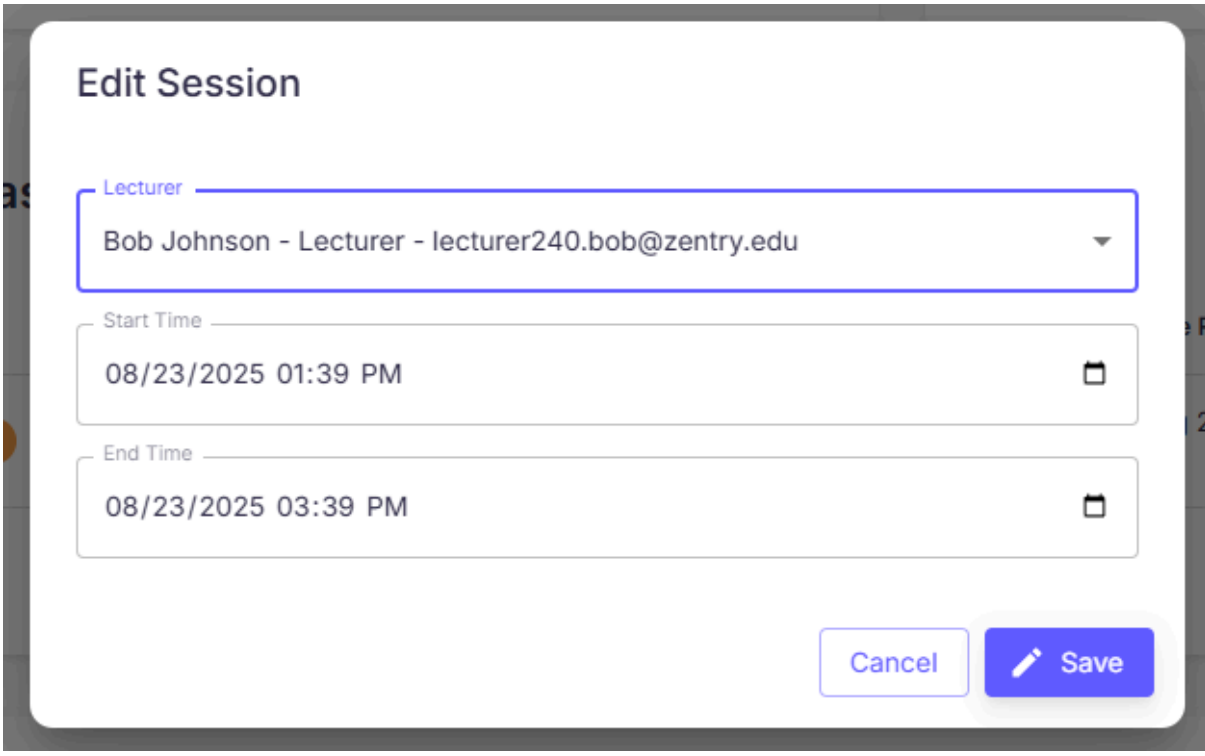
- 

Figure xx - Edit session dialog

- **Step 4**: Click **Save Changes** to confirm the edits, or **Cancel** to discard.
- **Step 5**: **System response:**
  - If the update is successful, the system refreshes the Sessions table to reflect the new information.
  - A green notification appears, e.g., “Session updated successfully.”

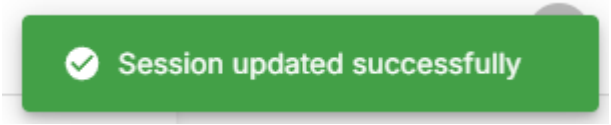


Figure xx - Edit session success message

  - If there’s a conflict (e.g., overlapping schedules), the system displays an error message in red, e.g., “Selected time conflicts with another session.”

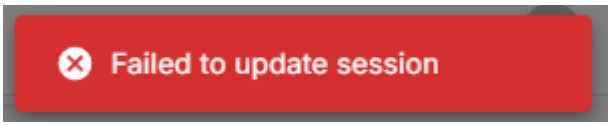


Figure xx - Edit session failure message

Delete Session

Description: This function allows users to permanently remove a scheduled session from a class when it is no longer needed (e.g., class is cancelled or rescheduled). The action is protected by a confirmation dialog and validation checks (e.g., cannot delete sessions that already have locked attendance unless you confirm or have permission).

Instruction:

- **Step 1**: On the **Class Details** ⟶ **Sessions** section, locate the target session in the table and click the **Delete**.

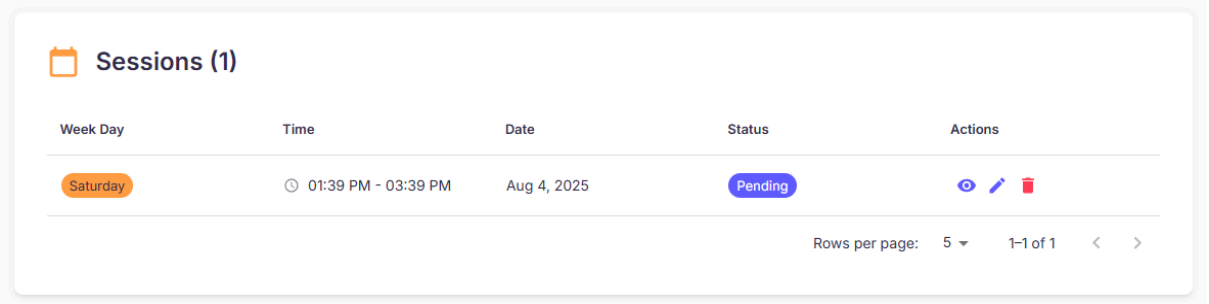


Figure xx - Edit session icon

- **Step 2**: A **Delete Session** confirmation dialog appears showing key details (Week Day, Time, Date, Room, Status). Review the information carefully.

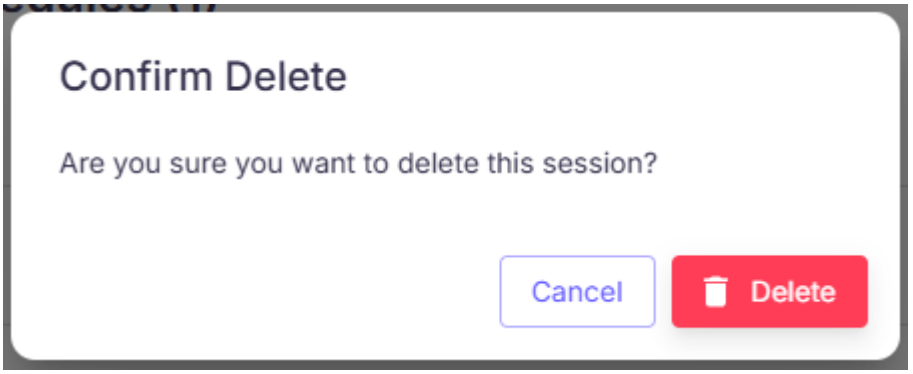


Figure xx - Edit session confirm dialog

- **Step 4**: Click **Delete** to proceed, or **Cancel** to abort.
- **Step 5**: **System response:**
  - **Successful:**
    - The dialog closes.
    - The **Sessions** table refreshes, and the session is removed.

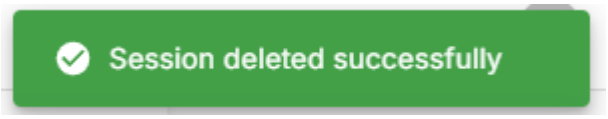
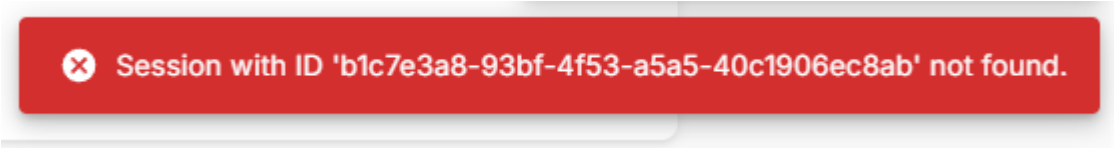


Figure xx - Edit session success message

  - **Unsuccessful:**
    - No records are changed.



3.1.6.7 Manage Session

View Session Details

Description: This function allows users to view complete information of a specific class session, including schedule details, room, related class/course, status, and student attendance summary. This page is used to verify session setup and quickly assess attendance statistics.

Instruction:

- **Step 1**: From the **Sessions** list in the **Class Details** page, locate the target session and click **View Details** in the **Actions** column.

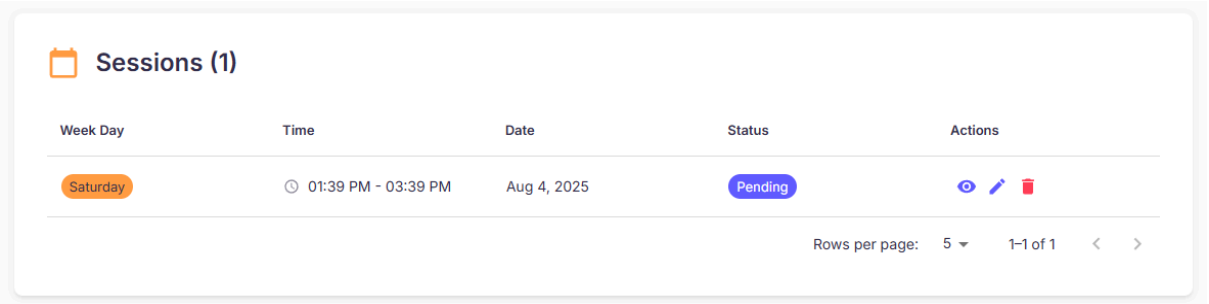


Figure xx - View session detail icon

- **Step 2**: **Session Details page layout:**

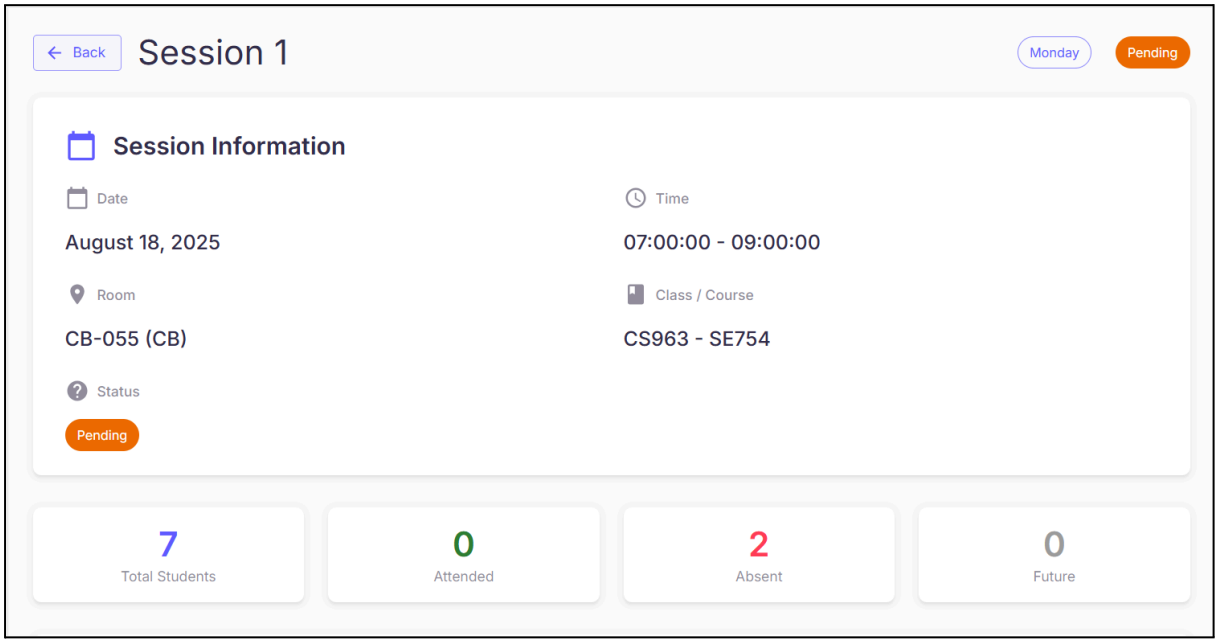


Figure xx - View session detail screen

The **Session Details** page is divided into two main sections:

1. **Session Information**
    - **Date**: Displays the scheduled date.
    - **Time**: Displays the session start and end time.
    - **Room**: Room code and name.
    - **Class / Course**: Associated class and course codes.

    - **Status**: Current session status (Pending, Ongoing, Completed, Cancelled).
2. **Attendance Summary**
    - **Total Students**: Number of students in this session.
    - **Attended**: Number of students marked as attended.
    - **Absent**: Number of students marked as absent.
    - **Future**: Number of students whose attendance status is not yet determined.

- **Step 3**: Click **Back** (top left) to return to the previous page (Sessions list). The weekday tag (e.g., Monday, Thursday) and session status tag are shown at the top right for quick identification.
- **Step 4**: **System response:**
    - The page loads session data in real-time from the system.
    - If the session is not found (invalid or deleted), the user is redirected to the Sessions list with an error message: *"Session not found or has been removed."*

3.1.6.8 Edit Attendance

Description: This function allows authorized users to update a student's attendance status for a specific session by switching between **Present** and **Absent**, with a confirmation step before saving.

Instruction:

- **Step 1**: On the **Session Details** page, scroll to the **Student Attendance** section.

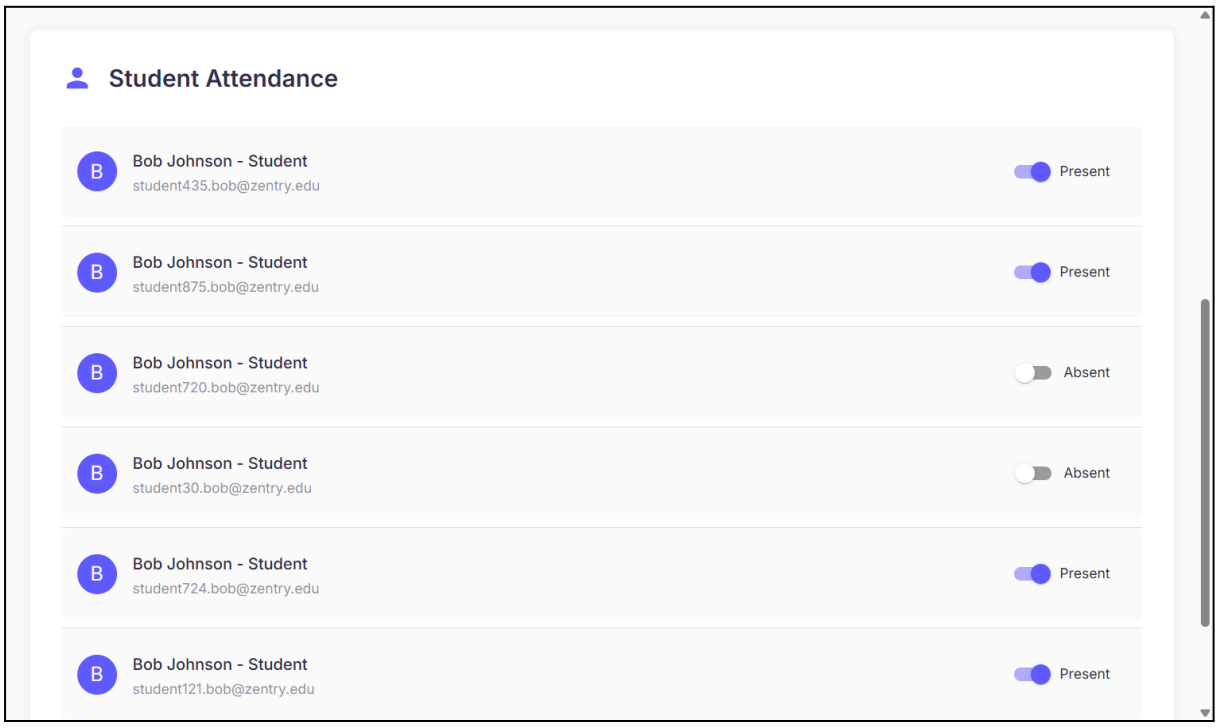


Figure xx - Edit attendance section

- **Step 2**: For the target student, toggle the switch to the desired status:
    - Turn **on** to mark **Present**.
    - Turn **off** to mark **Absent**.
- **Step 3**: In the confirmation dialog, review the new status and click **Confirm** to proceed (or **Cancel** to keep the current status).

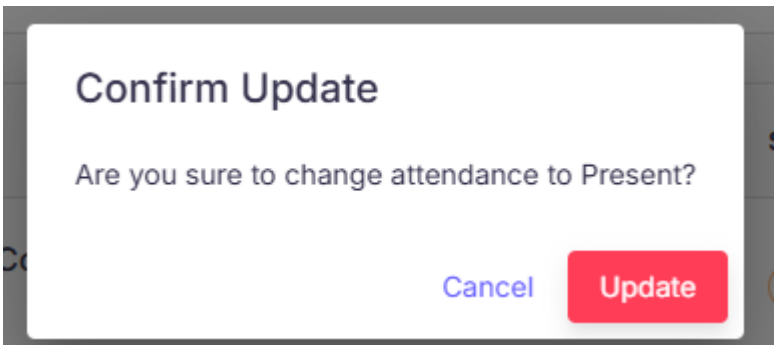


Figure xx - Edit attendance status - confirmation dialog

- **Step 4**: **System response:**
  - **If successful:**
    - The chip next to the student updates to **Present** or **Absent**.
    - The counters at the top (e.g., **Attended**, **Absent**) refresh automatically.

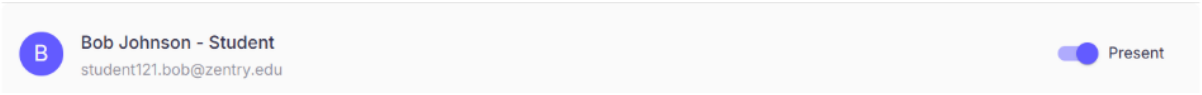
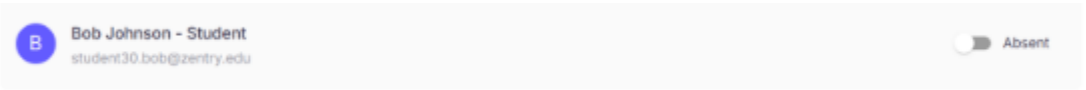


Figure xx - Edit attendance success message

  - **If unsuccessful:**
    - An error message appears (e.g., network issue or insufficient permission).
    - The toggle reverts to the previous state.



3.1.7 User Management

3.1.7.1 Manage Users

View Users

Description: This function allows authorized users to view and search for all users in the system, including their details such as full name, email, role, status, and account creation date.

Instruction:

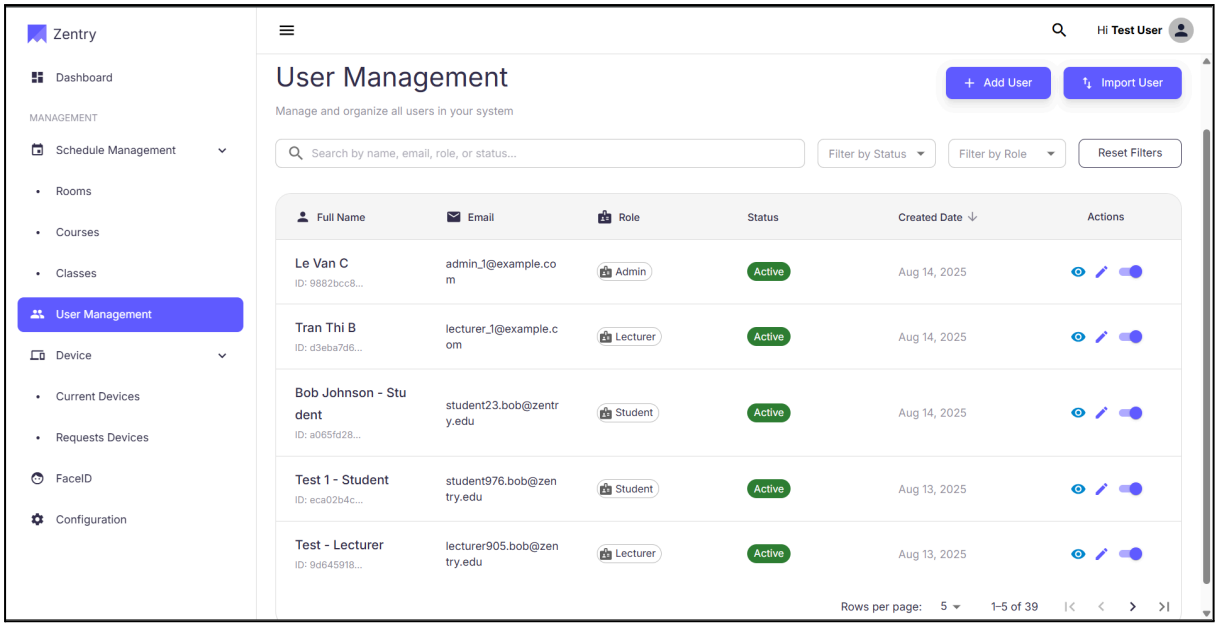


Figure xx - User management screen

- **Step 1**: From the **Dashboard**, navigate to **User Management** in the left sidebar menu.
- **Step 2**: The **User Management** page displays a table containing all users in the system. Each row shows:
    - **Full Name** & User ID
    - **Email**
    - **Role** (Admin, Lecturer, Student)
    - **Status** (Active/Inactive)
    - **Created Date**
    - **Actions** (View, Edit, Change Status toggle)
- **Step 3**: Use the search bar to find a user by name, email, role, or status.
- **Step 4**: Use the filters to narrow results:
    - **Filter by Status** (Active, Inactive)
    - **Filter by Role** (Admin, Lecturer, Student)

Add User

Description: This function allows admins to create a new user by entering an email, full name, selecting a role, and setting a password.

Instruction:

- **Step 1**: On the **User Management** page, click the **Add User** button in the top-right corner.



Figure xx - Add user button

- **Step 2**: In the **Add New User** dialog, fill in the fields:

- **Email:** Enter the user's email address.
- **Full Name:** Enter the user's full name.
- **Role:** Open the dropdown and select **Admin**, **Lecturer**, or **Student**.
- **Password:** Enter a temporary password that meets the security policy.

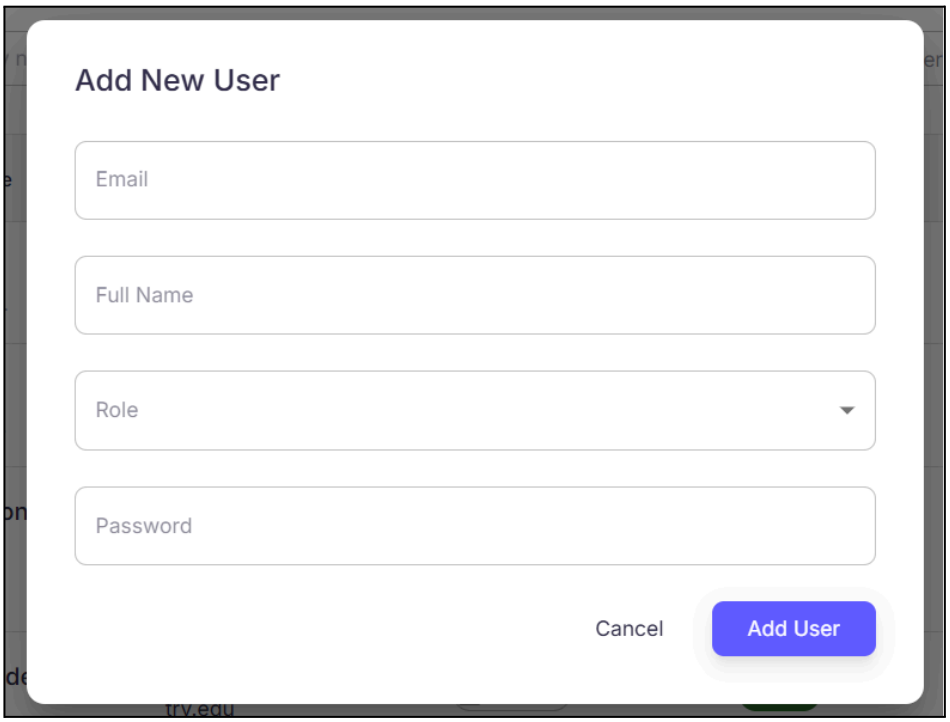


Figure xx - Add user dialog

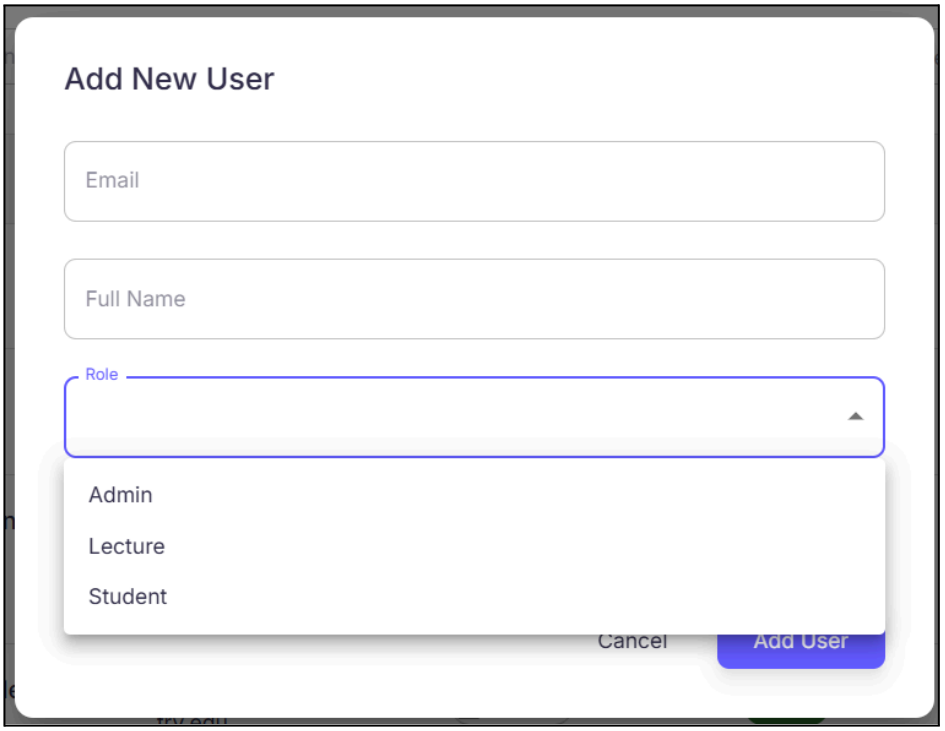


Figure xx - Add user - role selection

- **Step 3**: Click **Add User** to create the account (or **Cancel** to close without changes).
- **Step 4**: **System response:**
    - **If successful:** A green notification appears — “User added successfully”. The dialog closes, and the new user appears in the list (status: **Active** by default)

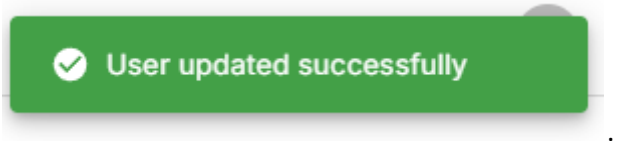
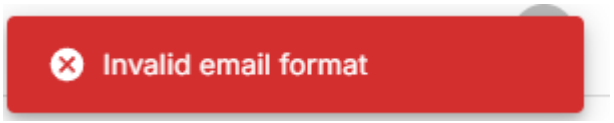


Figure xx - Add user success message

- **If unsuccessful:** An error message appears (e.g., duplicate email, invalid email format, weak password). The invalid fields are highlighted, and the dialog remains open for correction.



Import User

Description: This function allows admins to add multiple new users to the system by uploading a CSV file, saving time compared to manual entry.

Instruction:

- **Step 1**: On the User Management page, click the **Import User** button in the top-right corner.

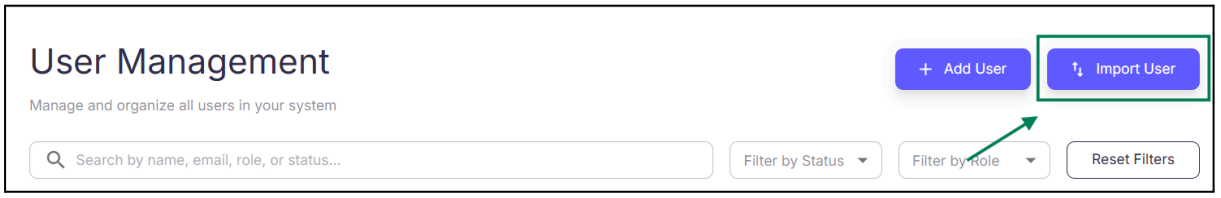


Figure xx - Import user button

- **Step 2**: In the Import Users dialog, click **Choose CSV File**.

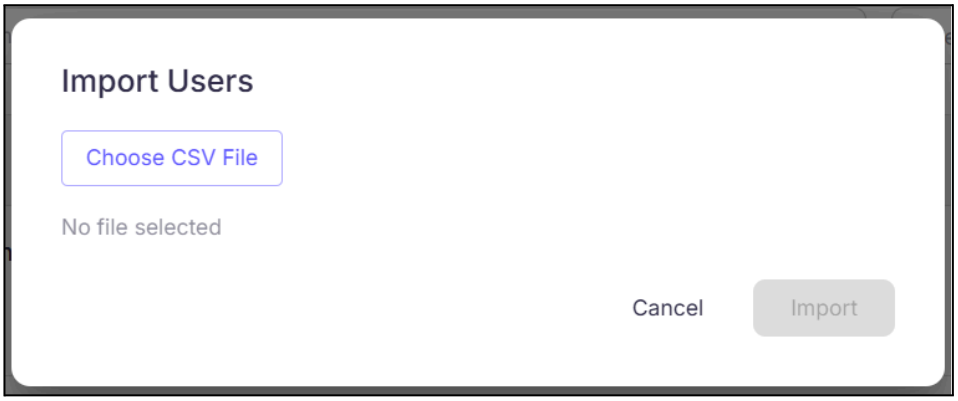


Figure xx - Import user dialog

- **Step 3**: In the file picker window, navigate to and select the desired CSV file from your computer. The file name will appear in the dialog once selected.

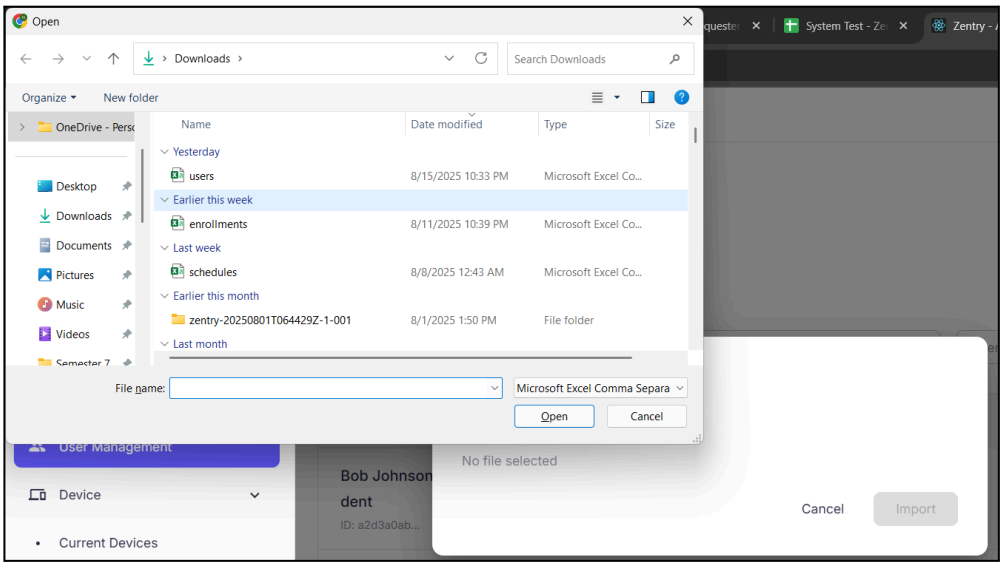


Figure xx - Import user - file selection

- **Step 4**: Click **Import** to upload the file (or **Cancel** to close without changes).



Figure xx - Import user dialog - file selected display

- **Step 5**: System response:
  - **If successful:**

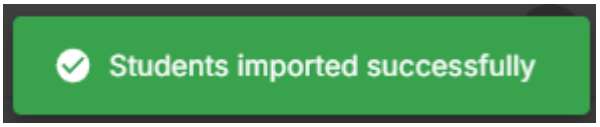
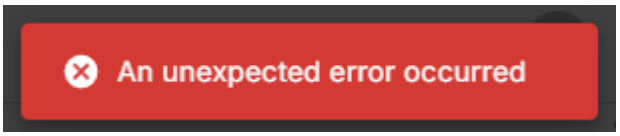


Figure xx - Import user success message

  - **If unsuccessful:**
    An error message appears (e.g., invalid file format, missing required columns, invalid data in some rows).
    - If the file is invalid, the system rejects the file and prompts to re-upload.

- If some rows are invalid, the system imports only the valid rows and displays a summary of the errors for correction.



Edit User

Description: This function allows admins to update an existing user's information such as email, full name, and role.

Instruction:

- **Step 1**: On the User Management page, locate the user you want to edit and click the **Edit User** icon (pencil).

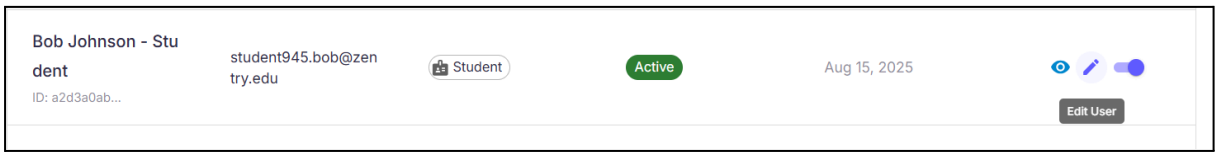


Figure xx - Edit user section

- **Step 2**: The **Edit User** dialog appears with the user's current details pre-filled:
  - **Email:** Update the user's email if needed.
  - **Full Name:** Modify the full name.
  - **Role:** Use the dropdown to change the role (Admin, Lecturer, Student).

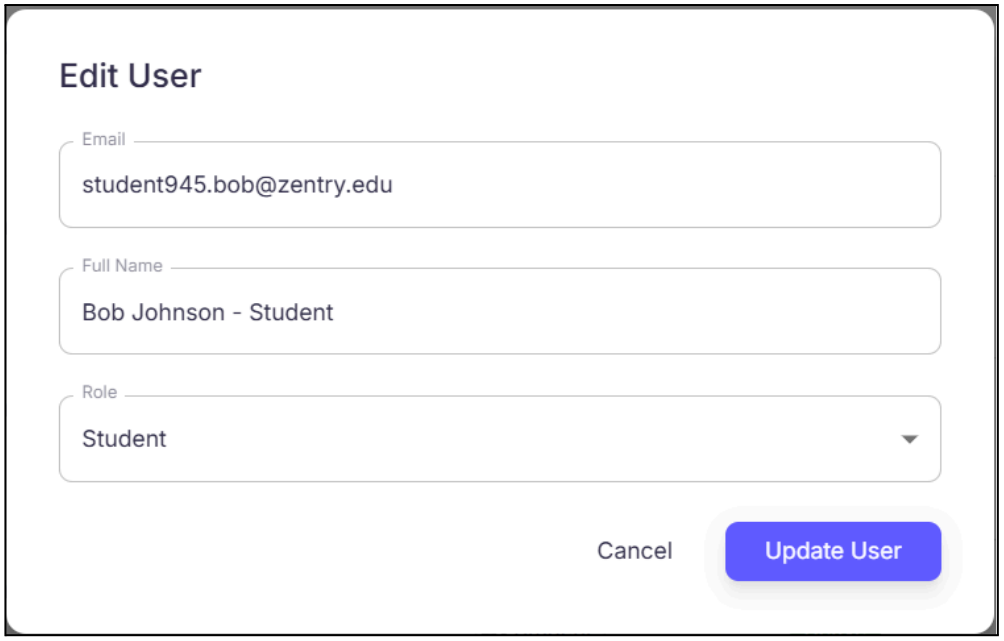


Figure xx - Edit user dialog

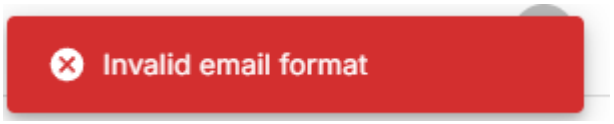
- **Step 3**: Click **Update User** to save changes (or **Cancel** to discard).
- **Step 4**: System response:

- **If successful:** A green notification appears — “User updated successfully”. The dialog closes and the updated details are shown in the user list.



Figure xx - Update user success message

- **If unsuccessful:** An error message appears (e.g., invalid email format, duplicate email). The problematic fields are highlighted and remain editable for correction.



Change User Status

Description: This function allows admins to change a user’s status between **Active** and **Inactive**, helping manage whether a user can access the system.

Instruction:

- **Step 1**: On the User Management page, locate the user whose status you want to change. Click the **Toggle status** button (switch icon on the right).



Figure xx - Change user status section

- **Step 2**: A confirmation dialog appears showing the current and new status.

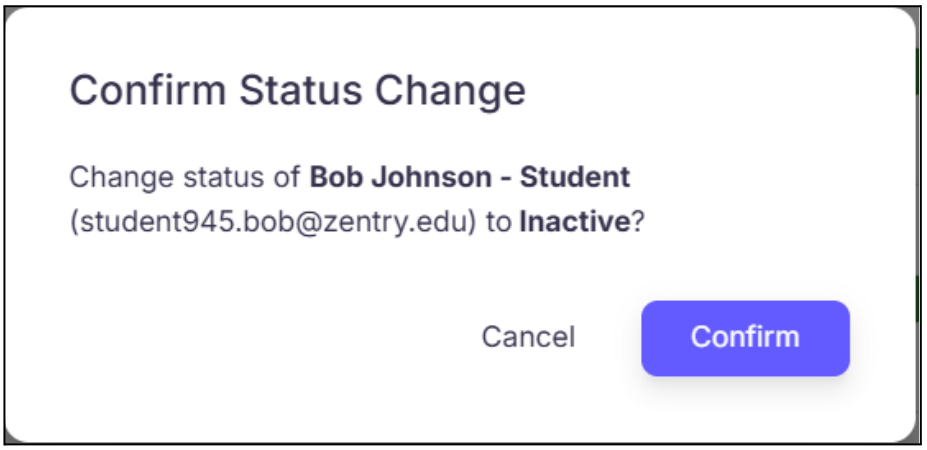


Figure xx - Change user status - confirmation dialog

- **Step 3**: In the dialog:
  - Click **Confirm** to proceed with the status change.

- Click **Cancel** to close the dialog without changes.

- **Step 4**: System response:
  - If successful: The user's status label updates immediately (e.g., Active → Inactive or Inactive → Active), and a green notification appears.

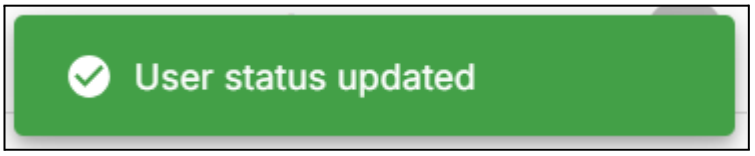
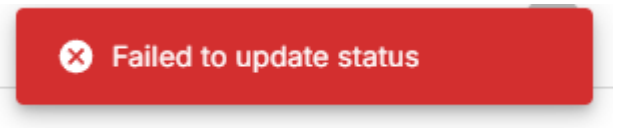


Figure xx - Change user status success message

  - If unsuccessful: An error message appears (e.g., network error). The status remains unchanged.



3.1.7.2 View User Details

Description: This function allows admins to view detailed information about a user, including their full name, email, role, account status, and creation date.

Instruction:

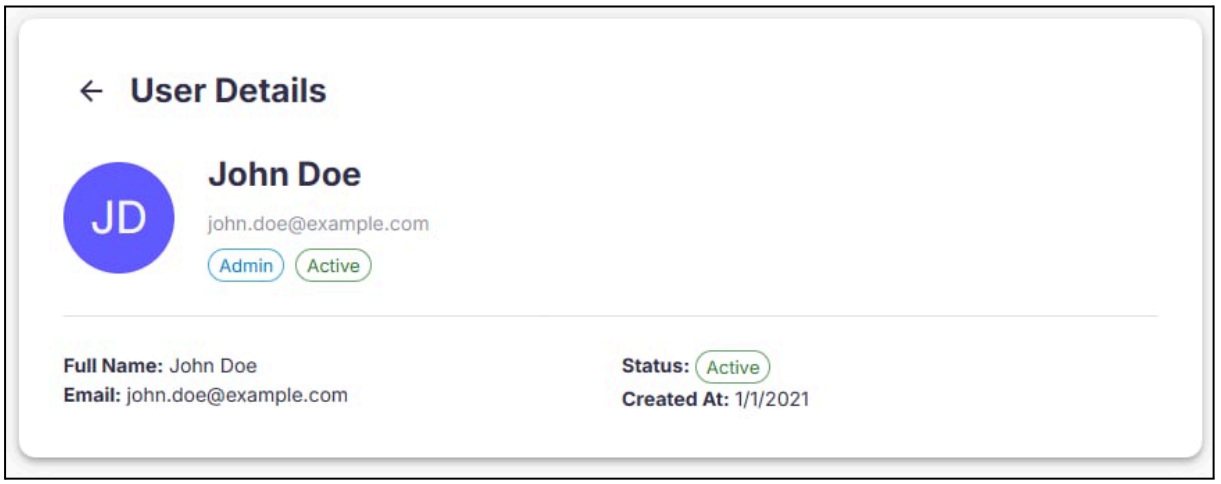


Figure xx - User detail section

- **Step 1**: On the User Management page, locate the user whose details you want to view. Click the **View User** icon (eye symbol) on the right side of the row.
- **Step 2**: The **User Details** page opens, displaying:
  - **Full Name:** User's full name.
  - **Email:** User's email address.
  - **Role:** Admin, Lecturer, or Student (shown as a badge).
  - **Status:** Current account status (Active/Inactive).
  - **Created At:** Date when the account was created.
- **Step 3**: Review the information.

- To return to the User Management list, click the **Back** arrow at the top-left corner of the page.

System Response:

- The system loads and displays the most up-to-date user information.
- If the user does not exist (e.g., deleted), the system shows an error message: *"User not found."*

3.1.8 Active Device Management

3.1.8.1 Manage Active Devices

View Active Devices

Description: This function allows admins to view and monitor all devices that are currently marked as **Active** in the system. Active devices are those successfully registered and approved for use.

Instruction:

- **Step 1**: On the sidebar menu, under **Device**, click **Current Devices**.

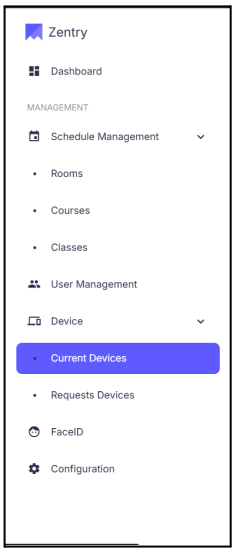


Figure xx - Current devices submenu item

- **Step 2**: In the **Current Devices Management** table, locate the **Status** column.

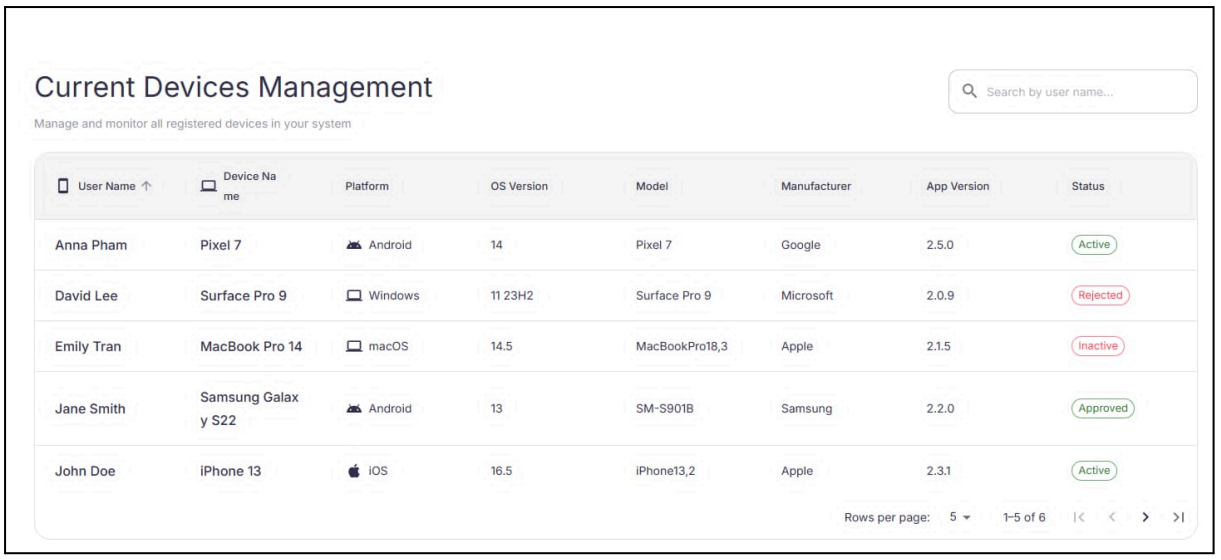
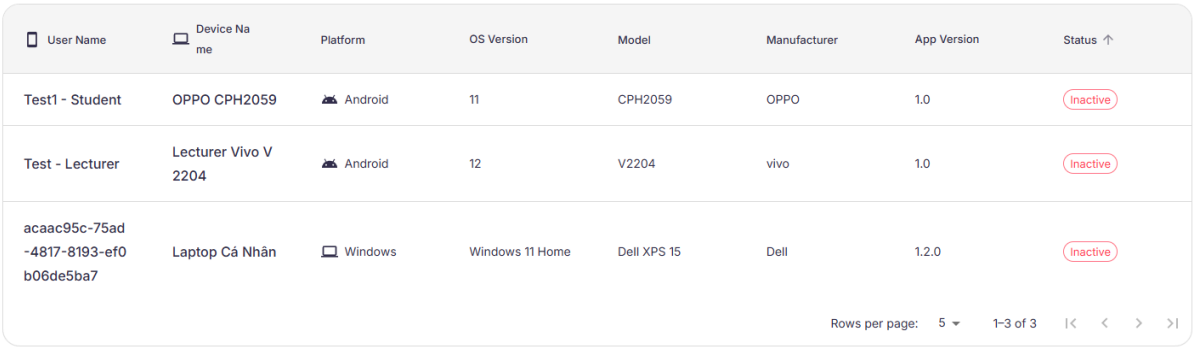


Figure xx - Current devices screen

- **Step 3**: Use the search bar or scroll to find devices. Identify those with the **Active** status badge (green).
- **Step 4**: Use the table filters or pagination to refine the list of displayed devices.
- **Step 5**: System response:
    - The table displays all devices with status **Active**, showing details such as **User Name, Device Name, Platform, OS Version, Model, Manufacturer, and App Version**.
    - If no devices are active, the table will appear empty or show a message like *"No active devices found."*

Inactive Device

Description: This function allows admins to view and manage all devices that are currently marked as **Inactive** in the system. Inactive devices are those that were previously registered but are no longer in use, disabled by admin, or disconnected from the system.



Instruction:

- **Step 1:** On the sidebar menu, under **Device**, click **Current Devices**.
- **Step 2:** In the **Current Devices Management** table, go to the **Status** column.
- **Step 3:** Use the search bar or filters to locate devices. Identify those with the **Inactive** status badge (gray or red outline).

- **Step 4:** (Optional) Click on a specific inactive device to view more details, including **last active date, registered user, platform, and OS version**.
- **Step 5:** System response:

- The table lists all devices with status **Inactive**.
- Admin may choose to either:
    - **Re-activate** the device (if eligible).
    - **Remove** the device permanently from the system.
- If no inactive devices exist, the system shows *"No inactive devices found."*

3.1.9 Device Request Management

View Device Requests

Description: This function allows administrators to review and manage device registration requests submitted by users. Admins can either **approve** or **reject** requests, with immediate updates reflected in the system.

Instruction:



Figure xx - Pending device requests screen

- **Step 1**: On the sidebar menu, navigate to **Device → Requests Devices**.
- **Step 2**: The system displays a list of **Pending Device Requests**, including details such as:
    - User Name
    - Device Name
    - Platform
    - OS Version
    - Model
    - Manufacturer
    - App Version
    - Current Status (active/inactive)
- **Step 3**: To approve or reject a device:
    - Click the ✓ **(Approve)** button to approve the request.
    - Click the ✕ **(Reject)** button to decline the request.
- **Step 4**: **System response:**

- If **approved**: The device is moved from **Requests Devices** to **Current Devices** list, and its status is updated accordingly.
- If **rejected**: The device request is removed from the pending list, and the user is notified of the rejection.

Edit Device Request

Description: This function allows administrators to confirm and finalize the approval or rejection of a pending device request. By clicking on the approve (✔) or reject (✖) icons, the admin can edit/confirm the decision before submitting.

Instruction:

- **Step 1**: On the **Pending Device Requests** page, click the ✔ **(Approve)** or ✖ **(Reject)** button in the *Actions* column for the corresponding device.
- **Step 2**: A confirmation dialog appears:
  - **Approve Device Dialog** – Confirms approval of the device (shows user ID + device info).
  - **Reject Device Dialog** – Confirms rejection of the device (shows user ID + device info).

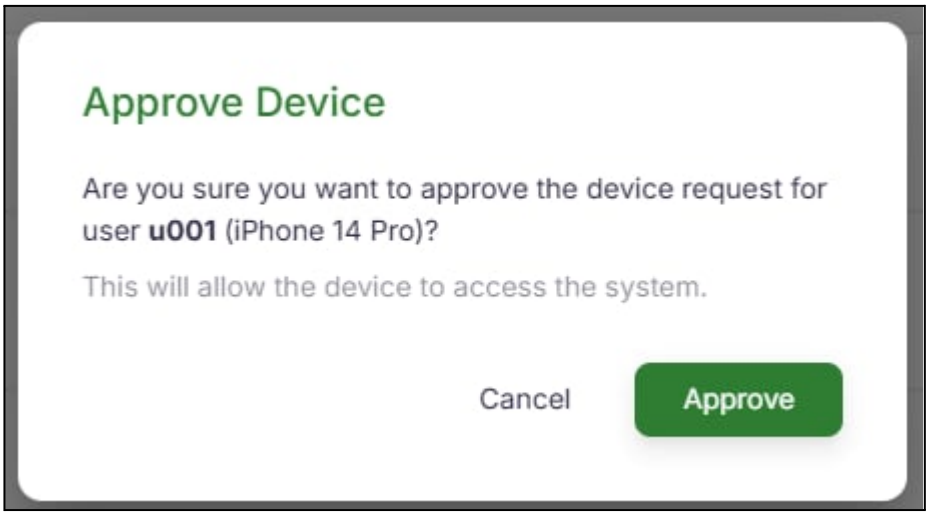


Figure xx - Edit device request - approval confirmation dialog

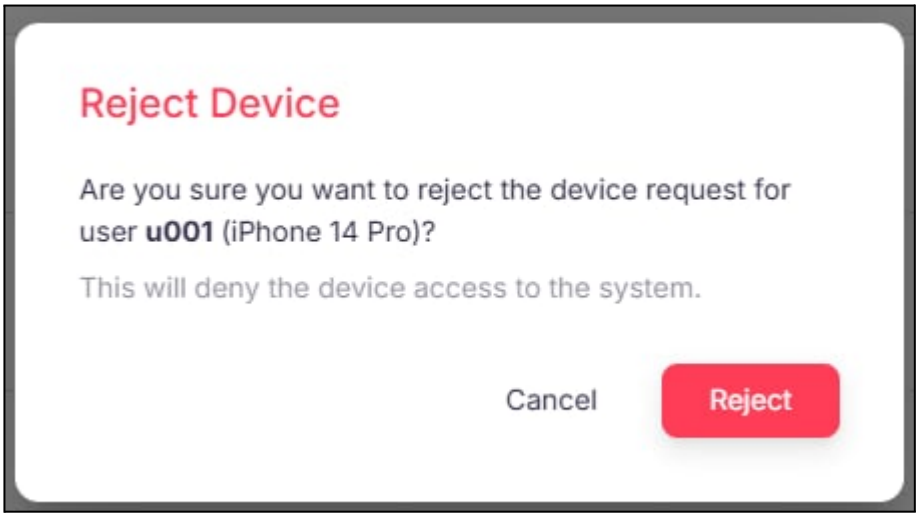


Figure xx - Edit device request - rejection confirmation dialog

- **Step 3**: In the dialog:
  - To continue, click **Approve** or **Reject**.
  - To cancel the action, click **Cancel**.
- **Step 4**: System Response:
  - If approval is successful: *"Device approved successfully"* (green notification).

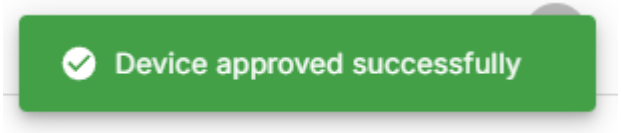


Figure xx - Edit device request - approved successfully

  - If rejection is successful: *"Device rejected successfully"* (red notification).

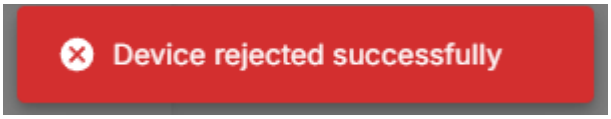


Figure xx - Edit device request - rejected successfully

  - If unsuccessful, an error message will appear.

3.1.10 FaceID Management

Description: This function allows administrators to view all active (registered) FaceIDs in the system. It provides a list of users with their FaceID registration status, registration date, and the latest update.

Instruction:

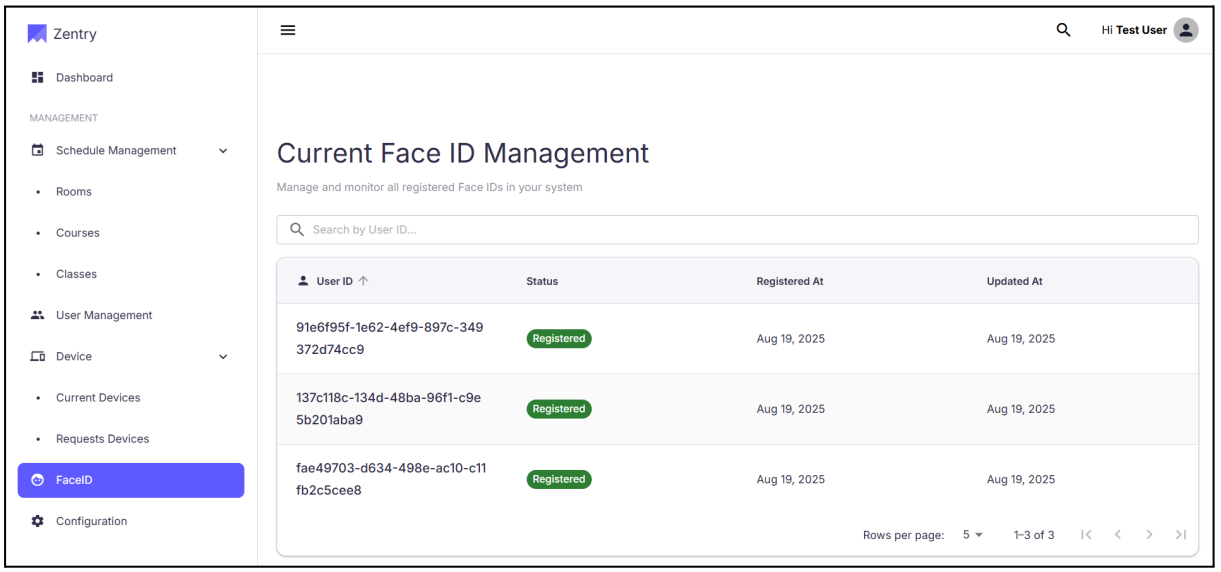


Figure xx - Current Face ID management screen

- **Step 1**: On the sidebar menu **FaceID**.
- **Step 2**: The system displays a table of all users with FaceID information, including:
  - **User ID**

- **Status** (Registered / Not yet)
- **Registered At** (the date the FaceID was registered)
- **Updated At** (the last modification date)

- **Step 3**: To focus on **active FaceIDs**, check the **Status** column and identify rows marked as **Registered** (green).
- **Step 4**: Use the **search box** at the top of the table to quickly find a specific user by User ID.
- **Step 5**: If the number of records exceeds one page, use the **pagination controls** at the bottom-right corner to navigate.

3.1.11 Configuration Management

3.1.11.1 Manage Attribute Definitions

View Attribute Definitions

Description: This function allows users to view detailed information about a specific attribute definition, including its key, display name, data type, allowed scopes, default value, and description.

Instruction:

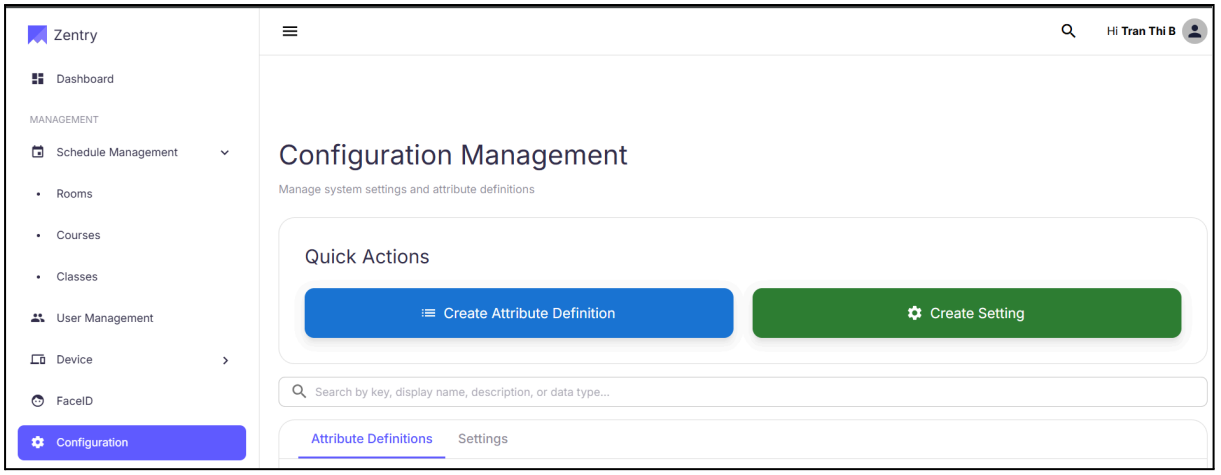


Figure xx - Configuration management screen

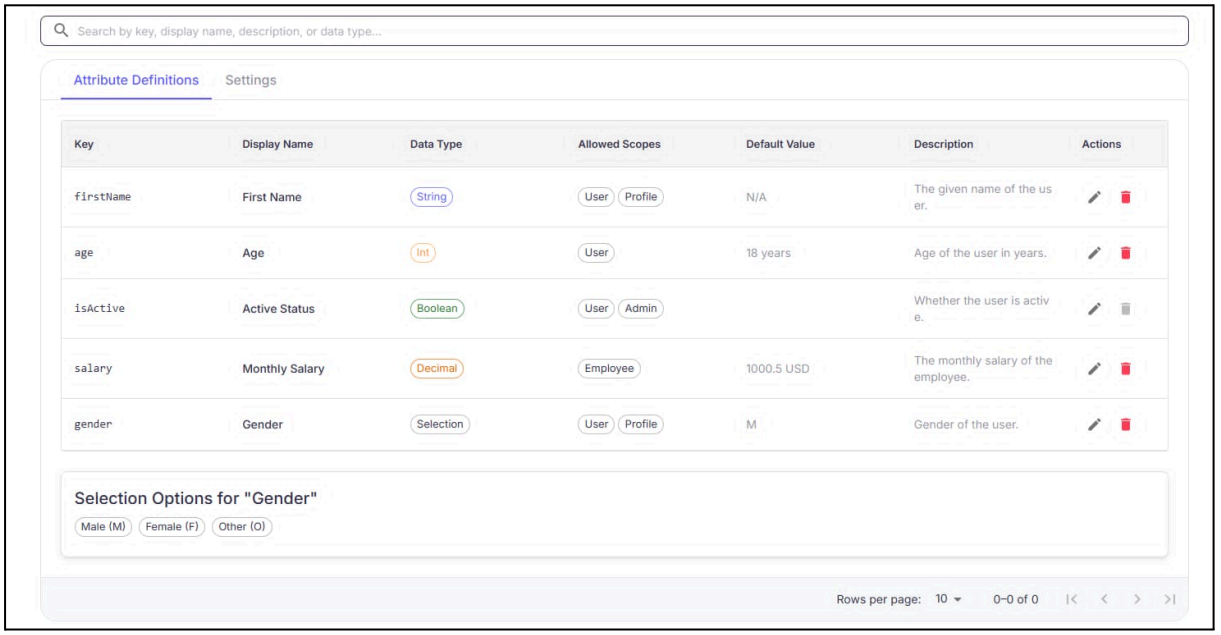


Figure xx - Attribute definitions screen

- **Step 1**: On the **Configuration Management** → **Attribute Definitions** page, click the **View (eye icon)** button in the **Actions** column corresponding to the attribute you want to see.
- **Step 2**: The system displays the **Attribute Definition Details** screen. This includes the following information:
    - **Key**: Internal system identifier of the attribute.
    - **Display Name**: Human-readable name shown in the UI.
    - **Data Type**: Type of data (e.g., String, Int, Boolean, Decimal, Selection).
    - **Allowed Scopes**: Scopes that can use this attribute (e.g., User, Profile, Employee, Admin).
    - **Default Value**: Default assigned value if none is provided.
    - **Description**: Explanation of the attribute's purpose.
    - **Selection Options** (if applicable): List of available choices (for Selection type).

Create Attribute Definition

Description: This function allows users to define a new system attribute with specific properties such as key, display name, data type, scope, unit, and default value.

Instruction:

- **Step 1**: On the *Configuration Management* page, click the **Create Attribute Definition** button.

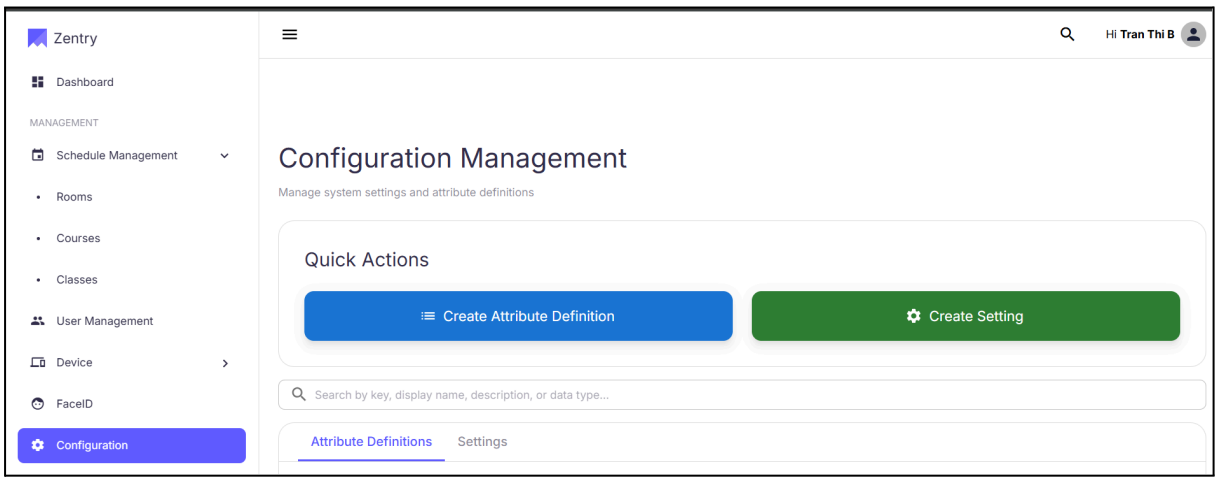


Figure xx - Configuration management screen - create attribution definition button

- **Step 2**: A dialog box will appear. Fill in the required fields:

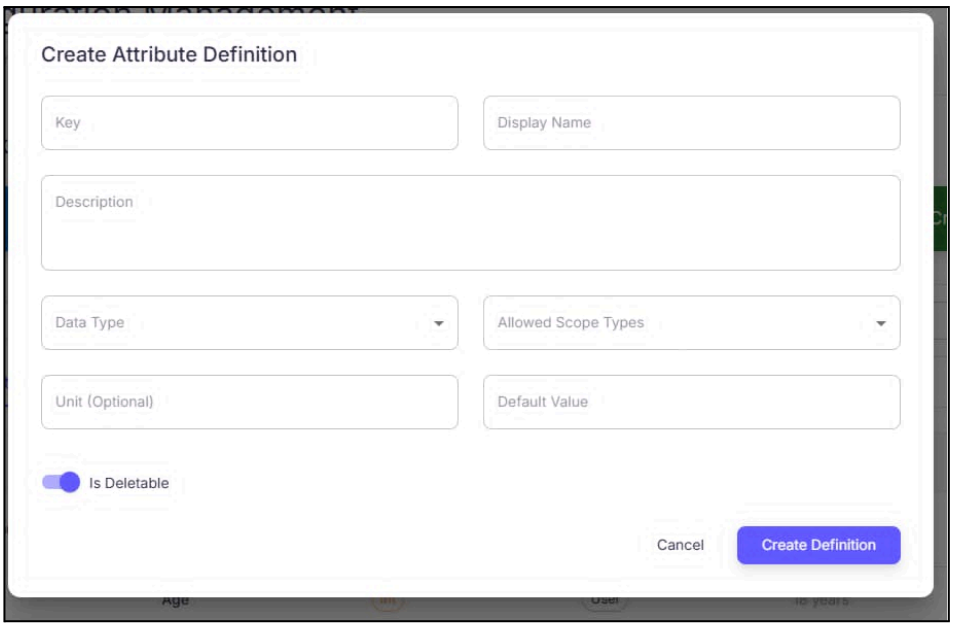


Figure xx - Create attribution definition dialog

- **Key**: Unique identifier of the attribute.
- **Display Name**: Human-readable name of the attribute.
- **Description**: Explanation of the attribute purpose.
- **Data Type**: Select from available types (*String, Int, Boolean, Decimal, Date, Json, Selection*).

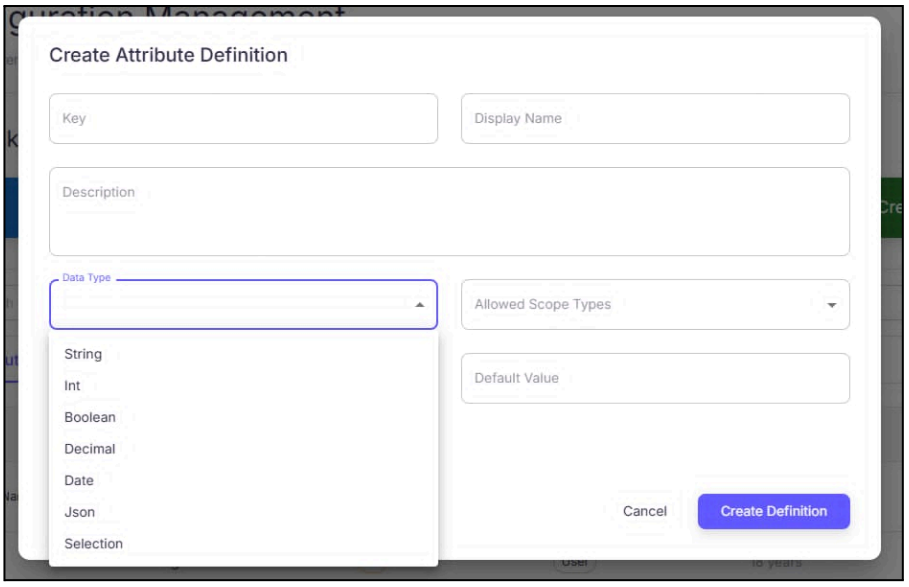


Figure xx – Create Attribute Definition – Select Data Type

- **Allowed Scope Types**: Choose where this attribute can be applied (*Global, Course, Session*).

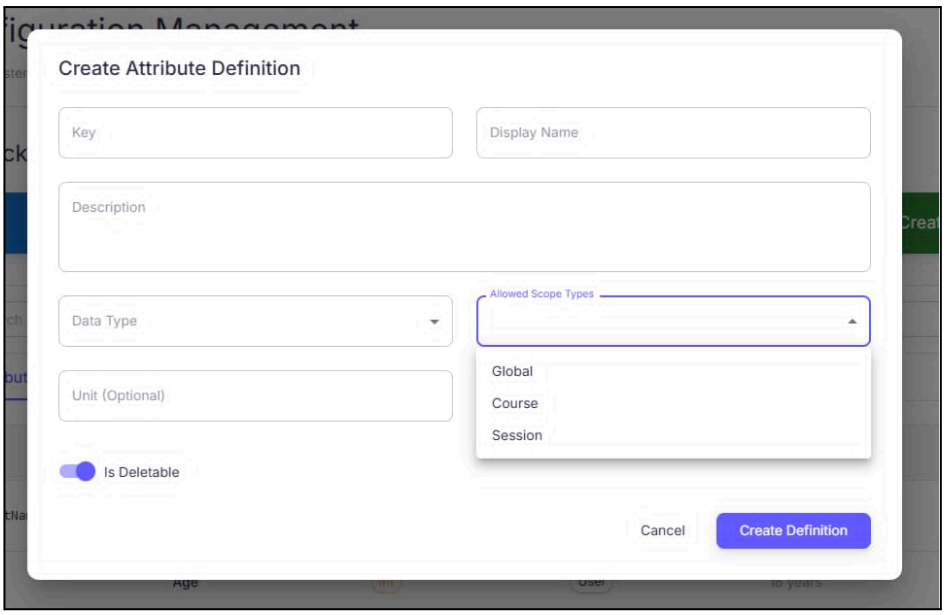


Figure xx – Create Attribute Definition – Select Scope Type

- **Unit (Optional)**: Measurement unit, if applicable.
- **Default Value**: Pre-filled value if not specified.
- **Is Deletable**: Toggle to allow or restrict deletion of this attribute.

- **Step 3**: After completing the form, click **Create Definition** to save the new attribute.
- **Step 4**: To cancel, click **Cancel,** and no changes will be saved.

System Response:

- If successful: A confirmation notification appears — *"Attribute Definition created successfully."* The new definition is added to the list under **Attribute Definitions**.

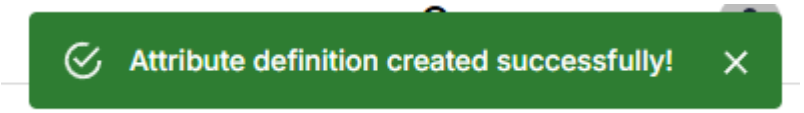
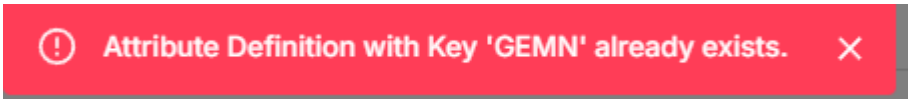


Figure xx – Create Attribute Definition success message

- If unsuccessful: An error message appears (e.g., *"Key already exists"* or *"Missing required field"*).



Edit Attribute Definition

Description: This function allows users to update details of an existing Attribute Definition such as display name, description, data type, allowed scope types, unit, default value, and deletable option.

Instruction:

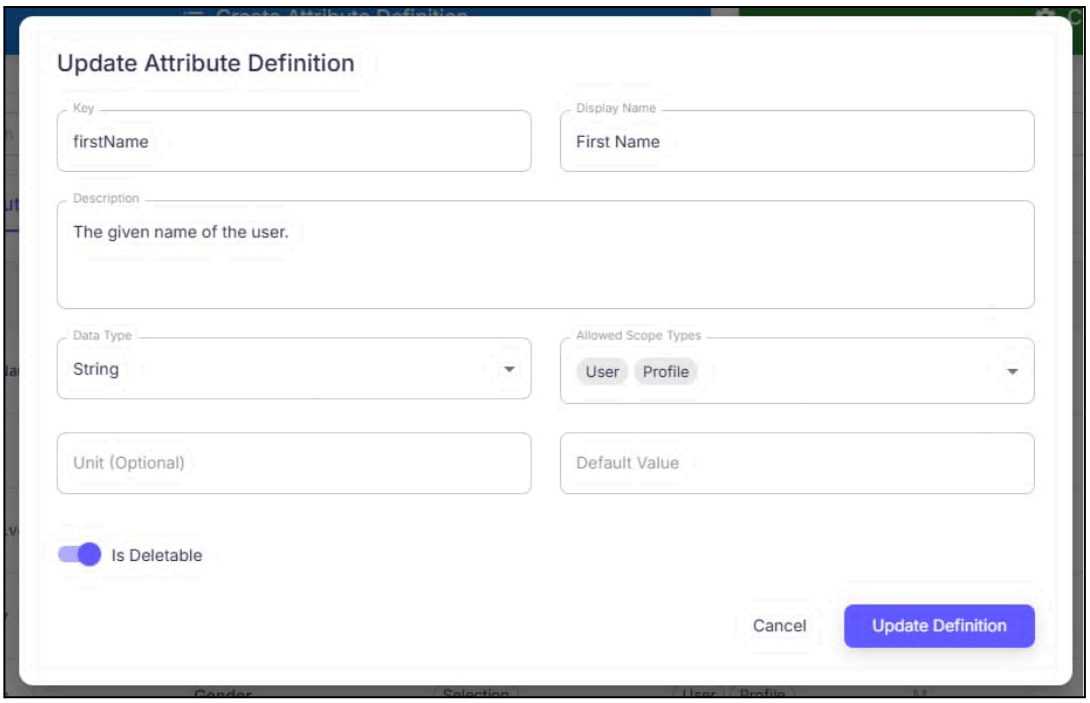


Figure xx - Edit attribution definition dialog

- **Step 1**: On the *Configuration Management* page, select the attribute definition you want to edit and click the **Edit** icon.
- **Step 2**: The **Update Attribute Definition** dialog will appear. Update the required fields:
    - **Key**: Unique identifier of the attribute (non-editable).
    - **Display Name**: Modify the human-readable name.
    - **Description**: Adjust the explanation if needed.
    - **Data Type**: Change the data type (*String, Int, Boolean, Decimal, Date, Json, Selection*).
      Figure xx – Update Attribute Definition – Select Data Type
    - **Allowed Scope Types**: Add or remove scope types (*User, Profile, Global, Course, Session*).
      Figure xx – Update Attribute Definition – Allowed Scope Types
    - **Unit (Optional)**: Update measurement unit.
    - **Default Value**: Modify default value if applicable.
    - **Is Deletable**: Toggle to allow or restrict deletion.
- **Step 3**: After making changes, click **Update Definition**.
- **Step 4**: To cancel, click **Cancel,** and no changes will be saved.

System Response:

- If successful: A confirmation notification appears — *"Attribute Definition updated successfully."* The list refreshes with updated values.

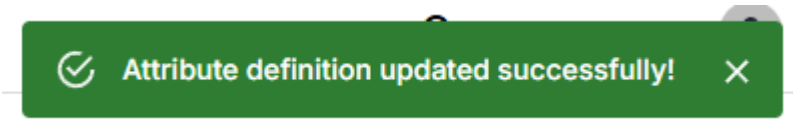
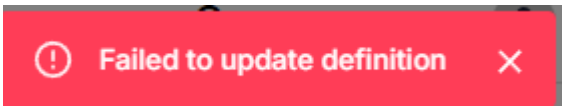


Figure xx – Update Attribute Definition success message

- If unsuccessful: An error message appears (e.g., *“Invalid data type”* or *“Missing required field”*).



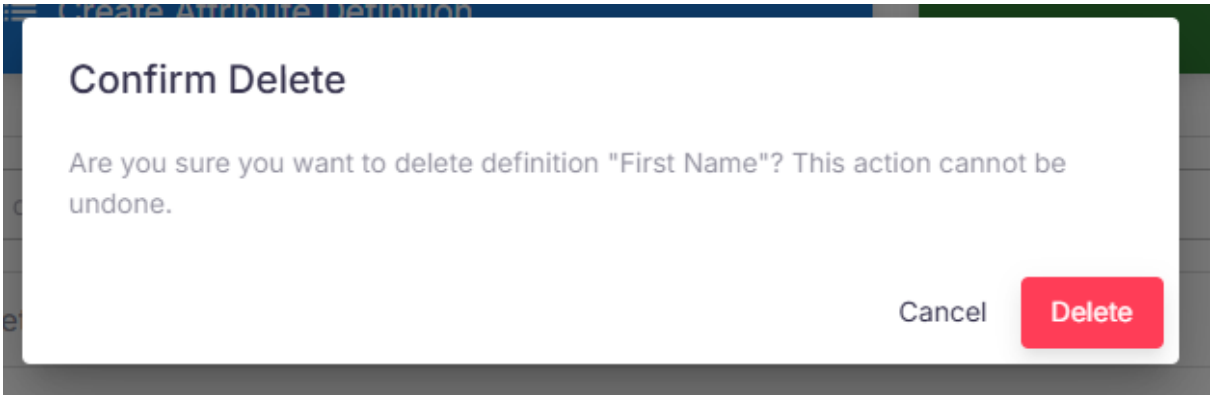
Delete Attribute Definition

Description: This function allows administrators to permanently delete an Attribute Definition from the system. Deletion is only available if the attribute is marked as **Deletable**.

Instruction:

- **Step 1**: On the *Configuration Management* page, locate the attribute definition you want to delete. Click the **Delete** icon (🗑️) in the action column.
- **Step 2**: A confirmation dialog will appear:

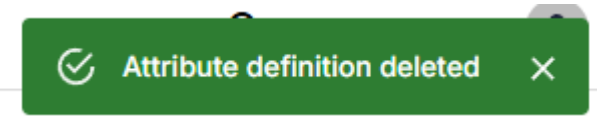
  "Are you sure you want to delete the attribute definition <AttributeName>? This action cannot be undone."



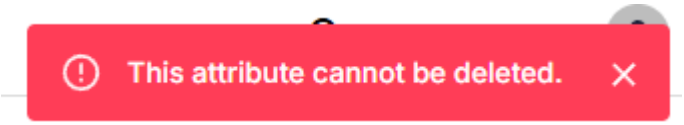
- **Step 3**: Choose one of the following:
  - Click **Delete** to confirm removal.
  - Click **Cancel** to abort the operation.

  System Response:

  - If successful: A green notification appears — *“Attribute Definition deleted successfully.”* The attribute is removed from the list.



  - If unsuccessful (e.g., attribute is not deletable or linked to critical data): An error message appears — *“This attribute cannot be deleted.”*



3.1.11.2 Manage Settings

View Settings

Description: This function allows administrators to view all system-level configuration settings (Attribute Definitions) in a structured table. Each setting includes details such as key, display name, data type, scope, value, and last update time.

Instruction:

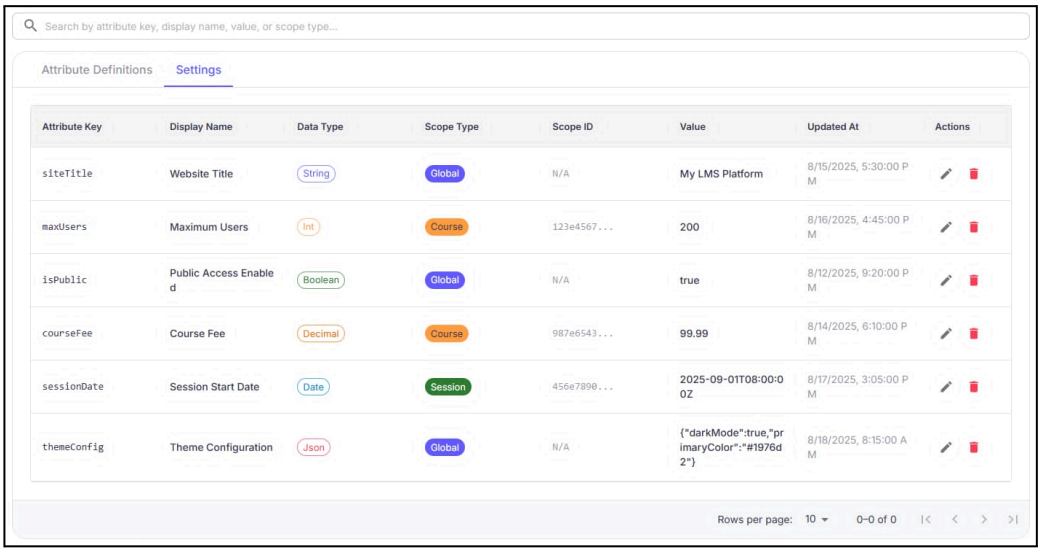


Figure xx - View settings screen

- **Step 1**: On the **Configuration Management** → **Settings**
- **Step 2**: The system displays a table containing all attribute definitions with the following columns:
    - **Attribute Key** – Unique identifier for the attribute (e.g., siteTitle, maxUsers).
    - **Display Name** – Human-readable name of the attribute.
    - **Data Type** – Type of data (String, Int, Boolean, Decimal, Date, JSON).
    - **Scope Type & Scope ID** – Defines where this attribute applies (Global, Course, Session, User).
    - **Value** – The current configured value.
    - **Updated At** – Last modification timestamp.
    - **Actions** – Edit (✏️) or Delete (🗑️).
- **Step 3**: Use the **search bar** at the top to filter settings by key, display name, value, or scope type.

  System Response:

    - The system dynamically updates the table based on search filters.
    - Pagination at the bottom allows navigating through large datasets (e.g., Rows per page: 10, 20, 50).

Create Setting

Description: This function allows users to create a new Setting by selecting an existing Attribute Definition, assigning a Scope Type (Global, Course, or Session), optionally providing a Scope ID, and specifying a Value.

Instruction:

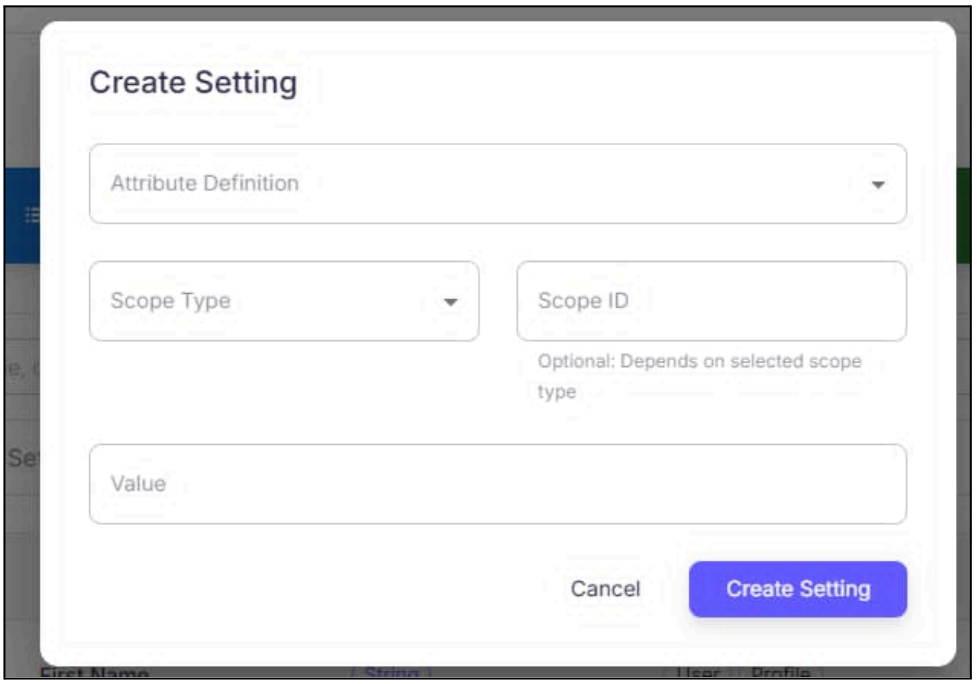
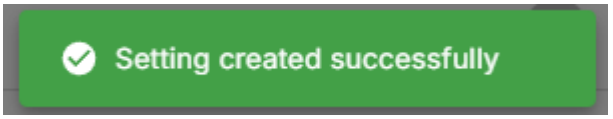


Figure xx - Create setting dialog

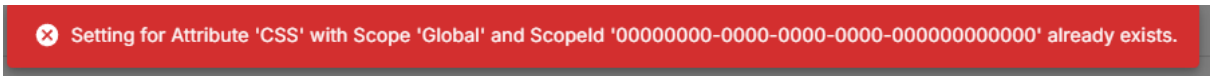
- **Step 1**: On the *Settings* page, click the **+ Create Setting** button.
- **Step 2**: In the **Create Setting** dialog, select the **Attribute Definition** from the dropdown list.
- **Step 3**: Choose a **Scope Type** (Global, Course, or Session).
    - If Course or Session is selected, enter the corresponding **Scope ID**.
- **Step 4**: Enter the **Value** for the Setting.
- **Step 5**: Click **Create Setting** to save.

System Response:

- If successful: The new Setting will appear in the Settings table with its Attribute Key, Scope, and Value.



- If unsuccessful: An error message is displayed (e.g., "Invalid value format").



Edit Setting

Description: This function allows users to edit an existing Setting. Users can change the Scope Type, Scope ID, and Value, while the Attribute Definition is fixed. The form will be pre-filled with the current Setting details.

Instruction:

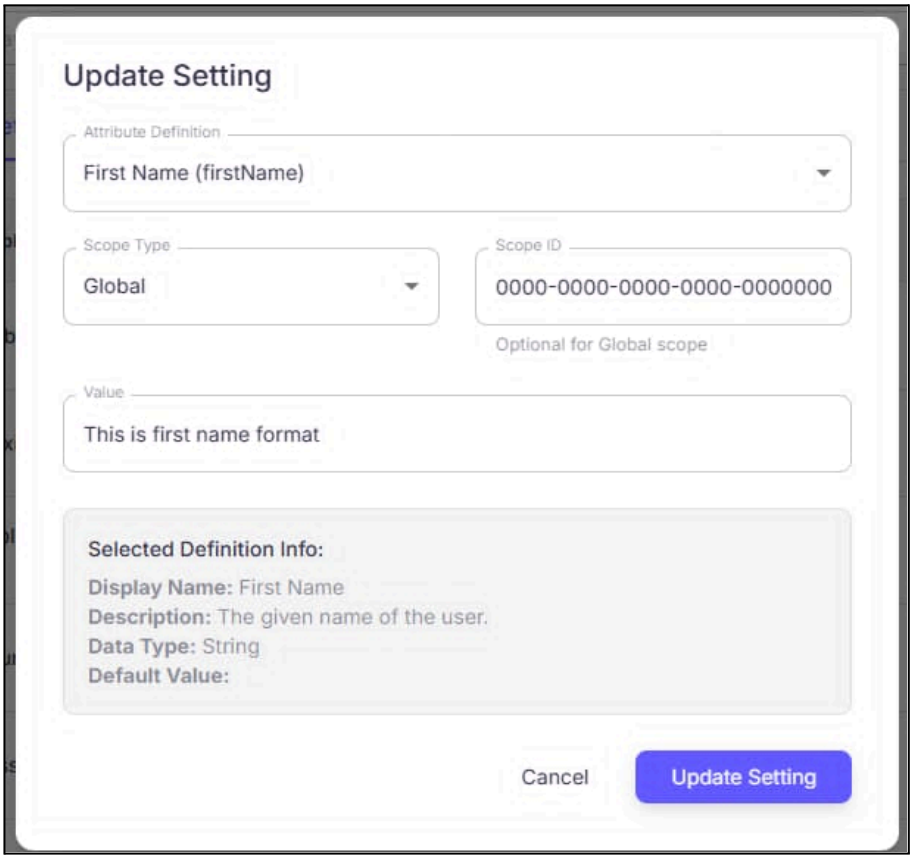
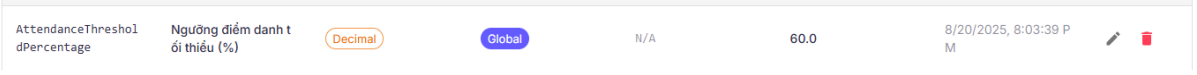


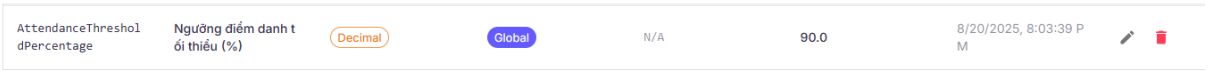
Figure xx - Edit setting dialog

- **Step 1**: On the *Settings* page, locate the Setting you want to edit and click the **Edit (✎)** button.
- **Step 2**: In the **Update Setting** dialog, review the pre-filled details:
  - **Attribute Definition** (fixed, not editable)
  - **Scope Type** and **Scope ID** (editable)
  - **Value** (editable)
- **Step 3**: Modify the **Scope Type**, **Scope ID**, or **Value** as needed.
- **Step 4**: Review the **Selected Definition Info** section, which displays reference details such as Display Name, Description, Data Type, and Default Value.
- **Step 5**: Click **Update Setting** to save changes.

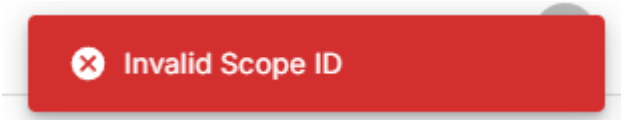
System Response:

- If successful: The Setting in the list is updated with the new values.





- If unsuccessful: An error message is displayed (e.g., "Invalid Scope ID").



Delete Setting

Description: This function allows users to delete an existing Setting from the system. A confirmation dialog will appear to ensure the user wants to proceed with the deletion.

Instruction:

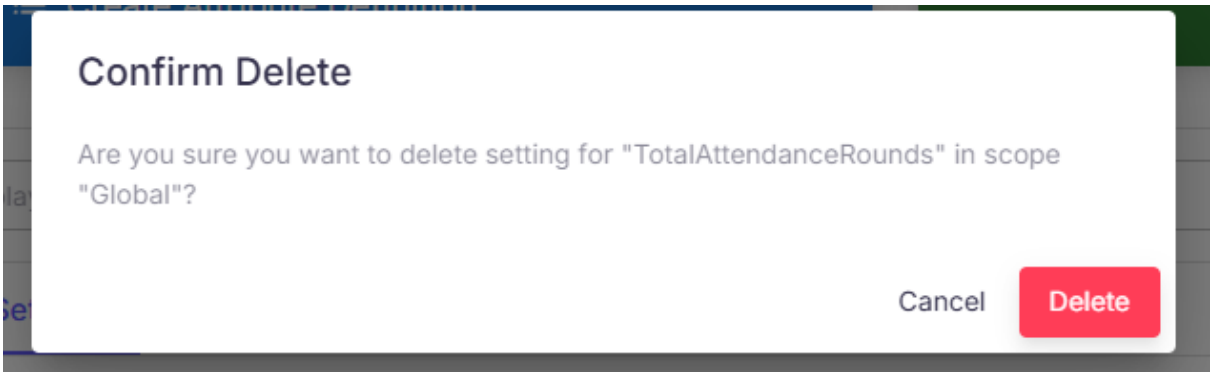
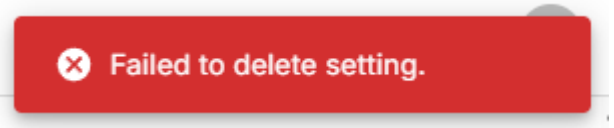


Figure xx - Delete setting confirmation dialog

- **Step 1**: On the *Settings* page, locate the Setting you want to remove and click the **Delete (🗑️)** button.
- **Step 2**: A **Confirm Delete** dialog will appear, showing the Setting's name and scope.
- **Step 3**: Click **Xóa (Delete)** to confirm, or **Hủy (Cancel)** to abort the action.





3.2 Application - Lecture

3.2.1 Overview

The **Zentry Mobile Applications** are developed separately for **Lecturers** and **Students**, supporting real-time attendance tracking and class participation.

- **Lecturer App**
    - View teaching schedule in list or calendar format.

    - Start and manage class sessions, including BLE-based check-in and FaceID verification.
    - Review attendance history and generate teaching reports.
    - Manage notifications and register devices.
- **Student App**
    - View personal class schedules and attendance history.
    - Perform attendance check-in via BLE scanning and FaceID confirmation.
    - Receive real-time notifications about class updates.
    - Manage personal profile, device registration, and FaceID enrollment.

Both apps ensure a seamless attendance process by integrating **BLE scanning** for proximity detection and **face recognition** for identity verification, thus reducing manual errors and enhancing security.

3.2.1 Authentication

3.2.1.1 Sign in

Description: The Sign In function allows lecturers and students to securely access the application using their registered email and password. This authentication step ensures that only authorized users can use the system's features, such as attendance management, schedule viewing, and reporting.

Instruction:

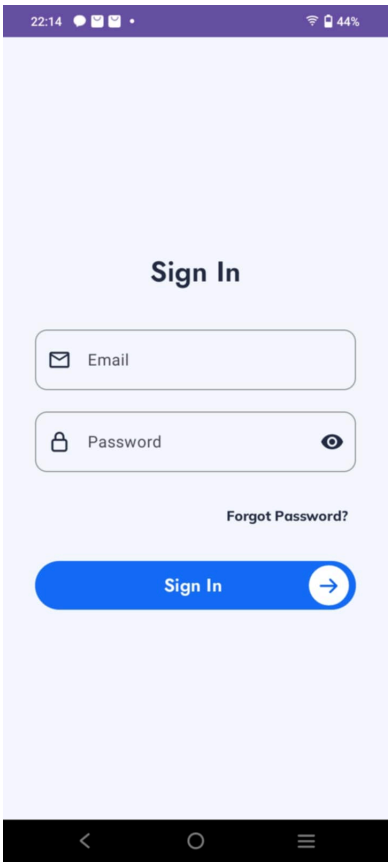


Figure xx - Sign in screen

Step 1: In the Email Address field, enter the email address registered with the system.

Step 2: In the Password field, enter the account password.

Step 3 (Optional): Select the Remember me option if you want the system to save your login information for the next session.

Step 4: Tap the Sign In button to submit your credentials.

- If the credentials are **valid**, the system will navigate to the Home Page.
- If the credentials are **invalid**, the system will display an error message and prompt you to re-enter.

3.2.1.2 Log out

Description: The **Sign Out** function allows lecturers to securely log out of the application. This ensures that no unauthorized access occurs on the same device once the lecturer has finished using the system.

Instruction:

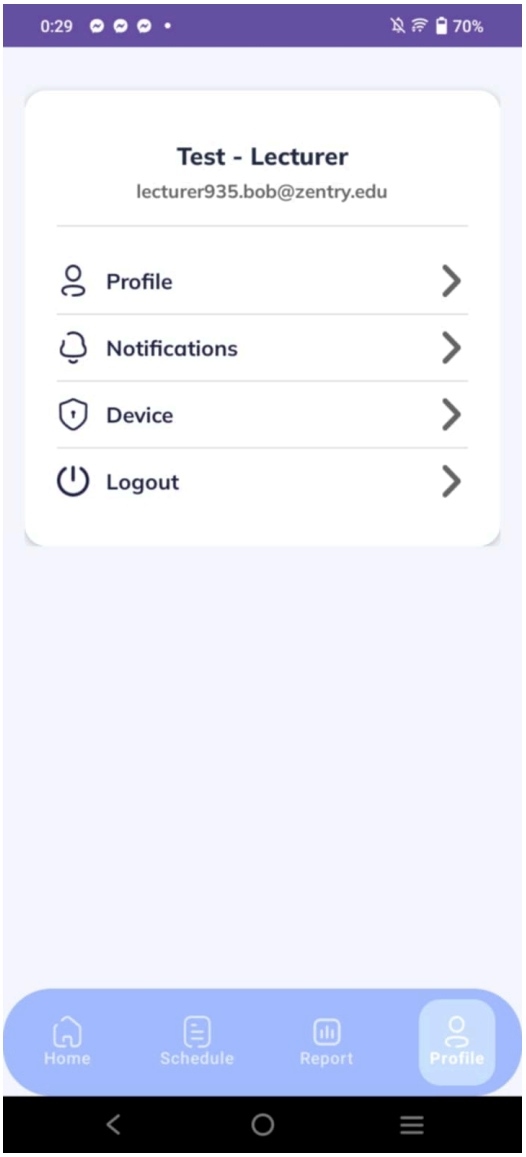


Figure xx - Logout screen

Step 1: From the bottom navigation bar, tap the **Profile** icon to open the Profile page.

Step 2: On the Profile page, locate the **Logout** option at the bottom of the menu.

Step 3: Tap **Logout**.

Step 4: The system will log out the current account and redirect to the **Sign In** page.

3.2.2 HomePage

3.2.2.1 View HomePage

Description: The Home Page is the default landing screen after successful login. It provides lecturers with a personalized dashboard including greeting messages, alerts, upcoming sessions, quick actions, and a weekly summary of teaching and attendance performance.

Instruction:

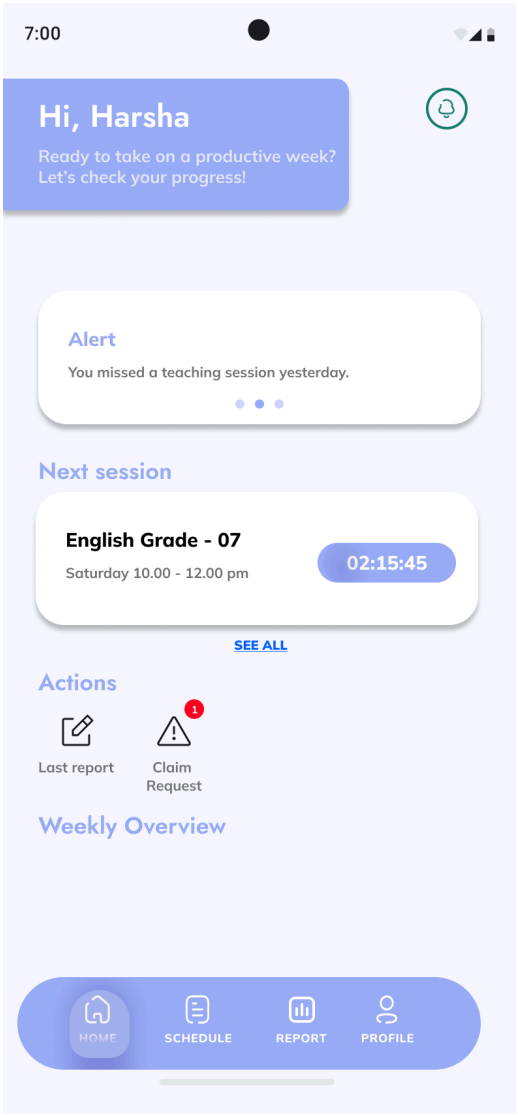


Figure xx - Home screen

Step 1: After signing in, the system navigates to the Home Page.

Step 2: View the personalized greeting message (e.g., “Hi, Harsha”).

Step 3: Review the Alert section for notifications such as missed sessions.

Step 4: Check the Next Session card, which displays:

- Subject and grade (e.g., English Grade – 07).
- Scheduled time (e.g., Saturday 10:00 – 12:00).
- A countdown timer until the session starts.

Step 5: Use the Actions section to quickly access:

- Last Report: review the most recent teaching/attendance report.
- Claim Request: submit or check responses to attendance-related claims.

Step 6: Scroll down to the **Weekly Overview**:

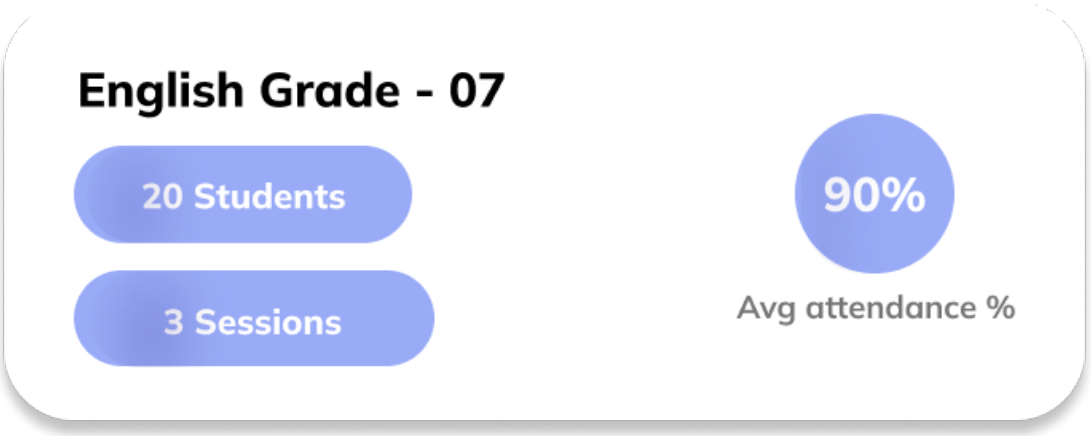


Figure xx - Weekly overview section

- In the initial state, this section may be collapsed or show only a label.
- In the detailed state, it displays per-class statistics including:
    - Number of students,
    - Number of sessions,
    - Average attendance percentage.

Step 7: Navigate between modules using the bottom navigation bar:

Home, Schedule, Report, Profile.

3.2.3 Schedule Management

3.2.3.1 View teaching schedule

Description: The Schedule page enables lecturers to view and manage their teaching timetable. Each scheduled class is displayed as a card containing subject, grade, classroom, time, and current status. Sessions are categorized as Start now, Upcoming, Missed, or View depending on their timing and progress.

Instruction:

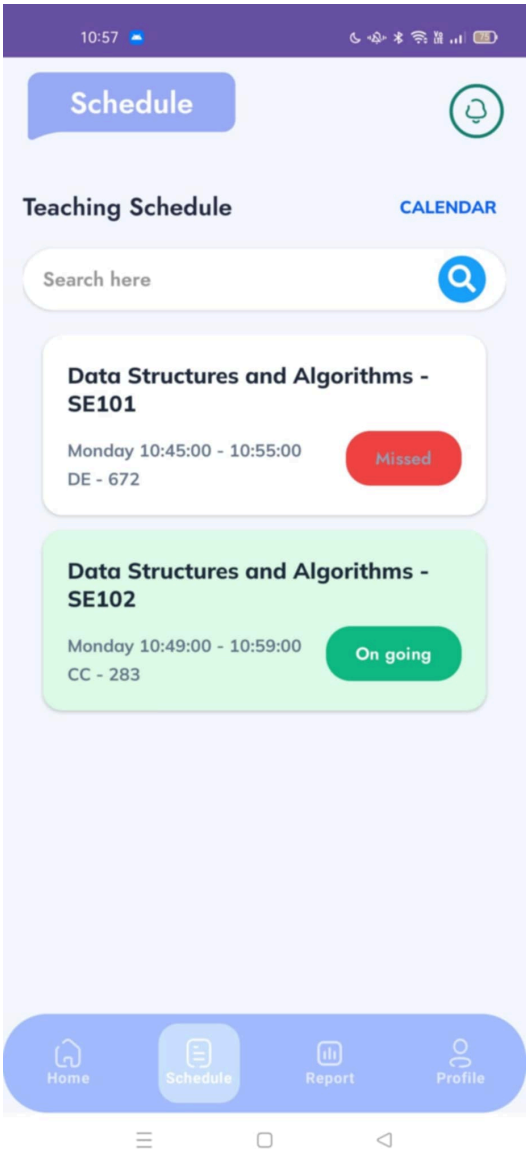


Figure xx - View teaching schedule screen

Step 1: From the bottom navigation bar, tap the Schedule icon to open the Teaching Schedule page.

Step 2: Review the list of scheduled sessions:

- Each card shows subject, grade, scheduled time, and assigned classroom.
- The right-side button indicates the session status:
    - **Start now** – the class is ready to begin.
    - **Upcoming** – the class is scheduled later.
    - **Missed** – the class has already passed without attendance.
    - **View** – the class is completed and details can be reviewed.

Step 3: Tap **Calendar** in the top-right corner to switch to calendar view.

Step 4: Select a session card to proceed with class details or attendance actions.

3.2.3.2 View Calendar

Description: The View Calendar function allows lecturers to visualize their teaching sessions in a calendar format. By selecting a specific date, lecturers can quickly see all scheduled classes for that day with corresponding time slots and classroom information.

Instruction:

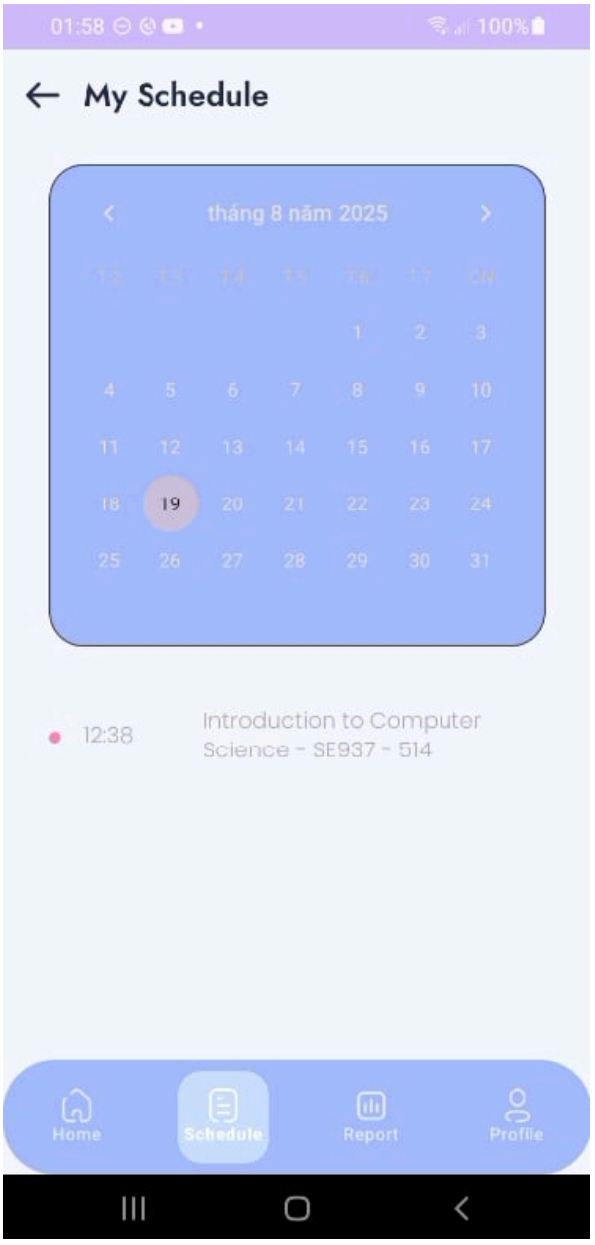


Figure xx - View calendar screen

Step 1: On the Schedule page, tap Calendar in the top-right corner.

Step 2: The system displays a monthly calendar view.

Step 3: Navigate between months using the left or right arrow icons.

Step 4: Select a specific date to view sessions scheduled for that day.

Step 5: The system displays the timeline of classes for the selected date, including:

- Subject and grade,
- Classroom (e.g., BETA – BE301),
- Start time (e.g., 07:30, 10:00).

Step 6: Tap on a session card in the timeline to access detailed information or start attendance actions.

Step 7: Tap the back arrow on the top-left corner to return to the list view of the Teaching Schedule.

3.2.3.3 View Class Session

Description: The View Class Session function provides lecturers with detailed information for a selected class. It allows reviewing session history, monitoring attendance in real-time, requesting FaceID verification, and ending the session once completed.

Instruction:

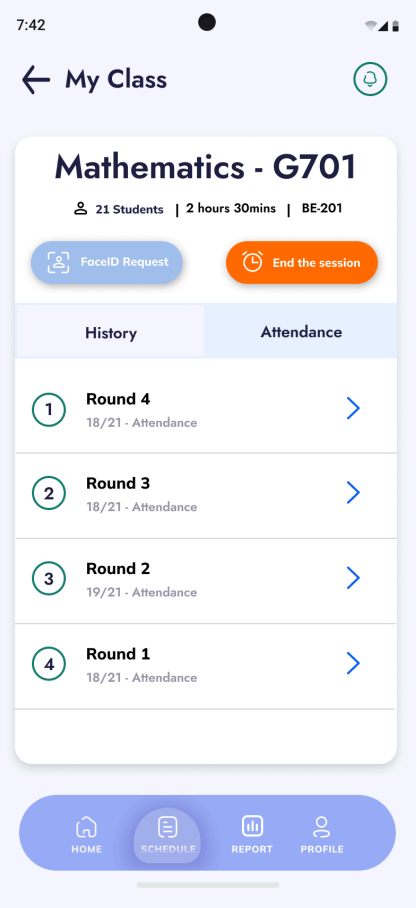


Figure xx - View class section history screen

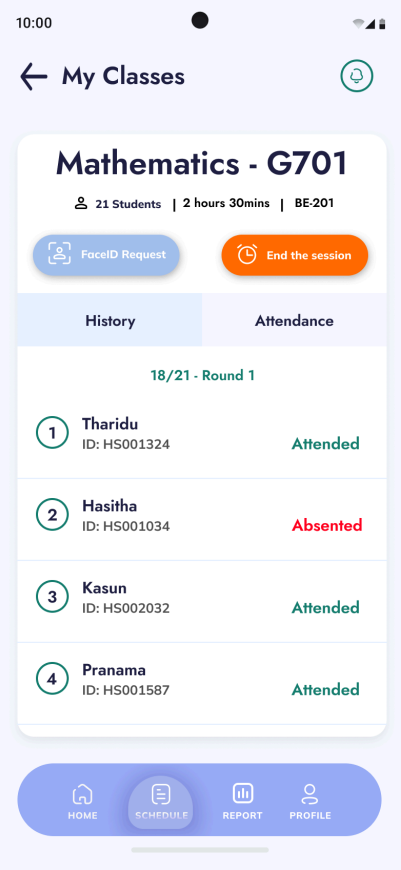


Figure xx - View class section attendance screen - round detail

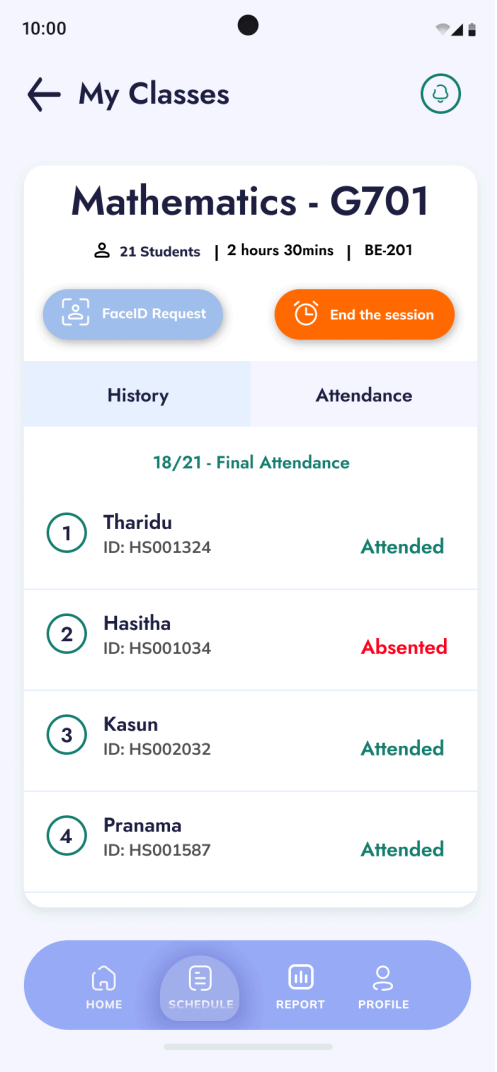


Figure xx - View class section attendance screen - final attendance

Step 1: From the Teaching Schedule list or calendar view, tap a session card (e.g., Mathematics – G701).

Step 2: The system opens the Class Session Detail page, displaying:

- Subject and grade,
- Number of students,
- Duration,
- Classroom information.

Step 3: Use the available actions:

- FaceID Request: send a face recognition request for attendance validation.
- End the session: close the current session and finalize attendance.

Step 4: Navigate between two tabs:

- History: view attendance records from previous rounds.
- Attendance: view real-time or final attendance results for the current session.

Step 5: In the Attendance tab, review the student list:

- Student name and ID,
- Attendance status (Attended in green, Absented in red).

Step 6: After ending the session, the final attendance record is saved and displayed.

Step 7: Tap the back arrow to return to the Teaching Schedule page.

3.2.3.4 FaceId request

Description: The FaceID Request function allows lecturers to initiate a face recognition attendance check during a session. The lecturer can configure a time limit for the request, and the system will record student responses within that period.

Instruction:

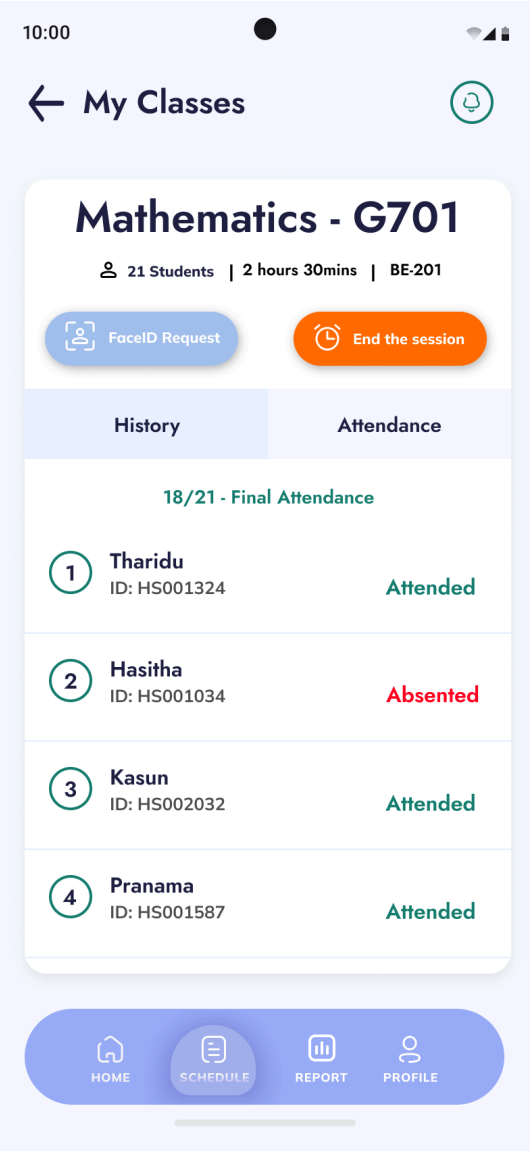


Figure xx - Face ID request screen

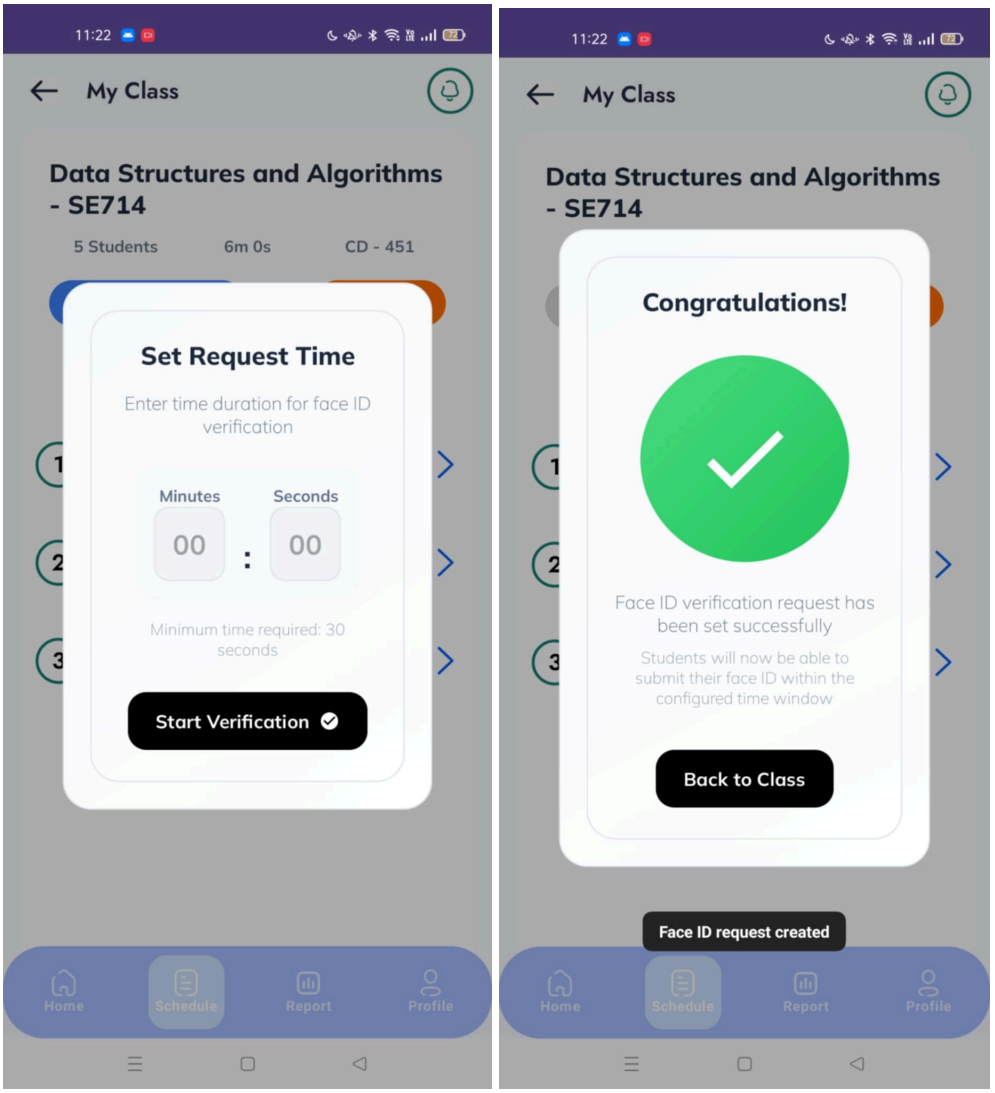


Figure xx - Create Face ID request screen

Step 1: On the **Class Session Detail** page, tap the **FaceID Request** button.

Step 2: The system displays a dialog titled *Set your request time*.

Step 3: Enter the desired time limit (e.g., *01:30 minutes*).

Step 4: Tap **Start Verification** to confirm.

Step 5: The system processes the request and adds it to the attendance history.

Step 6: Once successful, a confirmation screen appears:

- *"Congratulations – Your FaceID request has been set successfully"*

Step 7: Tap **Back to Class** to return to the Class Session Detail page.

3.2.3.5 End the session

Description: The **End the Session** function allows lecturers to manually stop an ongoing attendance session before the pre-set time limit. Once the lecturer taps this button, the system will immediately close the attendance window and finalize the attendance list.

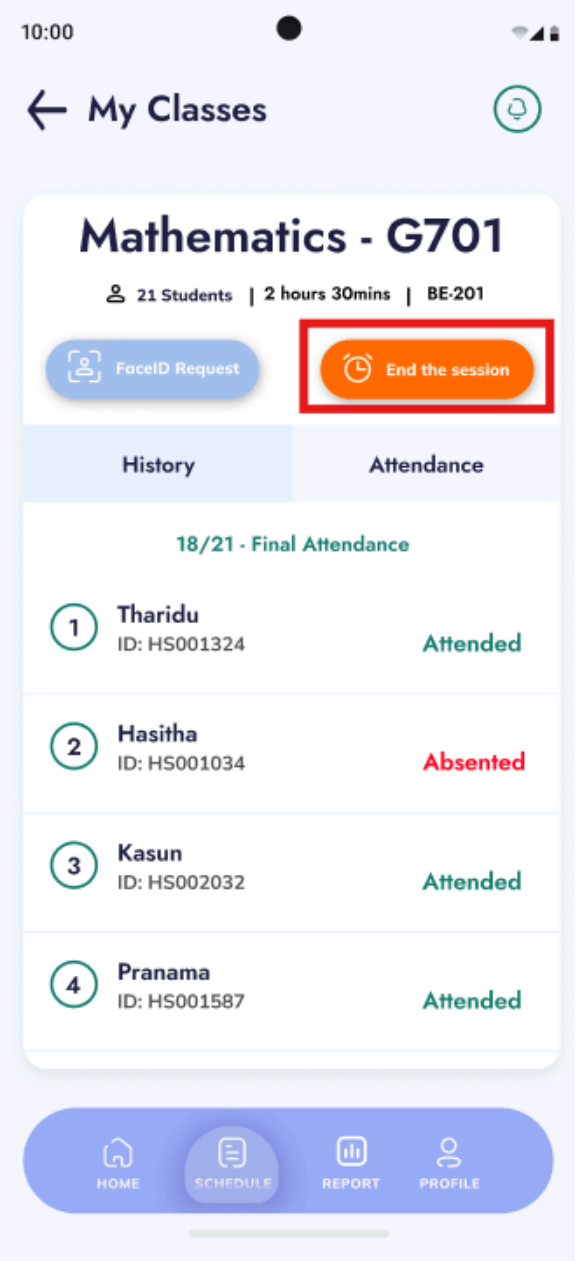


Figure xx - End session button

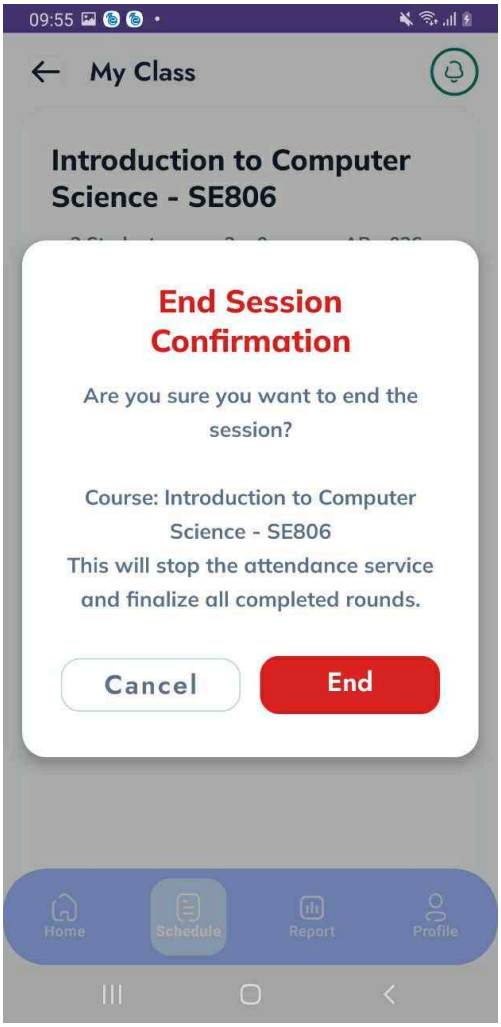


Figure xx - End session confirmation dialog

Step 1: On the **Class Session Detail** page, tap the **End the Session** button.

Step 2: The system displays a confirmation dialog titled **End Session Confirmation** with the course details.

Step 3: Tap **End** to confirm and stop the session (or tap **Cancel** to go back without ending).

Step 4: Once confirmed, the system stops the attendance process and finalizes the list of students as *Attended* or *Absent*.

Step 5: The lecturer is redirected to the **Attendance tab**, where the finalized attendance status of students is displayed.

3.2.3.6 History

Description: The **History** function allows lecturers to view a list of past or completed class sessions. Each record shows the course name, session time, and location. Lecturers can quickly check when a session was conducted and access detailed attendance information if needed.

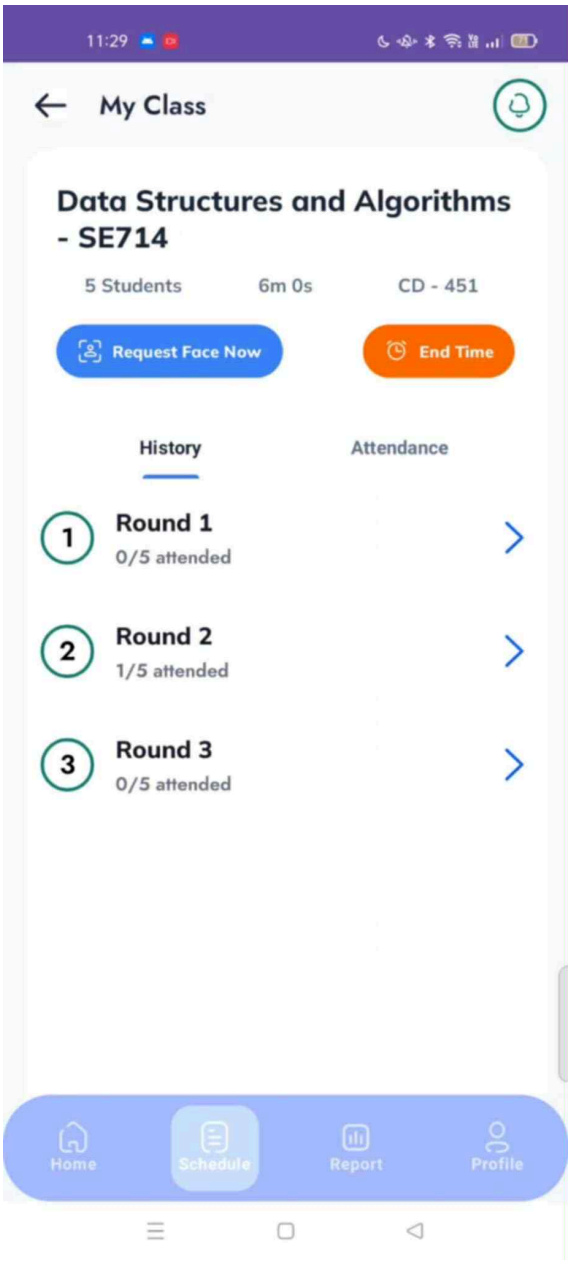


Figure xx - Class section history screen

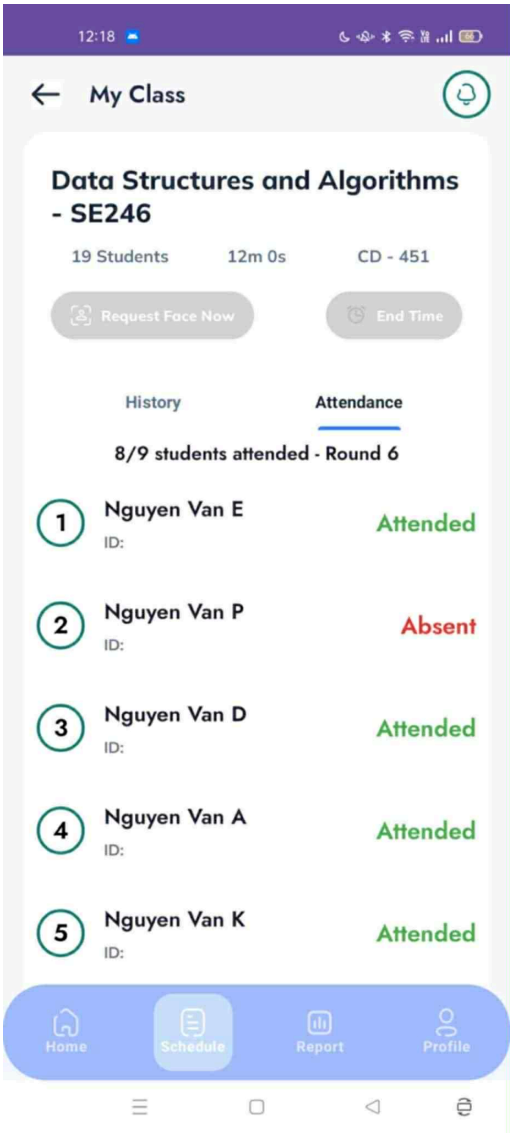


Figure xx - Class section attendance round detail screen

Step 1: On the **Class Session Detail** page, tap the **History** tab.

Step 2: The system redirects to the **History** screen.

Step 3: The system response attendance rounds of class

Step 4: Select a round from the list to view more details (attendance record, student status, etc.).

3.2.4 Report Management

3.2.4.1 View My report

The **Lecturer Report** function allows lecturers to view attendance reports for each course they are assigned. It provides an overview of the total number of students, total sessions, and detailed attendance records for each session. This feature helps lecturers track student participation throughout the semester.



Figure xx - My report screen

Step 1: From the bottom navigation bar, select **Report**.

Step 2: The system displays the **My Report** page, listing courses assigned to the lecturer.

Step 3: Select a course to view its attendance report.

Step 4: The course report displays:

- **Course name and code** (e.g., *Introduction to Computer Science – SE937*)
- **Number of students** enrolled
- **Total sessions** (e.g., *1/14 Sessions completed*)
- **Session List** – Each session entry includes:

- Session number (e.g., *Session 1*)
- Attendance summary (e.g., *0/7 – Attendance*)
- Session date (e.g., *19/08/2025*)

Step 5: Tap on a session to view detailed attendance for that specific class.

3.2.4.2 View report detail

The **Session Detail** function allows lecturers to view detailed attendance records for a specific class session. The screen displays the list of students, their attendance status (Attended/Absent), and student information (name, ID). Lecturers can also edit attendance within 24 hours of the session if necessary.

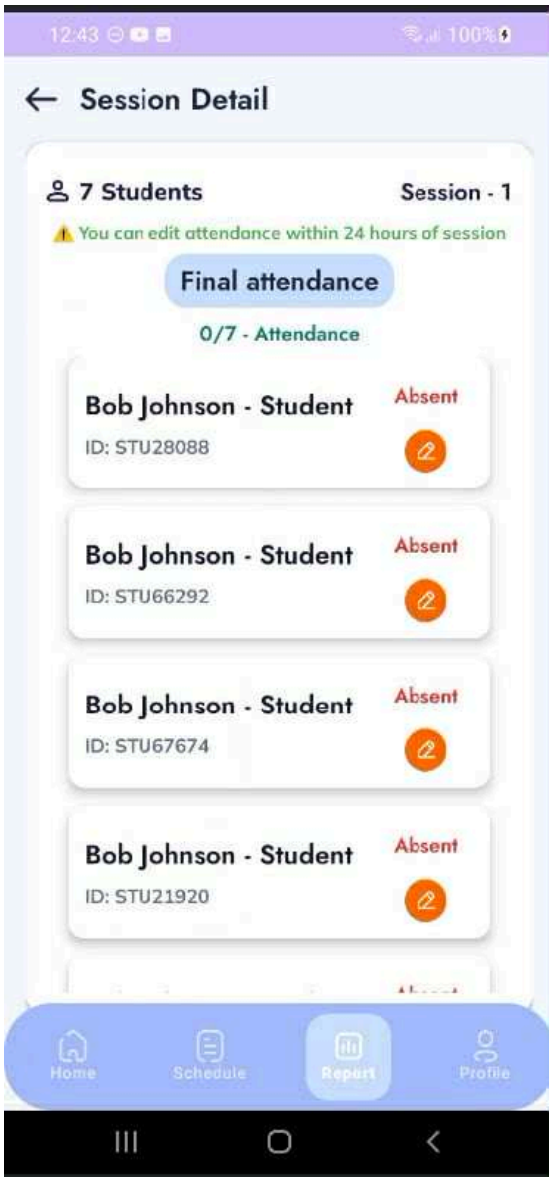


Figure xx - Session report detail screen

Step 1: On the **My Report** page, select a course.

Step 2: From the **Session List**, tap on a session (e.g., *Session – 1*).

Step 3: The system displays the **Session Detail** screen, including:

- **Total number of students** in the session.
- **Session number** and status.
- **Edit notice** (e.g., *You can edit attendance within 24 hours of session*).

- **Final Attendance summary** (e.g., *0/7 Attendance*).

Step 4: Below, the list of students is shown with:

- Student full name.
- Student ID.
- Attendance status (e.g., *Attended* or *Absent*).
- Edit option (pencil icon) to modify the status if still within the 24-hour edit window.

Step 5: Review the attendance records. If adjustments are needed, tap the **Edit** icon next to the student's status and update accordingly.

Step 6: Once edits are completed, the updated status is saved and reflected in the final report.

3.2.4.3 Update attendance

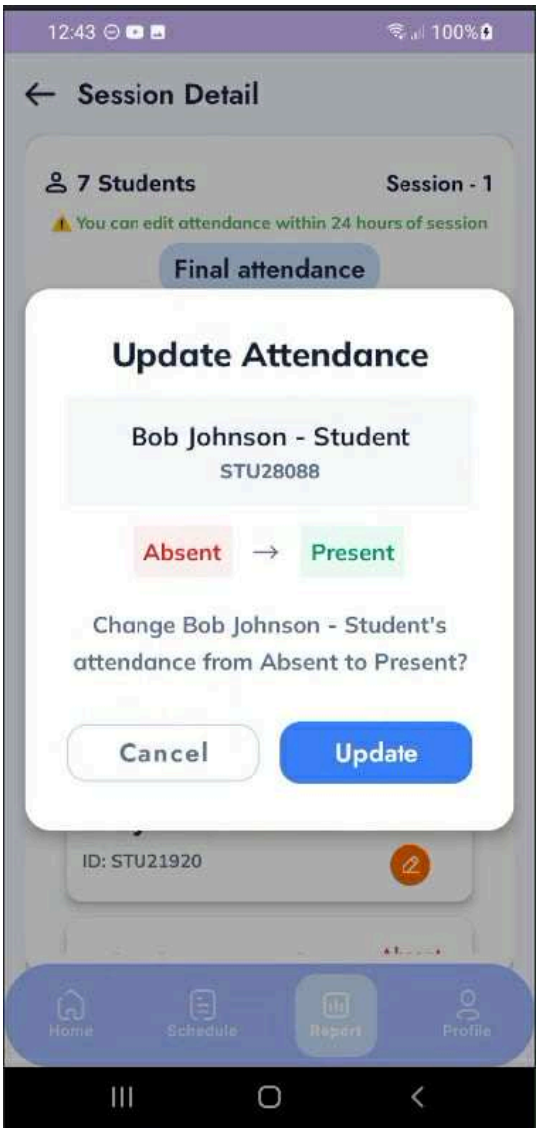


Figure xx - Update attendance confirmation dialog

The **Update Attendance** function allows lecturers to modify a student's attendance status within 24 hours after a session. This feature is useful for correcting errors or updating attendance records if a student was mistakenly marked *Absent* or *Present*.

Step 1: On the **Session Detail** screen, locate the student whose attendance needs updating.

Step 2: Tap the **Edit (pencil)** icon next to the student's status.

Step 3: The system displays the **Update Attendance** dialog, showing the student's name, ID, and current status.

Step 4: Select the new attendance status (e.g., change from *Absent* → *Present*).

Step 5: Tap **Update** to confirm the change (or **Cancel** to discard).

Step 6: The system saves the update and refreshes the student's attendance status in the session record.

Step 7: The updated status is reflected in both the **Session Detail** and the overall course report.

3.2.5 Profile Management

3.2.5.1 View lecturer profile

Description: The **View Lecturer Profile** function allows lecturers to access and review their personal account information stored in the system. This includes name, email, phone number, role, account status, and the date of registration.

The profile is **view-only** — lecturers **cannot edit or update** any information in this section.

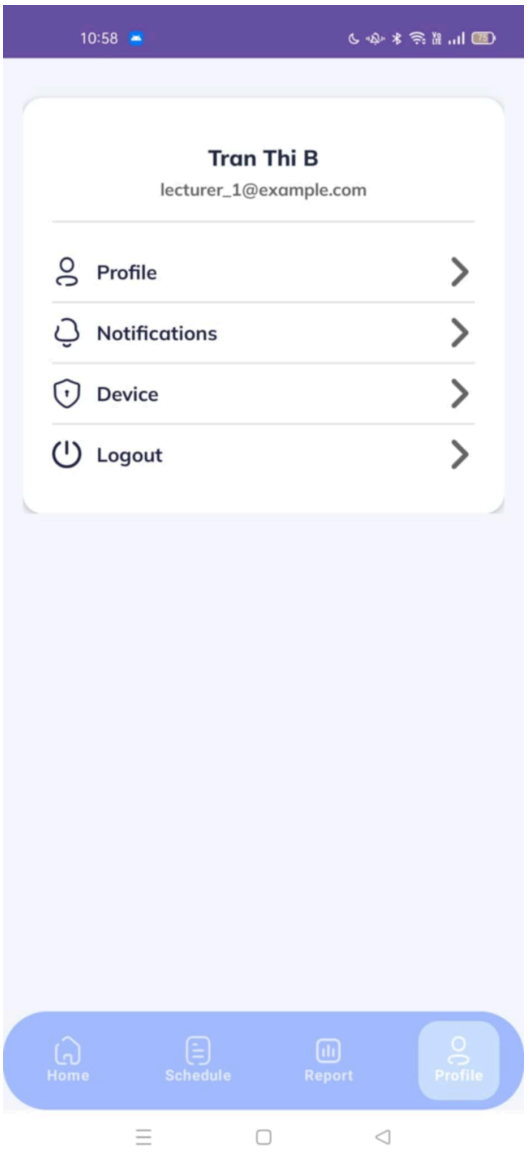


Figure xx - View lecturer profile button

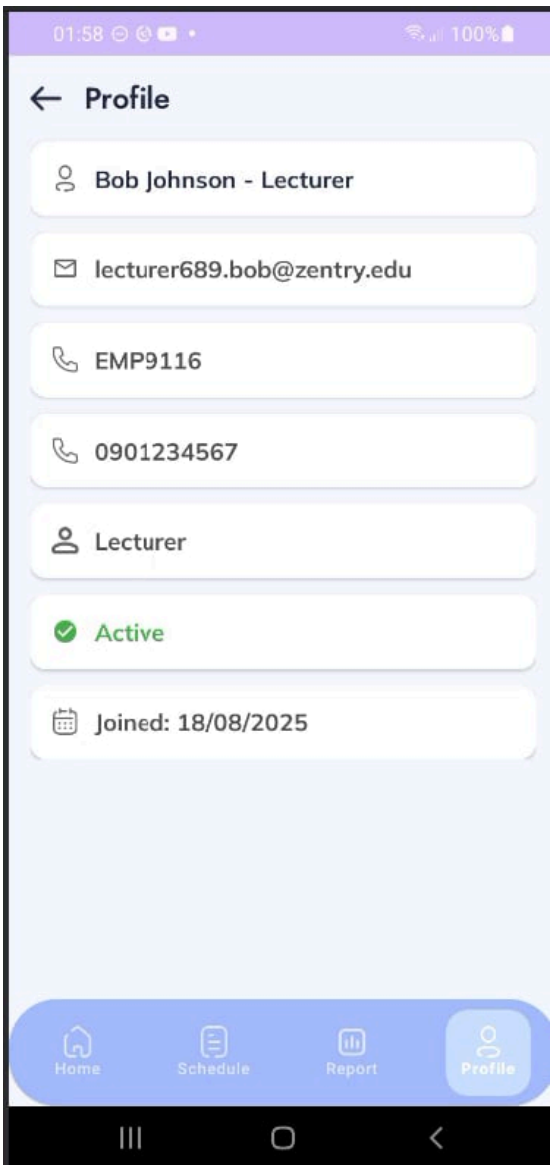


Figure xx - Lecture profile screen

Step 1: On the navigation bar, select **Profile**.

Step 2: The system displays the **Profile** menu, showing available options:

- Profile
- Notifications
- Device
- Logout

Step 3: Select **Profile** to view account details.

Step 4: The system displays **the Profile detail screen**, including:

- Lecturer's full name
- Email address
- Phone number
- Role (e.g., Lecturer)
- Code

- Account status (Active/Inactive)
- Joined date

Step 5: Review the information. This screen is read-only and does not allow modifications.

Step 6: To return, select the **Back** option at the top left.

3.2.5.2 Notification setting

Description: The **Notification Settings** function allows lecturers to manage how the application delivers notifications. Users can enable or disable sound, vibration, and general notifications according to their preference. These settings apply to all notifications generated by the system.

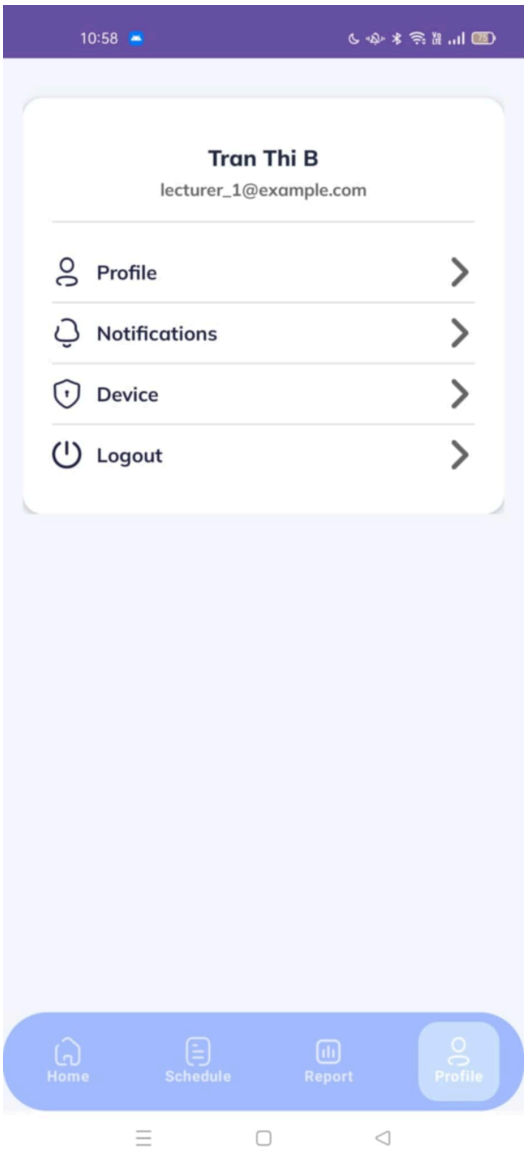


Figure xx - Notification setting button

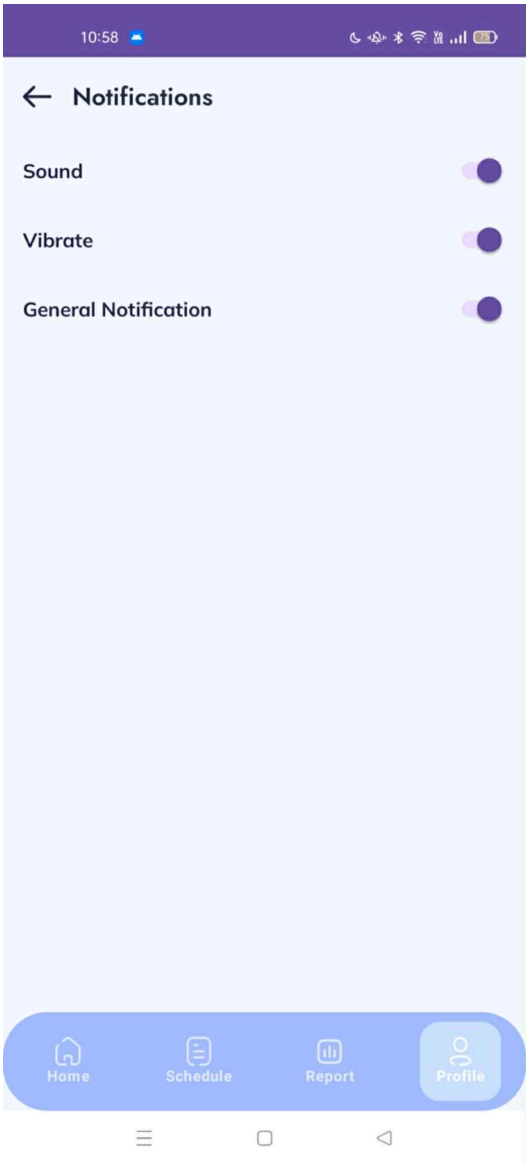


Figure xx - Notification setting screen

Step 1: From the Profile menu, select Notifications.

Step 2: The system displays the Notifications screen with the following options:

- Sound – Enable/disable notification sounds.
- Vibrate – Enable/disable vibration for notifications.
- General Notification – Turn all notifications on or off for the application.

Step 3: Use the toggle switches to turn each option on or off.

Step 4: Changes are saved automatically once a toggle is adjusted.

Step 5: To return, select the Back option at the top left.

3.2.5.3 Device Management

Description: The Device Management function allows lecturers to register their current mobile device with the system. Device registration is required in order to use the attendance features (such as starting and managing attendance sessions). Only registered devices can be used for attendance to ensure security and prevent unauthorized access.

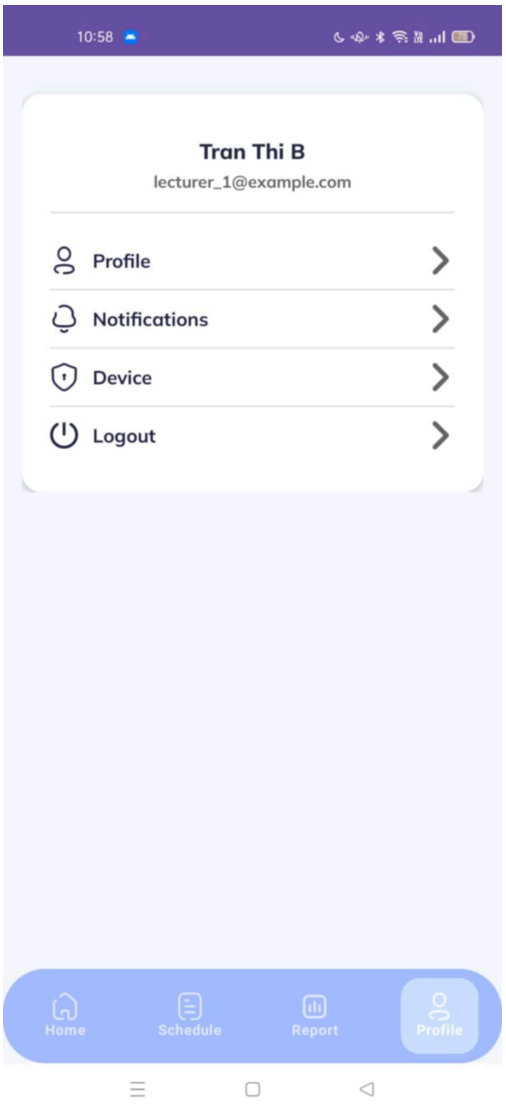


Figure xx - Device management button

Case 1 – No device registered

Step 1: From the Profile menu, select Device.

Step 2: The system displays a message:

“Device has not registered. Register now.”

Step 3: Tap Register Device.

Step 4: The system starts the registration process and shows the status:

“Registering…”

Step 5: Once successful, the device is linked to the lecturer’s account and becomes the active device for attendance.

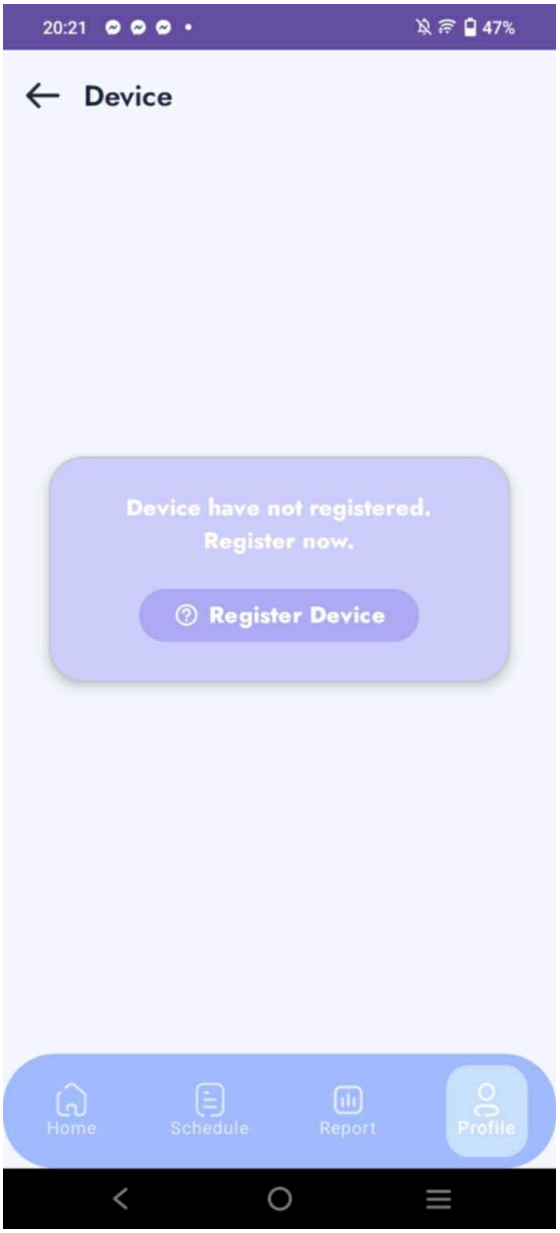


Figure xx - Device management screen - unregistered status

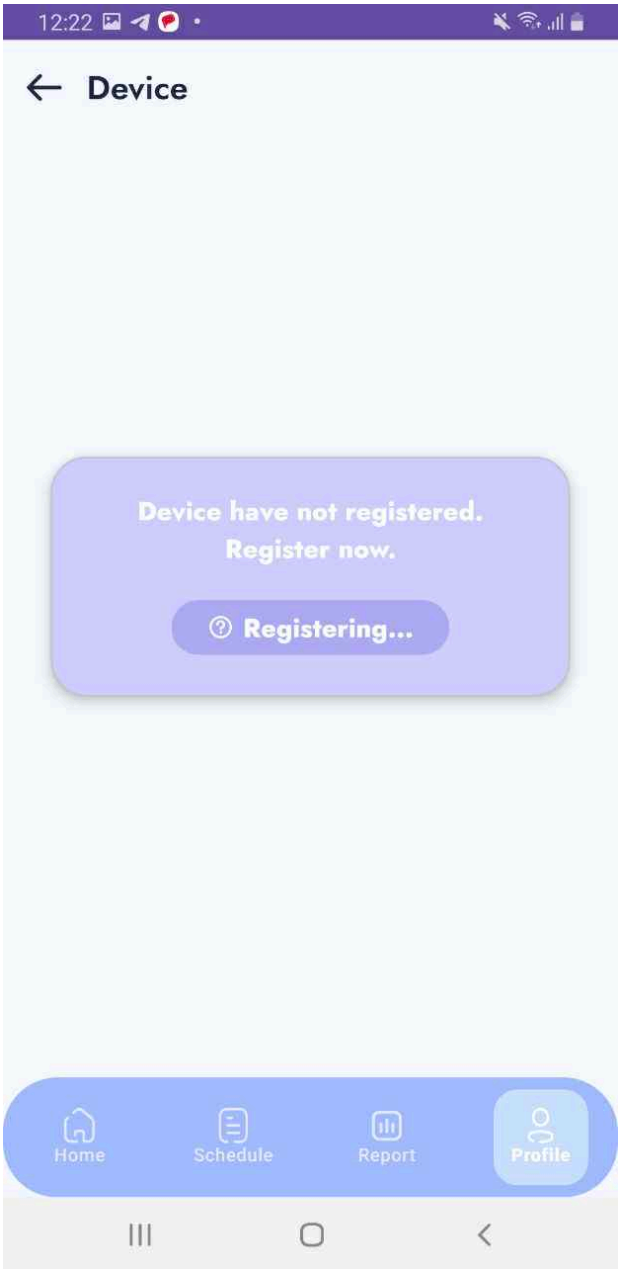


Figure xx - Device management screen - registering status

Case 2 – Device already registered

Step 1: From the Profile menu, select Device.

Step 2: The system displays the device information, including:

- Device Name
- Registered Date
- Model
- Manufacturer
- Platform (e.g., Android/iOS)
- OS Version
- App Version

Step 3: The status shows “Device Synced”, indicating the device is successfully linked.



Figure xx - Device detail screen

3.3 Application - Student

3.3.1 Authentication

3.3.1.1 Sign in

Description: The Sign In function allows lecturers and students to securely access the application using their registered email and password. This authentication step ensures that only authorized users can use the system's features, such as attendance management, schedule viewing, and reporting.

Instruction:

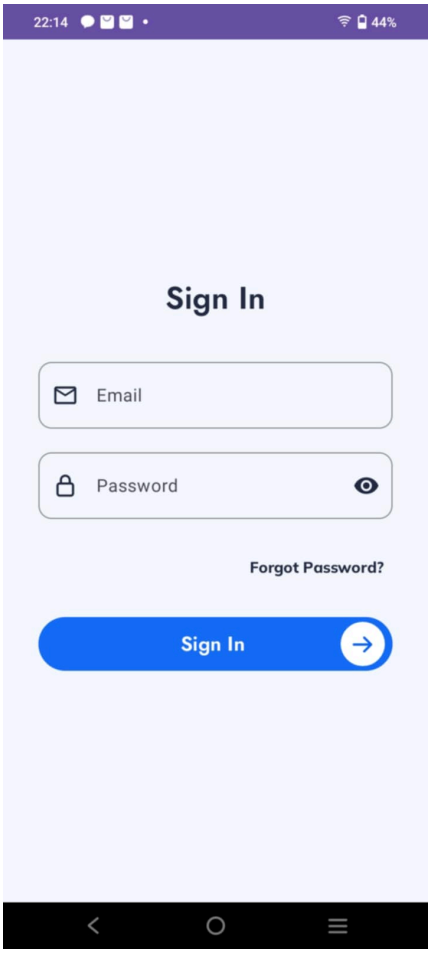


Figure xx - Sign in screen

Step 1: In the Email Address field, enter the email address registered with the system.

Step 2: In the Password field, enter the account password.

Step 3 (Optional): Select the Remember me option if you want the system to save your login information for the next session.

Step 4: Tap the Sign In button to submit your credentials.

- If the credentials are **valid**, the system will navigate to the Home Page.
- If the credentials are **invalid**, the system will display an error message and prompt you to re-enter.

3.2.1.2 Log out

Description: The **Sign Out** function allows lecturers to securely log out of the application. This ensures that no unauthorized access occurs on the same device once the lecturer has finished using the system.

Instruction:

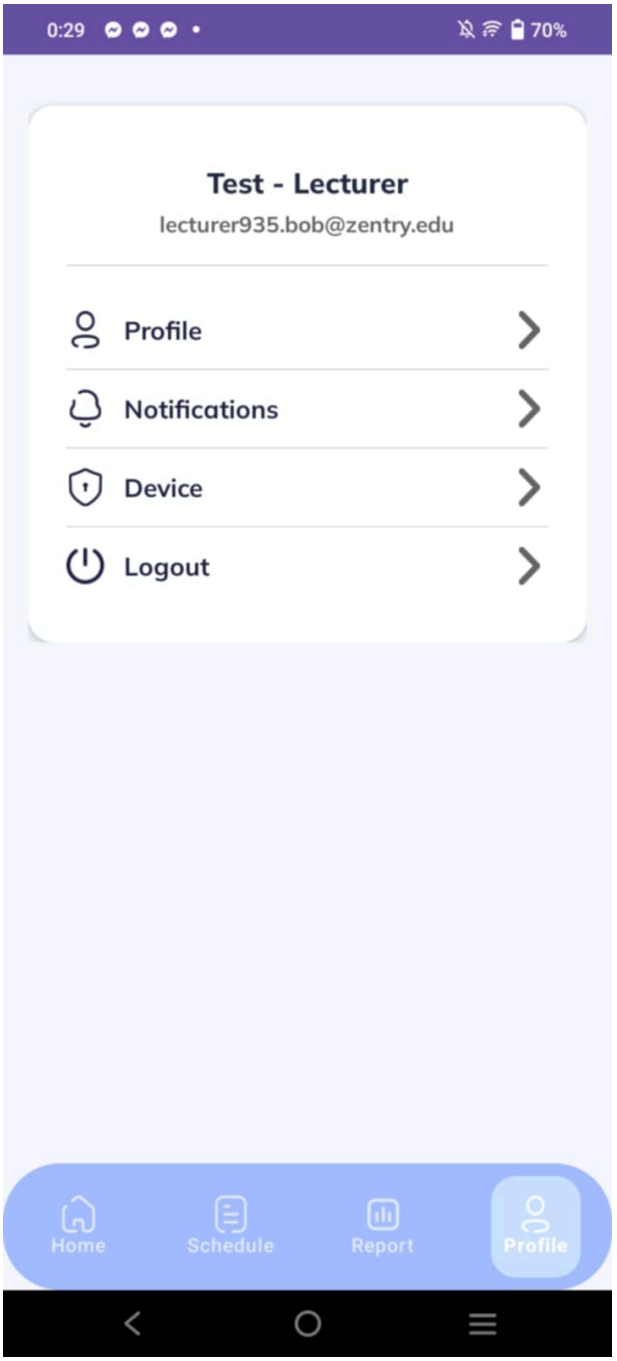


Figure xx - Logout button

Step 1: From the bottom navigation bar, tap the **Profile** icon to open the Profile page.

Step 2: On the Profile page, locate the **Logout** option at the bottom of the menu.

Step 3: Tap **Logout**.

Step 4: The system will log out the current account and redirect to the **Sign In** page.

3.2.2 HomePage

Description: The Student Home Page provides students with a personalized dashboard after signing in. It displays upcoming sessions, exam information, recent actions, and a weekly overview of courses. This page acts as the central hub for students to quickly access essential academic activities.

Instruction:

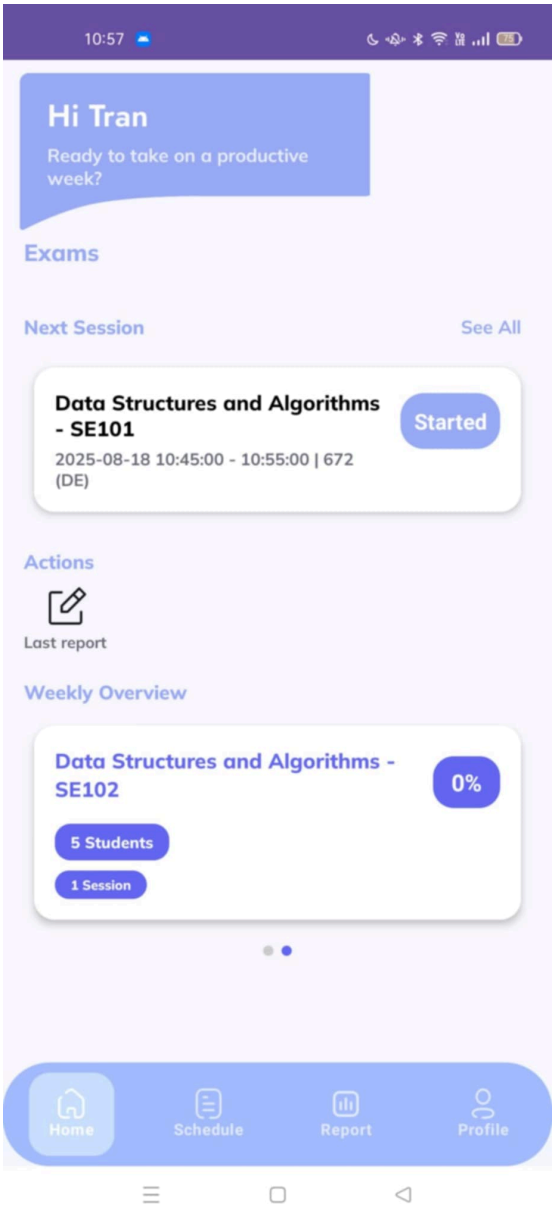


Figure xx - Home screen

Step 1: After signing in, the system automatically redirects students to the Home Page.

Step 2: The Home Page displays the following sections:

- Greeting Section – A personalized message for the student.
- Exams – Information about scheduled exams.

- Next Session – Displays the upcoming class with details such as course name, schedule, and status (e.g., Started).
- Actions – Provides quick access to the latest report or related functions.
- Weekly Overview – Shows an overview of student activities for the week, including number of students in a session, progress percentage, and session details.

Step 3: Students can tap on any section (e.g., Next Session or Weekly Overview) to view detailed information.

Step 4: Use the bottom navigation bar to move between Home, Schedule, Report, and Profile.

3.2.3 Schedule Management

3.2.3.1 View schedule

Description: The Student Schedule function allows students to view their list of upcoming and past classes. It provides detailed information about each course session, including course name, schedule time, and classroom location. Students can also join an active class session directly from this screen.

Instruction:

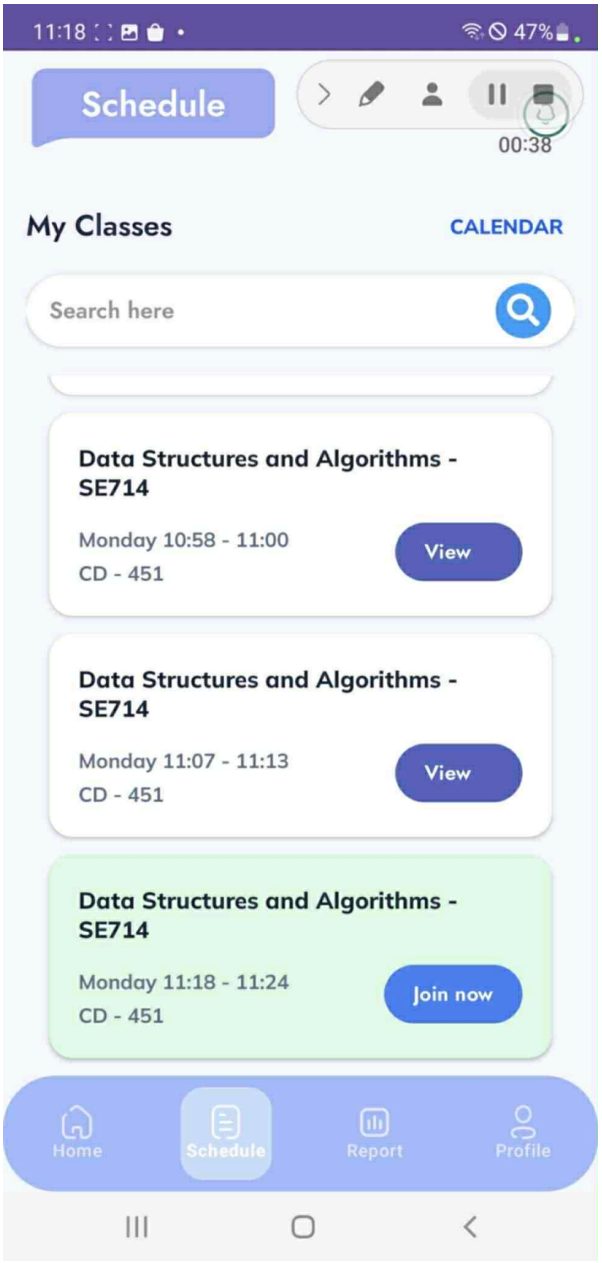


Figure xx - Student schedule screen

Step 1: From the bottom navigation bar, select **Schedule**.

Step 2: The system displays the **My Classes** list with all scheduled sessions.

Step 3: Each session card shows:

- **Course name and code** (e.g., *Data Structures and Algorithms – SE714*)
- **Date and time** of the session (e.g., *Monday 11:18 – 11:24*)
- **Classroom** (e.g., *CD – 451*)
- **Action button**:
    - **Join now** – if the session is currently active.
      View – for upcoming or past sessions.

Step 4: Tap **Join now** to enter an active session or **View** to see more session details.

Step 5: Use the search bar to quickly find a specific class by name or code.

3.2.3.2 View Final attendance

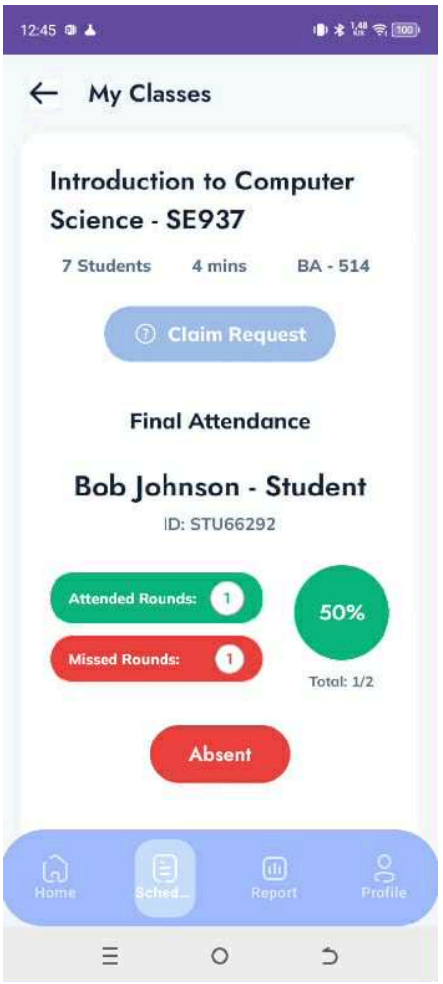


Figure xx - View final attendance screen

The **View Final Attendance** function allows students to check the finalized attendance result of a specific class session. The screen provides detailed statistics including the number of attended rounds, missed rounds, attendance percentage, and the student's final status (*Attended* or *Absent*).

Step 1: From the **Schedule** tab, select a class session.

Step 2: The system displays the **Final Attendance** screen with:

- **Course name and code** (e.g., *Introduction to Computer Science – SE937*)
- **Number of students** in the class
- **Session duration**
- **Classroom** information

Step 3: Review the **Final Attendance result** including:

- Student name and ID
- **Attended Rounds** count
- **Missed Rounds** count
- **Attendance percentage** (e.g., *50%*)
- Final status (*Attended* or *Absent*)

Step 4: If the result is correct, no further action is needed.

3.2.3.3 View Calender

Description: The View Calendar function allows students to visualize their sessions in a calendar format. By selecting a specific date, students can quickly see all scheduled classes for that day with corresponding time slots and classroom information.

Instruction:

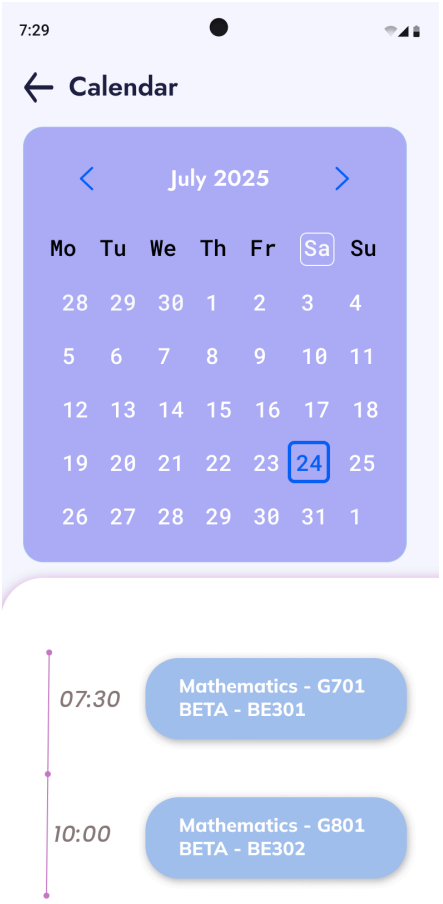


Figure xx - View calendar screen

Step 1: On the Schedule page, tap Calendar in the top-right corner.

Step 2: The system displays a monthly calendar view.

Step 3: Navigate between months using the left or right arrow icons.

Step 4: Select a specific date to view sessions scheduled for that day.

Step 5: The system displays the timeline of classes for the selected date, including:

- Subject and grade,
- Classroom (e.g., BETA – BE301),
- Start time (e.g., 07:30, 10:00).

3.2.3.4 FaceId request

Description: The Respond to FaceID Request function allows students to verify their identity and mark attendance when a lecturer initiates a FaceID check. Upon receiving the request notification, the student must complete face recognition within the given time limit for their attendance to be recorded.

Instruction:

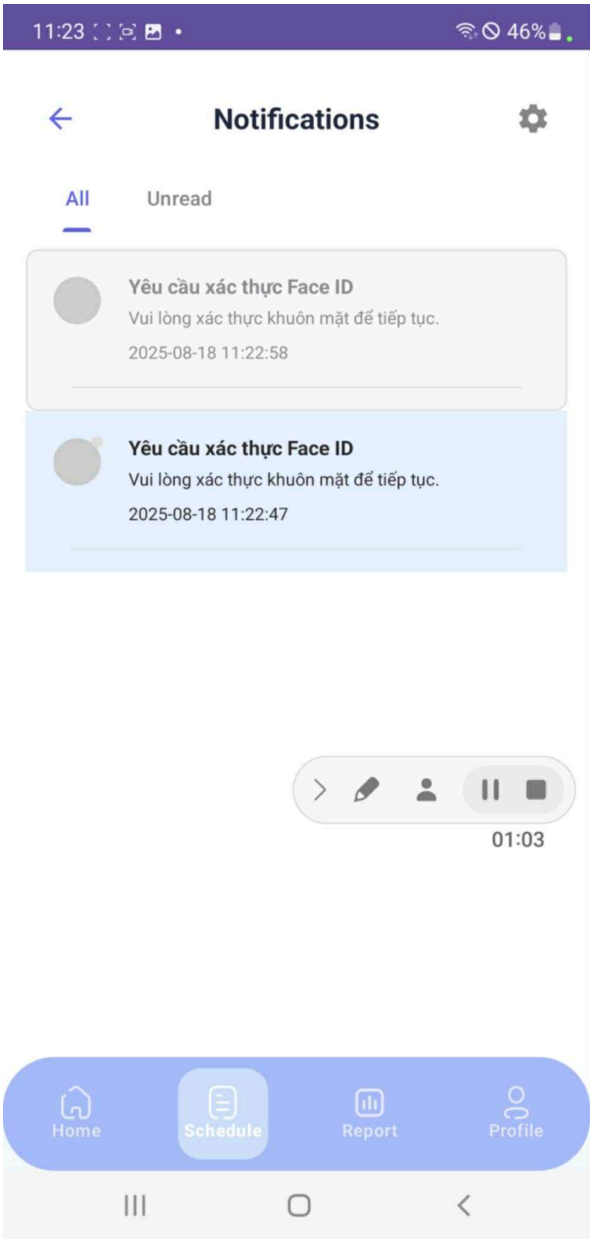


Figure xx - Face ID requests screen

3.2.4 Report Management

3.2.4.1 View report

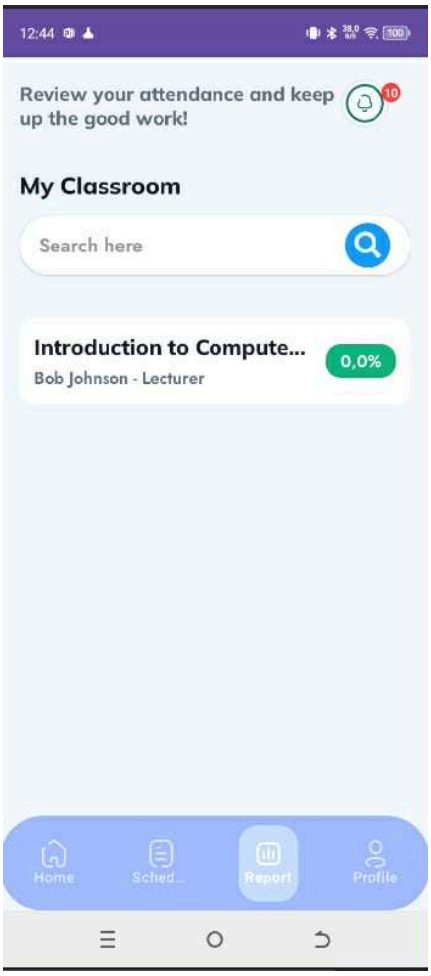


Figure xx - Report screen

The View Student Report function allows students to track their attendance across enrolled courses. It provides a summary of attendance percentage per course and enables students to review detailed attendance records for each session. This helps students monitor their participation and ensure they meet the required attendance rate.

Step 1: From the bottom navigation bar, select Report.

Step 2: The system displays the My Classroom list with all courses the student is enrolled in.

Step 3: Each course entry shows:

- Course name (e.g., Introduction to Computer Science)
- Lecturer name (e.g., Bob Johnson – Lecturer)
- Attendance percentage (e.g., 0.0%)

Step 4: Use the search bar to quickly locate a specific course.

Step 5: Tap on a course to view the detailed attendance report, including individual sessions and attendance status (Attended/Absent).

3.2.4.2 View report detail

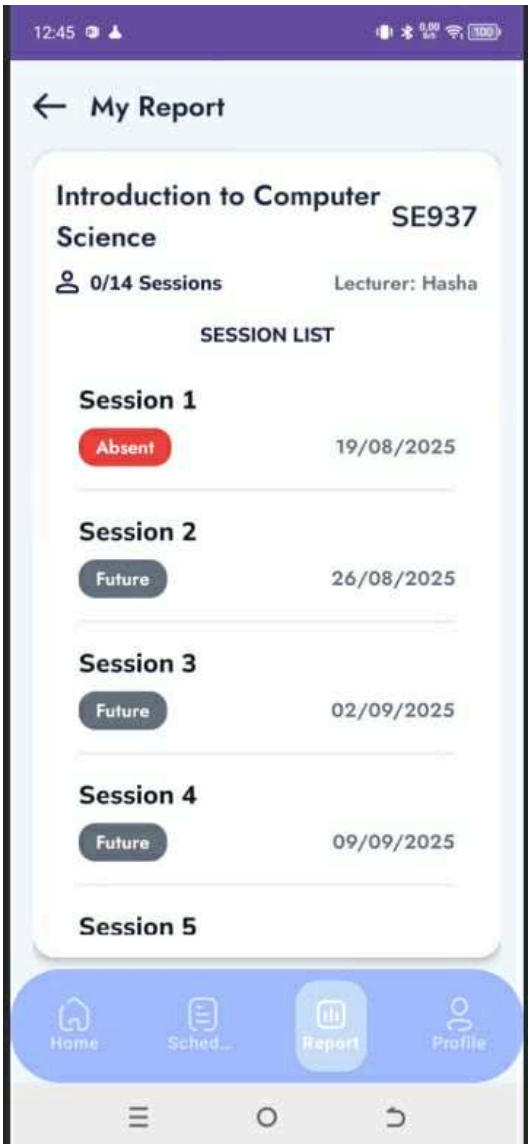


Figure xx - Report detail screen

The **View Report Detail** function allows students to see their attendance history for a specific course. It displays all sessions in the course, indicating whether the student was *Attended*, *Absent*, or if the session is scheduled for the *Future*. This helps students keep track of their participation and plan for upcoming classes.

Step 1: On the **Report** page, select a course from the **My Classroom** list.

Step 2: The system opens the **Report Detail** screen for the selected course.

Step 3: The top section displays:

- **Course name and code** (e.g., *Introduction to Computer Science – SE937*)
- **Lecturer name** (e.g., *Hasha*)
- **Attendance summary** (e.g., *0/14 Sessions*)

Step 4: The **Session List** shows each session with:

- **Session number** (e.g., *Session 1*)
- **Attendance status** (e.g., *Absent, Attended, Future*)
- **Date** of the session (e.g., *19/08/2025*)

Step 5: Scroll to view all sessions of the course.

Step 6: Tap on a session (if allowed) to see more detailed attendance information.

3.2.5 Profile Management

3.2.5.1 View student profile

Description: The **View Student Profile** function allows students to access and review their personal account information stored in the system. This includes name, email, phone number, role, account status, and the date of registration.

The profile is **view-only** — students **cannot edit or update** any information in this section.

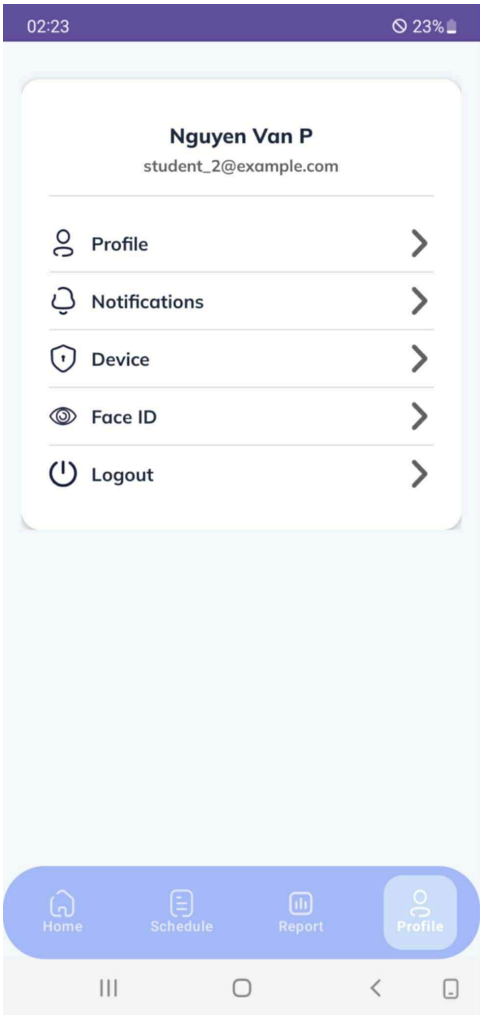


Figure xx - Student profile screen

Step 1: On the navigation bar, select **Profile**.

Step 2: The system displays the **Profile** menu, showing available options:

- Profile
- Notifications
- Device
- Logout

Step 3: Select **Profile** to view account details.

Step 4: The system displays **the Profile detail screen**, including:

- Lecturer's full name
- Email address
- Phone number
- Role (e.g., Lecturer)
- Account status (Active/Inactive)
- Joined date

Step 5: Review the information. This screen is read-only and does not allow modifications.

Step 6: To return, select the **Back** option at the top left.

3.2.5.2 Notification setting

Description: The **Notification Settings** function allows lecturers to manage how the application delivers notifications. Users can enable or disable sound, vibration, and general notifications according to their preference. These settings apply to all notifications generated by the system.

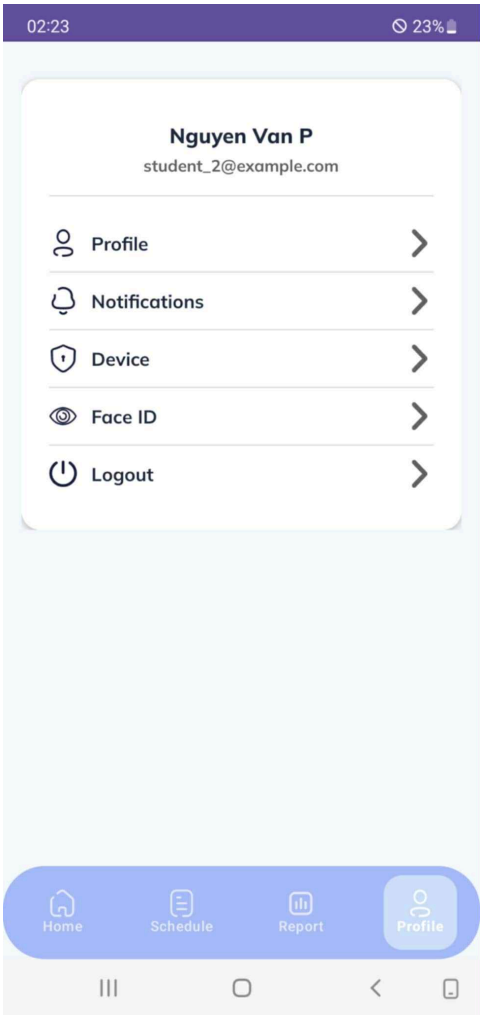


Figure xx - Notification setting button

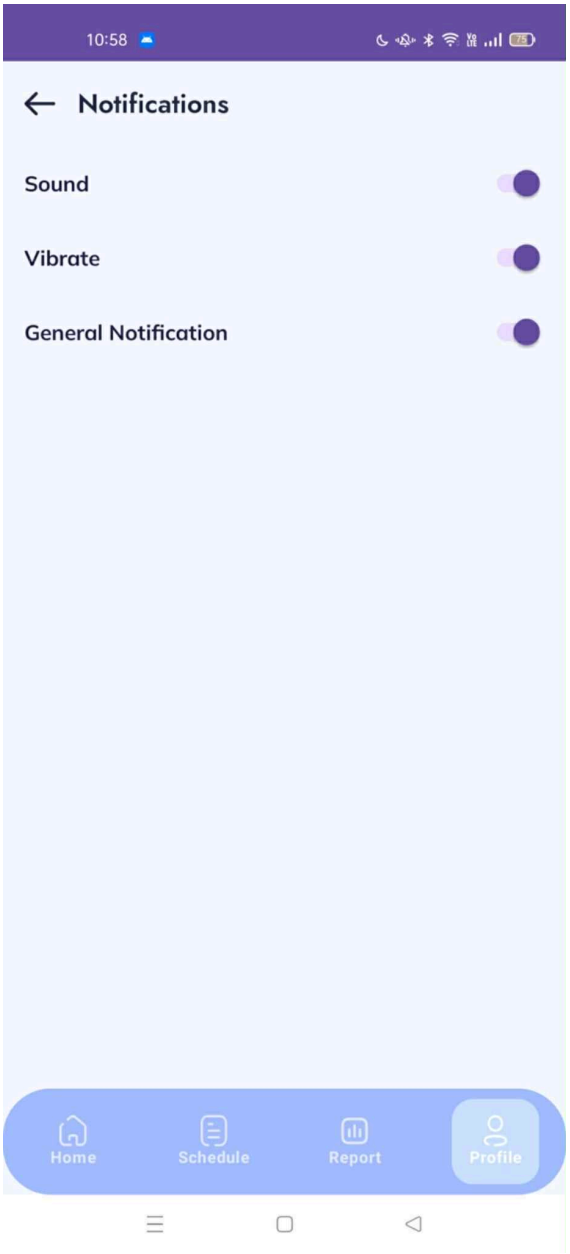


Figure xx - Notification setting screen

Step 1: From the Profile menu, select Notifications.

Step 2: The system displays the Notifications screen with the following options:

- Sound – Enable/disable notification sounds.
- Vibrate – Enable/disable vibration for notifications.
- General Notification – Turn all notifications on or off for the application.

Step 3: Use the toggle switches to turn each option on or off.

Step 4: Changes are saved automatically once a toggle is adjusted.

Step 5: To return, select the Back option at the top left.

3.2.5.3 Device Management

Description: The Device Management function allows lecturers to register their current mobile device with the system. Device registration is required in order to use the attendance features (such as starting and managing attendance sessions). Only registered devices can be used for attendance to ensure security and prevent unauthorized access.

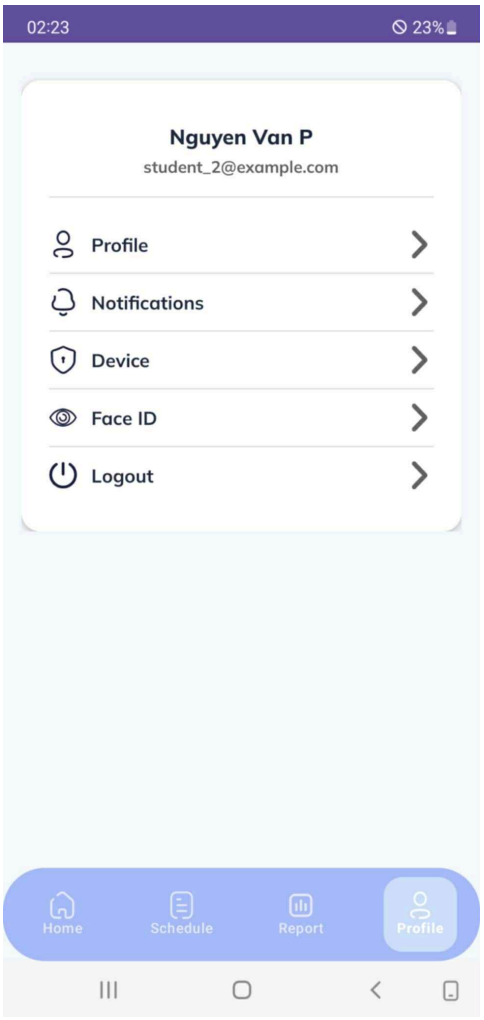


Figure xx - Device management button

Case 1 – No device registered

Step 1: From the Profile menu, select Device.

Step 2: The system displays a message:

“Device has not registered. Register now.”

Step 3: Tap Register Device.

Step 4: The system starts the registration process and shows the status:

“Registering…”

Step 5: Once successful, the device is linked to the lecturer’s account and becomes the active device for attendance.

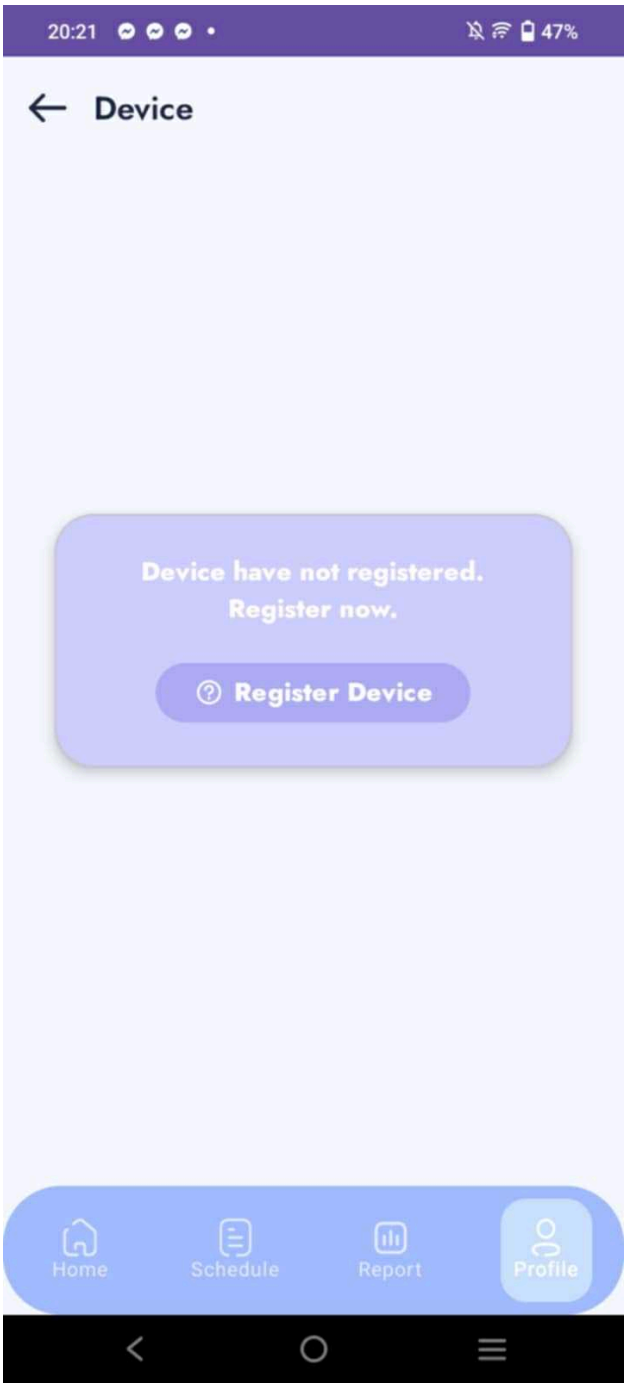


Figure xx - Device management screen - unregistered status

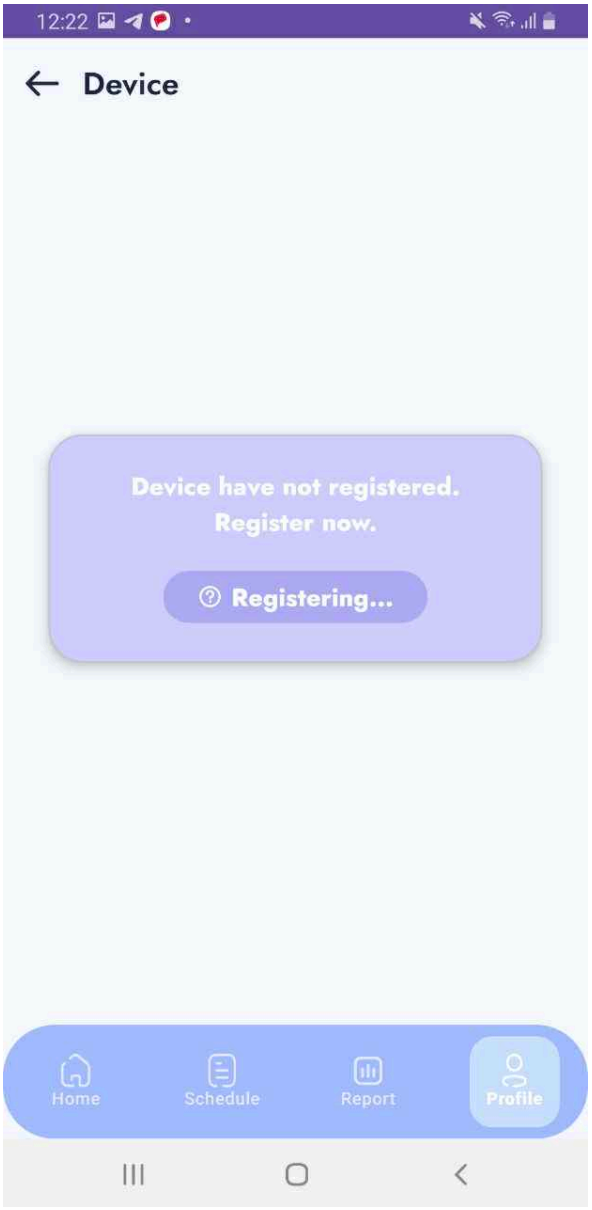


Figure xx - Device management screen - registering status

Case 2 – Device already registered

Step 1: From the Profile menu, select Device.

Step 2: The system displays the device information, including:

- Device Name
- Registered Date
- Model
- Manufacturer
- Platform (e.g., Android/iOS)
- OS Version
- App Version

Step 3: The status shows “Device Synced”, indicating the device is successfully linked.



Figure xx - Device management screen - synced status

3.2.5.4 Face Id

The **Register/Update Face ID** function allows students to scan and save their facial identity in the system. This biometric authentication is required for attendance verification. Students can register Face ID for the first time or update it when needed (e.g., appearance changes, device change).

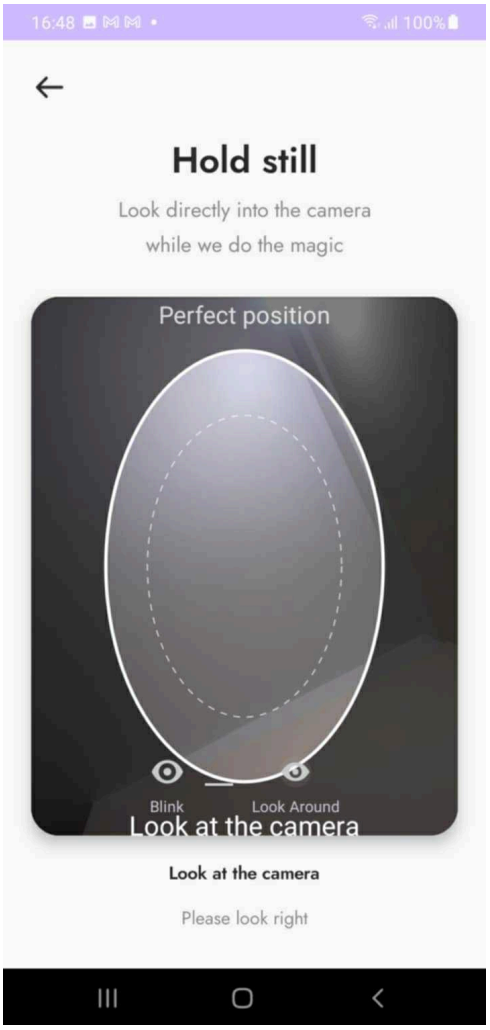


Figure xx - Register Face ID screen

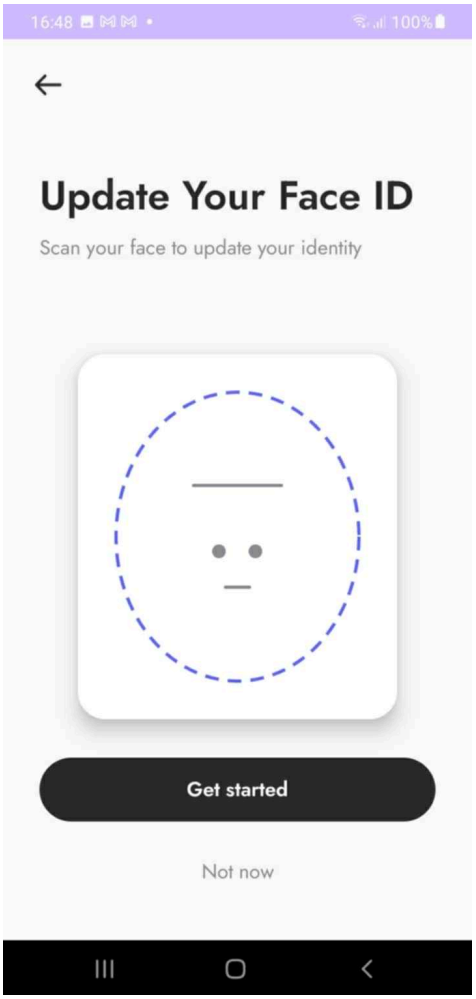


Figure xx - Update Face ID screen



Figure xx - Register Face ID success screen

Step 1: From the application menu, select **Register/Update Face ID**.

Step 2: The system displays the **Face ID Verification** screen with instructions to position the face in the camera frame.

Step 3: Tap **Get started** to begin scanning.

Step 4: Follow the on-screen guidance:

- Look directly at the camera
- Hold still
- Blink and look around if requested

Step 5: The system captures and processes the student's face data.

Step 6: Once successful, a confirmation screen appears:
"Register Face ID successfully!"

Step 7: The Face ID is now stored and synced with the student's account for use in attendance sessions.